



California Regional Water Quality Control Board

Santa Ana Region

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Agency Secretary

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Arnold Schwarzenegger
Governor

<http://www.waterboards.ca.gov/santaana>

ORDER NO. R8-2006-0003 NPDES NO. CA0105376

The following Discharger is authorized to discharge in accordance with the waste discharge requirements set forth in this Order:

Discharger	City of Beaumont
Name of Facility	Wastewater Treatment Plant No. 1
Facility Address	715 W. 4 th Street
	Beaumont, CA 92223
	Riverside County

The Discharger is authorized to discharge from the following discharge points as set forth below:

Discharge Point	Effluent Description	Discharge Point Latitude	Discharge Point Longitude	Receiving Water
001	Tertiary treated and disinfected	33 °, 55', 24" N	116°, 59', 34" W	Cooper's Creek; San Timoteo Management Zone (STMZ)
002	Recycled Water	33 °, 55', 25" N	116°, 59', 35" W	Beaumont Groundwater Management Zone
003	Stormwater	33 °, 55', 25" N	116°, 59', 31" W	Cooper's Creek; STMZ
004	Stormwater	33 °, 55', 24" N	116°, 59', 38" W	Cooper's Creek; STMZ
005	Stormwater	33 °, 55', 23" N	116°, 59', 42" W	Cooper's Creek, STMZ
006	Stormwater	33 °, 55', 25" N	116°, 59', 24" W	Cooper's Creek, STMZ

This Order was adopted by the Regional Water Board on:	January 18, 2006
This Order shall become effective on:	January 18, 2006
This Order shall expire on:	January 18, 2011
The U.S. Environmental Protection Agency (U.S. EPA) and the Regional Water Board have classified this discharge as a major discharge.	
The Discharger shall file a Report of Waste Discharge in accordance with Title 23, California Code of Regulations, <u>not later than 180 days in advance of the Order expiration date</u> as application for issuance of new waste discharge requirements.	

IT IS HEREBY ORDERED, that Order No. 00-10 is superseded upon the effective date of this Order except for enforcement purposes, and, in order to meet the provisions contained in Division 7 of the California Water Code (CWC) and regulations adopted therein, and the provisions of the federal Clean Water Act (CWA), and regulations and guidelines adopted therein, the Discharger shall comply with the requirements in this Order.

I, Gerard J. Thibeault, Executive Officer, do hereby certify that Order No. R8-2006-0003 with all attachments is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Santa Ana Region, on January 18, 2006.

Gerard J. Thibeault, Executive Officer

**CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
REGION 8, SANTA ANA REGION**

**ORDER NO. R8-2006-0003
NPDES NO. CA0105376**

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I. FACILITY INFORMATION

The following Discharger is authorized to discharge in accordance with the Waste Discharge Requirements set forth in this Order:

Discharger	City of Beaumont
Name of Facility	Wastewater Treatment Plant No. 1, Beaumont
Facility Address	715 W. 4 th Street
	Beaumont, CA 92223
	Riverside County
Facility Contact, Title, and Phone	Alan Kapanicas, City Manager, (951) 769-8534
Mailing Address	550 E. 6 th Street, Beaumont, CA 92223
Type of Facility	POTW
Facility Design Flow	4 MGD

II. FINDINGS

The California Regional Water Quality Control Board, Santa Ana Region (hereinafter Regional Water Board), finds:

- A. Background.** The City of Beaumont (hereinafter Discharger) is currently discharging under Order No. 00-10 and National Pollutant Discharge Elimination System (NPDES) Permit No. CA0105376. The Discharger submitted a Report of Waste Discharge, dated October 28, 2004, and applied for a NPDES permit renewal to discharge up to 4 MGD of tertiary treated wastewater from the City of Beaumont Wastewater Treatment Plant No. 1, hereinafter Facility.
- B. Facility Description.** The Discharger operates Wastewater Treatment Plant No. 1 through Urban Logic Consultants, a private contractor. The treatment system consists of bar screens, aeration/equalization, clarification, sand filtration, UV disinfection, sludge thickening/drying, and aerobic digestion. Wastewater is discharged from Discharge point 001 to Cooper's Creek, which leads to San Timoteo Creek, Reach 3, a tributary to the Santa Ana River, Reach 5. Reach 3 of San Timoteo Creek is unlined and flows in the Creek recharge the underlying San Timoteo Management Zone. While the discharge is to Cooper's Creek, it is considered a *de facto* discharge to San Timoteo Creek and the San Timoteo Management Zone. Most of the wastewater produced at Treatment Plant No. 1 will be recycled for uses overlying the Beaumont Management Zone, including landscape irrigation. Attachment B shows the location of the facility. Attachment C provides a flow schematic of the treatment process at the facility.
- C. Legal Authorities.** This Order is issued pursuant to section 402 of the Federal Clean Water Act (CWA) and implementing regulations adopted by the U.S. Environmental Protection Agency (USEPA) and Chapter 5.5, Division 7 of the California Water Code (CWC). It shall serve as a NPDES permit for point source discharges from this facility to surface waters. This Order also serves as Waste Discharge Requirements (WDRs) pursuant to Article 4, Chapter 4 of the CWC.

- D. Background and Rationale for Requirements.** The Regional Water Board developed the requirements in this Order based on information submitted as part of the application and through monitoring and reporting programs. Attachment F, which contains background information and rationale for Order requirements, is hereby incorporated into this Order and constitutes part of the Findings for this Order. Attachments A through E and G through L are also incorporated into this Order.
- E. California Environmental Quality Act (CEQA).** This action to adopt an NPDES permit is exempt from the provisions of the California Environmental Quality Act (Public Resources Code Section 21100, et seq.) in accordance with Section 13389 of the CWC.
- F. Technology-based Effluent Limitations.** The Code of Federal Regulations (CFR) at 40 CFR §122.44(a) requires that permits include applicable technology-based limitations and standards. This Order includes technology-based effluent limitations based on tertiary treatment or equivalent requirements that meet both the technology-based secondary treatment standards for publicly owned treatment works (POTWs) and protect the beneficial uses of the receiving waters. The Regional Water Board has considered the factors listed in CWC §13241 in establishing these requirements. A discussion of the development of the technology-based effluent limitations is included in the Fact Sheet (Attachment F).
- G. Water Quality-Based Effluent Limitations.** Section 122.44(d) of 40 CFR requires that permits include water quality-based effluent limitations (WQBELs) to attain and maintain applicable numeric and narrative water quality objectives to protect the beneficial uses of the receiving water. Where numeric water quality objectives have not been established, 40 CFR §122.44(d) specifies that WQBELs may be established using USEPA criteria guidance under CWA section 304(a), proposed State objectives or a State policy interpreting narrative objectives supplemented with other relevant information, or an indicator parameter. EPA and the State Water Board have determined that for toxic pollutants discharges to non-ocean waters, it is not practicable to express water quality-based effluent limitations as an average weekly and an average monthly, and recommend using a maximum daily and an average monthly effluent limitation for such discharges. This Order implements this recommendation.
- H. Water Quality Control Plans.** The Regional Water Board adopted a revised Water Quality Control Plan for the Santa Ana Region (hereinafter Basin Plan) that became effective on January 24, 1995. The Basin Plan designates beneficial uses, establishes water quality objectives, and contains implementation programs and policies to achieve those objectives for all waters in the Santa Ana Region addressed through the plan. More recently, the Basin Plan was amended significantly to incorporate revised boundaries for groundwater subbasins, now termed "management zones", new nitrate-nitrogen and TDS objectives for the new management zones, and new nitrogen and TDS management strategies applicable to both surface and ground waters. This Basin Plan Amendment was adopted by the Regional Water Board on January 22, 2004. The State Water Resources Control Board (State Water Board) and Office of Administrative Law (OAL) approved the Amendment on September 30, 2004 and December 23, 2004, respectively. The surface water provisions of the Amendment are awaiting approval by the U.S. Environmental Protection Agency. This Order implements those provisions, which, for the City of Beaumont, are more stringent than those in the Basin Plan. In addition, State Water Board Resolution No. 88-63 requires that, with certain exceptions, the Regional Water Board assign the

municipal and domestic supply use to water bodies that do not have beneficial uses listed in the Basin Plan. Beneficial uses applicable to receiving waters are as follows:

Discharge Point	Receiving Water Name	Beneficial Use(s)
001, 003, 004, 005, and 006	Copper's Creek and San Timoteo Creek, Reach 3 ¹	a. Groundwater recharge, b. Water contact recreation, c. Non-contact water recreation, d. Warm freshwater habitat, and e. Wildlife habitat,.
001, 003, 004, 005, and 006	Santa Ana River, Reach 5 ²	a. Agricultural supply, b. Groundwater recharge, c. Water contact recreation, d. Non-contact water recreation, e. Warm freshwater habitat, f. Wildlife habitat, and g. Rare, threatened, or endangered species.
001, 003, 004, 005, and 006	San Timoteo Groundwater Management Zone	a. Municipal and domestic supply, b. Agricultural Supply, c. Industrial process supply, and d. Industrial service supply
002	Beaumont Groundwater Management Zone	a. Municipal and domestic supply, b. Agricultural Supply, c. Industrial process supply, and d. Industrial service supply

Requirements of this Order specifically implement the applicable Water Quality Control Plans.

- I. Stormwater.** On April 17, 1997, the State Board adopted the General Industrial Storm Water Permit, Order No. 97-03-DWQ, NPDES No. CAS000001. This General Permit implements the Final Regulations (40 CFR 122, 123, and 124) for storm water runoff published on November 16, 1990 by EPA in compliance with Section 402(p) of the Clean Water Act (CWA). This Order includes pertinent provisions of the General Industrial Storm Water permit appropriate for this discharge. The Regional Water Board has determined that pollution prevention is necessary to achieve water quality objectives. Consequently, this Order requires the Discharger to establish, update as necessary and implement a pollution prevention plan and stormwater monitoring.
- J. National Toxics Rule (NTR) and California Toxics Rule (CTR).** USEPA adopted the NTR on December 22, 1992, which was amended on May 4, 1995 and November 9, 1999, and the CTR on May 18, 2000, which was amended on February 13, 2001. These rules include water quality criteria for priority pollutants and are applicable to this discharge.

¹ Excepted from municipal and domestic supply (MUN)

² Excepted from municipal and domestic supply downstream of Orange Avenue (Redlands)

- K. State Implementation Policy.** On March 2, 2000, the State Water Board adopted the *Policy for Implementation of Toxics Standards for Inland Surface Waters, Enclosed Bays, and Estuaries of California* (State Implementation Policy or SIP). The SIP became effective on April 28, 2000, with respect to the priority pollutant criteria promulgated for California by the USEPA through the NTR and to the priority pollutant objectives established by the Regional Water Boards in their basin plans, with the exception of the provision on alternate test procedures for individual discharges that have been approved by USEPA Regional Administrator. The alternate test procedures provision was effective on May 22, 2000. The SIP became effective on May 18, 2000. The SIP includes procedures for determining the need for and calculating WQBELs and requires Dischargers to submit data sufficient to do so. On February 24, 2005, the State Water Board amended the SIP. The Office of Administrative Law (OAL) approved the amendments on May 31, 2005. On July 13, 2005, the United States Environmental Protection Agency (USEPA) approved the amendments.
- L. Antidegradation Policy.** Section 131.12 of 40 CFR requires that State water quality standards include an antidegradation policy consistent with the federal policy. The State Water Board established California's antidegradation policy in State Water Board Resolution 68-16, which incorporates the requirements of the federal antidegradation policy. Resolution 68-16 requires that existing quality of waters be maintained unless degradation is justified based on specific findings. As discussed in the Fact Sheet (Attachment F), the permitted discharge is consistent with the antidegradation provision of 40 CFR §131.12 and State Water Board Resolution 68-16.
- M. Anti-Backsliding Requirements.** Sections 402(o)(2) and 303(d)(4) of the CWA and federal regulations at 40 CFR § 122.44(l) prohibit backsliding in NPDES permits. These anti-backsliding provisions require effluent limitations in a reissued permit to be as stringent as those in the previous permit, with some exceptions where limitations may be relaxed. All effluent limitations in this Order are at least as stringent as the effluent limitations in the previous Order No. 00-10.
- N. Monitoring and Reporting.** Section 122.48 of 40 CFR requires that all NPDES permits specify requirements for recording and reporting monitoring results. Sections 13267 and 13383 of the CWC authorize the Regional Water Boards to require technical and monitoring reports. The Monitoring and Reporting Program establishes monitoring and reporting requirements to implement federal and State requirements. This Monitoring and Reporting Program is provided in Attachment E.
- O. Standard and Special Provisions.** Standard Provisions, which in accordance with 40 CFR §§122.41 and 122.42, apply to all NPDES discharges and must be included in every NPDES permit, are provided in Attachment D. The Regional Water Board has also included in this Order special provisions applicable to the Discharger. A rationale for the special provisions contained in this Order is provided in the attached Fact Sheet (Attachment F).

- P. Notification of Interested Parties.** The Regional Water Board has notified the Discharger and interested agencies and persons of its intent to prescribe Waste Discharge Requirements for the discharge and has provided them with an opportunity to submit their written comments and recommendations. Details of notification are provided in the Fact Sheet (Attachment F) of this Order.
- Q. Consideration of Public Comment.** The Regional Water Board, in a public meeting, heard and considered all comments pertaining to the discharge. Details of the Public Hearing are provided in the Fact Sheet (Attachment F) of this Order.

III. DISCHARGE PROHIBITIONS

- A. Wastes discharged shall be limited to tertiary treated and disinfected effluent.
- B. Discharge of wastewater at a location or in a manner different from that described in the Findings is prohibited.
- C. The bypass or overflow of untreated wastewater or wastes to surface waters or surface water drainage courses is prohibited, except as allowed in Standard Provision I.H. of Attachment D, Federal Standard Provisions.
- D. The discharge of any substances in concentrations toxic to animal or plant life in the affected receiving water is prohibited.
- E. The discharge of any radiological, chemical, or biological warfare agent or high level radiological waste is prohibited.

IV. EFFLUENT LIMITATIONS AND DISCHARGE SPECIFICATIONS

A. Effluent Limitations:

1. Final Effluent Limitations

- a. The discharge of tertiary treated wastewater shall maintain compliance with the following effluent limitations, with compliance measured at effluent box monitoring location (M-001), as described in the attached Monitoring and Reporting Program (Attachment E):

Parameter	Units	Effluent Limitations				
		Average Monthly	Average Weekly	Maximum Daily	Instantaneous Minimum	Instantaneous Maximum
Biochemical Oxygen Demand 5-day @ 20°C	mg/L	20	30	--	--	--
	lbs/day	667	1001	--	--	--
Total Suspended Solids	mg/L	20	30	--	--	--
	lbs/day	667	1001	--	--	--
PH	standard units	--	--	--	6.5	8.5
Selenium	ug/L	4	--	8	--	--
	lbs/day	0.13	--	0.27	--	--

- b. Percent Removal: The average monthly percent removal of BOD 5-day 20°C and total suspended solids shall not be less than 85 percent.
- c. TDS Limitations: The following TDS limitations apply to surface water discharges, with compliance measured at effluent box monitoring location (M-001), as described in the attached Monitoring and Reporting Program (Attachment E).
 - (1) Provided that maximum benefit is demonstrated (see Special Provisions C.1.a and C.1.b), TDS limits are specified as follows. The lower of the two total dissolved solids limits specified in (a) or (b), below, is the limit.
 - (a) The 12-month running average total dissolved solids concentration and mass emission rate shall not exceed 490 mg/l and 16,346 lbs/day³, respectively, and
 - (b) The 12-month total dissolved solids concentration shall not exceed the 12-month average total dissolved solids concentration in the water supply by more than 250 mg/l.
 - (2) If maximum benefit is not demonstrated (see Special Provisions C.1.c), the 12-month running average total dissolved solids concentration and mass emission rate shall not exceed 320 mg/l and 10,675 lbs/day⁴, respectively,
- d. Total Inorganic Nitrogen (TIN) Limitations: The following TIN limitations apply to surface water discharges, with compliance measured at effluent box monitoring location (M-001), as described in the attached Monitoring and Reporting Program (Attachment E).
 - (1) Provided that maximum benefit is demonstrated (see Provision 1a and b), the 12-month running average TIN concentration and mass emission rate shall not exceed 6 mg/l and 200 lbs per day⁵, respectively.
 - (2) If maximum benefit is not demonstrated (see Special Provisions C.1.c), the 12-month running average TIN concentration and mass emission rate shall not exceed 4.1 mg/l and 137 lbs per day⁶, respectively.
- e. The discharge shall at all times be adequately oxidized, filtered, and disinfected tertiary treated wastewater and shall meet the following limitations:
 - (1) The turbidity of the filter effluent shall not exceed any of the following:
 - (a) Average of 2 Nephelometric Turbidity Unit (NTU) within any 24-hour period;
 - (b) 5 NTU more than 5 percent of the time in any 24-hour period; and

³ Calculated from 4 mgd x 8.34 x 490 mg/l.

⁴ Calculated from 4 mgd x 8.34 x 320 mg/l.

⁵ Calculated from 4 mgd x 8.34 x 6 mg/l.

⁶ Calculated from 4 mgd x 8.34 x 4.1 mg/l.

- (c) 10 NTU at any time.
 - (2) The 7-day median number of total coliform shall not exceed a Most Probable Number (MPN) of 2.2 total coliform bacteria per 100 milliliters (ml).
 - (3) The number of total coliform organism shall not exceed an MPN of 23 total coliform bacteria per 100 ml in more than one sample in any 30-day period.
 - (4) No total coliform sample shall exceed an MPN of 240 total coliform bacteria per 100 ml.
- 2. Toxicity Requirements:
 - a. There shall be no acute or chronic toxicity in the plant effluent nor shall the plant effluent cause any acute or chronic toxicity in the receiving water. All waters shall be maintained free of toxic substances in concentrations which are toxic to, or which produce detrimental physiological responses in human, plant, animal, or indigenous aquatic life. This Order contains no numeric limitation for toxicity. However, the discharger shall conduct chronic toxicity monitoring.
 - b. The discharger shall implement the accelerated monitoring specified in Attachment E when the result of any single chronic toxicity test of the effluent exceeds 1.0 TUc.
 - c. The discharger shall develop an Initial Investigation Toxicity Reduction Evaluation (IITRE) work plan that describes the steps the discharger intends to follow if required by Toxicity Requirement No. 2.d., below. The work plan shall include at a minimum:
 - (1) A description of the investigation and evaluation techniques that will be used to identify potential causes/sources of the exceedance, effluent variability, and/or efficiency of the treatment system in removing toxic substances. This shall include a description of an accelerated chronic toxicity testing program.
 - (2) A description of the methods to be used for investigating and maximizing in-house treatment efficiency and good housekeeping practices.
 - (3) A description of the evaluation process to be used to determine if implementation of a more detailed TRE/TIE is necessary.
 - d. The discharger shall implement the IITRE work plan whenever the results of chronic toxicity tests of the effluent exceed:
 - (a) A two month median value of 1.0 TUc for survival or reproduction endpoint or,
 - (b) Any single test value of 1.7 TUc for survival endpoint.
 - e. The discharger shall develop a detailed Toxicity Reduction Evaluation and Toxicity Identification Evaluation (TRE/TIE) work plan that shall describe the steps the discharger intends to follow if the implemented IITRE fails to identify the cause of, or rectify, the toxicity.

- f. The discharger shall use as guidance, at a minimum, EPA manuals EPA/600/2-88/070 (industrial), EPA/600/4-89-001A (municipal), EPA/600/6-91/005F (Phase I), EPA/600/R-92/080 (Phase II), and EPA-600/R-92/081 (Phase III) to identify the cause(s) of toxicity. If during the life of this Order the aforementioned EPA manuals are revised or updated, the revised/updated manuals may also be used as guidance. The detailed TRE/TIE work plan shall include:
 - (a) Further actions to investigate and identify the cause of toxicity;
 - (b) Actions the discharger will take to mitigate the impact of the discharge and to prevent the recurrence of toxicity; and
 - (c) A schedule for these actions.
- g. The discharger shall implement the TRE/TIE workplan if the IITRE fails to identify the cause of, or rectify, the toxicity, or if in the opinion of the Executive Officer the IITRE does not adequately address an identified toxicity problem.
- h. The discharger shall assure that adequate resources are available to implement the required TRE/TIE.

B. **Land Discharge Specifications:** Limitations necessary to protect groundwater recharged by surface water discharges are specified in IV.A. - Effluent Limitations, above. Limitations necessary to protect groundwater as the result of recycled water use are specified in IV.C. Reclamation Specifications, below.

C. **Reclamation Specifications:**

1. Beginning February 1, 2006, the use of recycled water for landscape irrigation or other similar uses shall maintain compliance with the following limitations. Compliance is to be measured at monitoring location Rec-001 where representative samples of recycled water can be obtained for laboratory testing and analysis as described in the attached Monitoring and Reporting Program (Attachment E).

a. Physical/Biological Limitations:

Parameter	Units	Recycled Water Limitations				
		Average Monthly	Average Weekly	Maximum Daily	Instantaneous Minimum	Instantaneous Maximum
Biochemical Oxygen Demand 5-day @ 20°C	mg/L	30	45	--	--	--
Total Suspended Solids	mg/L	30	45	--	--	--
pH	standard units	--	--	--	6	9

- b. TDS Limitations: The following TDS limitations apply to recycled water use sites overlying the Beaumont management zone:

- (1) Provided that maximum benefit is demonstrated (see Special Provisions VI.C.1.a. and VI.C.1.b.), the 12-month running average TDS concentration of recycled water shall not exceed 490 mg/L and the 10-year volume weighted rolling average TDS concentration of the commingled non-potable water supply and recycled water shall not exceed 390 mg/l.
 - (2) If maximum benefit is not demonstrated (see Special Provisions VI.C.1.c.), the 12-month running average total dissolved solids concentration shall not exceed 230 mg/l
- c. Total Inorganic Nitrogen (TIN) Limitations: The following TIN limitations apply to the non-potable water supply (recycled water blended with untreated State Project water) to be used at non-potable water use sites overlying the Beaumont management zone:
 - (1) Provided that maximum benefit is demonstrated (see Special Provisions VI.C.1.a. and VI.C.1.b.), the 12-month running average TIN concentration shall not exceed 6.67 mg/l.
 - (2) If maximum benefit is not demonstrated (see Special Provisions VI.C.1.c.), the 12-month running average TIN concentration shall not exceed 2 mg/l.
- d. Selenium: The 12-month running average Selenium concentration shall not exceed 10 ug/L
- e. Recycled water shall at all times be adequately oxidized, filtered, and disinfected tertiary treated wastewater and shall meet the following limitations:
 - (1) The turbidity of the filter effluent shall not exceed any of the following:
 - (a) Average of 2 Nephelometric Turbidity Unit (NTU) within any 24-hour period;
 - (b) 5 NTU more than 5 percent of the time in any 24-hour period; and
 - (c) 10 NTU at any time.
 - (2) The 7-day median number of total coliform shall not exceed a Most Probable Number (MPN) of 2.2 total coliform bacteria per 100 milliliters (ml).
 - (3) The number of total coliform organism shall not exceed an MPN of 23 total coliform bacteria per 100 ml in more than one sample in any 30-day period.
 - (5) No total coliform sample shall exceed an MPN of 240 total coliform bacteria per 100 ml.
2. The use of recycled water shall only commence after final approval for such use is granted by the California Department of Health Services (CDHS). The Discharger shall provide the Regional Water Board with a copy of the CDHS approval letter within 30 days of the approval notice.

3. The Discharger shall be responsible for assuring that recycled water is delivered and utilized in conformance with this Order, the recycling criteria contained in Title 22, Division 4, Chapter 3, Sections 60301 through 60355, California Code of Regulations, and the "Guidelines for Use of Reclaimed Water" by the California Department of Health Services. The discharger shall conduct periodic inspections of the facilities of the recycled water users to monitor compliance by the users with this Order.
4. The Discharger shall establish and enforce Rules and Regulations for Recycled Water users, governing the design and construction of recycled water use facilities and the use of recycled water in accordance with the uniform statewide recycling criteria established pursuant to the California Water Code Section 13521.
 - a. Use of recycled water by the discharger shall be consistent with its Rules and Regulations for Recycled Water Use.
 - b. Any revisions made to the Rules and Regulations shall be subject to the review of the Regional Water Board, the California Department of Health Services, and the County of Riverside Department of Environmental Health. The revised Rules and Regulations or a letter certifying that the discharger's Rules and Regulations contain the updated provisions in this Order, shall be submitted to the Regional Water Board within 60 days of adoption of this Order by the Regional Water Board.
5. The Discharger shall, within 60 days of the adoption of this Order, review and update as necessary its program to conduct compliance inspections of recycled water reuse sites. Inspections shall determine the status of compliance with the discharger's Rules and Regulations for Recycled Water Use.
6. The storage, delivery, or use of recycled water shall not individually or collectively, directly or indirectly, result in a pollution or nuisance, or adversely affect water quality, as defined in the California Water Code
7. Prior to delivering recycled water to any new user, the discharger shall submit to the Regional Water Board, the California Department of Health Services and the Riverside County Health Department a report containing the following information for review and approval:
 - a. The average number of persons estimated to be served at each use site area on a daily basis.
 - b. The specific boundaries of the proposed use site area including a map showing the location of each facility, drinking water fountain, and impoundment to be used.
 - c. The person or persons responsible for operation of the recycled water system at each use area.
 - d. The specific use to be made of the recycled water at each use area.

- e. The methods to be used to assure that the installation and operation of the recycled system will not result in cross connections between the recycled water and potable water piping systems. This shall include a description of the pressure, dye or other test methods to be used to test the system.
- f. Plans and specifications which include following:
 - 1) Proposed piping system to be used.
 - 2) Pipe locations of both the recycled and potable systems.
 - 3) Type and location of the outlets and plumbing fixtures that will be accessible to the public.
 - 4) The methods and devices to be used to prevent backflow of recycled water into the potable water system.
 - 5) Plan notes relating to specific installation and use requirements.
8. The Discharger shall require the user(s) to designate an on-site supervisor responsible for the operation of the recycled water distribution system within the recycled water use area. The supervisor shall be responsible for enforcing this Order, prevention of potential hazards, the installation, operation and maintenance of the distribution system, maintenance of the distribution and irrigation system plans in "as-built" form, and for the distribution of the recycled wastewater in accordance with this Order.

V. RECEIVING WATER LIMITATIONS

A. Surface Water Limitations

1. Receiving water limitations are based upon water quality objectives contained in the Basin Plan. As such, they are a required part of this Order. The discharge shall not cause the following in Cooper's Creek, San Timoteo Creek, in the Santa Ana River, Reach 5, or in downstream Reaches of the Santa Ana River:
 - a. Coloration of the receiving waters, which causes a nuisance or adversely affects beneficial uses.
 - b. Deposition of oil, grease, wax or other materials in the receiving waters in concentrations which result in a visible film or in coating objects in the water, or which cause a nuisance or affect beneficial uses.
 - c. An increase in the amounts of suspended or settleable solids in the receiving waters, which will cause a nuisance or adversely affect beneficial uses as a result of controllable water quality factors.
 - d. Taste or odor-producing substances in the receiving waters at concentrations, which cause a nuisance or adversely affect beneficial uses.
 - e. The presence of radioactive materials in the receiving waters in concentrations, which are deleterious to human, plant or animal life.
 - f. The depletion of the dissolved oxygen concentration below 5.0 mg/l.
 - g. The temperature of the receiving waters to be raised above 90°F (32°C) during the period of June through October, or above 78°F (26°C) during the rest of the year.

- h. The concentration of pollutants in the water column, sediments, or biota to adversely affect the beneficial uses of the receiving water. The discharge shall not result in the degradation of inland surface water communities and populations, including vertebrate, invertebrate, and plant species.
 2. The discharge of wastes shall not cause a violation of any applicable water quality standards for receiving waters adopted by the Board or State Board, as required by the Clean Water Act and regulations adopted thereunder.
 3. Pollutants not specifically mentioned and limited in this Order shall not be discharged at levels that will bioaccumulate in aquatic resources to levels, which are harmful to human health.
 4. The discharge shall not contain constituent concentrations of mercury that will result in the bioaccumulation of methylmercury in fish flesh tissue greater than 0.3 milligram methylmercury/kilogram. (See also Section VI.C.2.g. and VI.C.3., below)
- B. Groundwater Limitations** for recycled water use at sites overlying the Beaumont Groundwater Management Zone.
1. The use of recycled water shall not cause the underlying groundwater to be degraded, unreasonably affect beneficial uses, or cause a condition of pollution or nuisance.

VI. PROVISIONS

A. Standard Provisions

1. **Standard Provisions.** The Discharger shall comply with all Standard Provisions included in Attachment D of this Order.
2. **Regional Water Board Standard Provisions.** The Discharger shall comply with the following provisions:
 - a. Neither the treatment nor the discharge of waste shall create, or threaten to create, a nuisance or pollution as defined by Section 13050 of the California Water Code.
 - b. The discharger shall optimize chemical additions needed in the treatment process to meet waste discharge requirements so as to minimize total dissolved solid increases in the recycled water.

- c. The discharger shall conduct a Pollutant Minimization Program (PMP) when there is evidence that the priority pollutant is present in the effluent above an effluent limitation (e.g., sample results reported as detected but not quantified (DNQ) when the effluent limitation is less than the MDL, sample results from analytical methods more sensitive than those methods included in the permit, presence of whole effluent toxicity, health advisories for fish consumption, results of benthic or aquatic organism tissue sampling) and either: (i) A sample result is reported as DNQ and the effluent limitation is less than the reported ML; or (ii) A sample result is reported as ND and the effluent limitation is less than the MDL. The PMP shall include, but not be limited to, the following actions and submittals acceptable to the Regional Water Board:
- (1) An annual review and semi-annual monitoring of potential sources of the reportable priority pollutant(s), which may include fish tissue monitoring and other bio-uptake sampling;
 - (2) Quarterly monitoring for the reportable priority pollutant(s) in the influent to the wastewater treatment system;
 - (3) Submittal of a control strategy designed to proceed toward the goal of maintaining concentrations of the reportable priority pollutant(s) in the effluent at or below the effluent limitation;
 - (4) Implementation of appropriate cost-effective control measures for the reportable priority pollutant(s), consistent with the control strategy; and
 - (5) An annual status report that shall be sent to the Regional Water Board including:
 - (a) All PMP monitoring results for the previous year;
 - (b) A list of potential sources of the reportable priority pollutant(s);
 - (c) A summary of all actions undertaken pursuant to the control strategy; and
 - (d) A description of actions to be taken in the following year.
- d. The discharger shall maintain a copy of this Order at the site so that it is available to site operating personnel at all times. Key operating personnel shall be familiar with its content.
- e. The discharger shall take all reasonable steps to minimize any adverse impact to receiving waters resulting from noncompliance with any requirements specified in this Order, including such accelerated or additional monitoring as necessary to determine the nature and impact of the noncomplying discharge.
- f. The provisions of this Order are severable, and if any provision of this Order, or the application of any provisions of this Order to any circumstance, is held invalid, the application of such provision to other circumstances, and the remainder of this Order shall not be affected thereby.
- g. Collected screenings, sludge, and other solids removed from liquid wastes shall be disposed of in a manner approved by the Regional Water Board's Executive Officer.

- h. If the discharger demonstrates a correlation between the biological oxygen demand (BOD) and total organic carbon (TOC) concentrations in the effluent to the satisfaction of the Executive Officer, compliance with the BOD limits contained in this Order may be determined based on analyses of the TOC of the effluent.
- i. In the event of any change in control or ownership of land or waste discharge facility presently owned or controlled by the discharger, the discharger shall notify the succeeding owner or operator of the existence of this Order by letter, a copy of which shall be forwarded to the Regional Water Board.
- j. The treatment facilities shall be designed, constructed, operated, and maintained to prevent inundation or washout due to floods with a 100-year return frequency.

B. Monitoring and Reporting Program Requirements

The Discharger shall comply with the Monitoring and Reporting Program, and future revisions thereto, in Attachment E of this Order. This monitoring and reporting program may be modified by the Executive Officer at any time during the term of this Order, and may include an increase in the number of parameters to be monitored, the frequency of the monitoring or the number and size of samples to be collected. Any increase in the number of parameters to be monitored, the frequency of the monitoring or the number and size of samples to be collected may be reduced back to the levels specified in the original monitoring and reporting program at the discretion of the Executive Officer.

C. Special Provisions

1. Reopener Provisions

- a. This Order may be reopened to address any changes in State or federal plans, policies or regulations that would affect the quality requirements for the discharges.
- b. This Order may be reopened once the 2004 Basin Plan amendment is approved by EPA, to include revised waste load allocations for discharges to Cooper's Creek as specified in the 2004 amended Basin Plan.
- a. This Order may be reopened to include effluent limitations for pollutants determined to be present in the discharge in concentrations that pose a reasonable potential to cause or contribute to violations of water quality objectives.
- b. This Order may be reopened and modified in accordance with the requirements set forth at 40 CFR 122 and 124, to include the appropriate conditions or limits to address demonstrated effluent toxicity based on newly available information, or to implement any EPA-approved new State water quality standards applicable to effluent toxicity.
- c. This Order may be reopened to incorporate appropriate biosolids requirements if the State Water Resources Control Board and the Regional Water Quality Control Board are given the authority to implement regulations contained in 40 CFR 503.

- d. This Order may be reopened to include an appropriate bioaccumulation based effluent limit for mercury if test results (as required in Attachment E of this Order) show that the concentration levels of methylmercury in the fish tissue are at or above 0.3 milligrams per kilogram.

2. Special Studies, Technical Reports and Additional Monitoring Requirements

By March 1, 2006, the discharger shall notify the Executive Officer of its continuous involvement with the comprehensive mercury investigation program currently being conducted by a group of Santa Ana River system dischargers. If the discharger discontinues its involvement with this comprehensive program, the discharger shall, within 60 days of that date, submit for the approval of the Executive Officer its plan for the annual testing of mercury levels in fish flesh samples collected from the Santa Ana River, upstream of, at, and downstream of the point of the River's confluence with San Timoteo Creek. Upon approval, the discharger shall implement the plan.

3. Best Management Practices and Pollution Prevention

- a. Storm water discharges shall not result in noncompliance with the lawful requirements of municipalities, counties, drainage districts, and other local agencies on storm water discharges into storm drain systems or other courses under their jurisdiction.
- b. Stormwater Pollution Prevention Plan - The discharger must update and implement the Storm Water Pollution Prevention Plan for the treatment facility in accordance with Attachment "J" of this Order.
- c. Best Management Practices Plan. The Discharger shall develop, notify the Regional Water Board of completion, and implement within 90 days of the effective date of this Order, a Best Management Practices Plan (BMPP). If necessary, the plan, or any existing plan, shall be updated to address any changes in operation and/or management of the facility. Notification that a plan has been updated shall be submitted to the Regional Water Board within 30 days of revision.

The BMPP shall be consistent with the general guidance contained in the EPA *Guidance Manual for Developing Best Management Practices (BMPs)* (EPA 833-B-93-004). In particular, a risk assessment of each area identified by the Discharger shall be performed to determine the potential for hazardous or toxic waste/material discharge to surface waters.

4. Compliance Schedules (Not Applicable)

5. Construction, Operation and Maintenance Specifications

- a. The discharger's wastewater treatment plant shall be supervised and operated by persons possessing certificates of appropriate grade pursuant to Title 23, Division 3, Chapter 14, California Code of Regulations.

- b. The discharger shall provide safeguards to assure that should there be reduction, loss, or failure of electric power, the discharger will comply with the requirements of this Order.
 - c. The discharger shall update as necessary, the "Operation and Maintenance Manual (O&M Manual)" which it has developed for the treatment facility to conform to latest plant changes and requirements. The O&M Manual shall be readily available to operating personnel onsite. The O&M Manual shall include the following:
 - (1) Description of the treatment plant table of organization showing the number of employees, duties and qualifications and plant attendance schedules (daily, weekends and holidays, part-time, etc). The description should include documentation that the personnel are knowledgeable and qualified to operate the treatment facility so as to achieve the required level of treatment at all times.
 - (2) Detailed description of safe and effective operation and maintenance of treatment processes, process control instrumentation and equipment.
 - (3) Description of laboratory and quality assurance procedures.
 - (4) Process and equipment inspection and maintenance schedules.
 - (5) Description of safeguards to assure that, should there be reduction, loss, or failure of electric power, the discharger will be able to comply with requirements of this Order.
 - (6) Description of preventive (fail-safe) and contingency (response and cleanup) plans for controlling accidental discharges, and for minimizing the effect of such events. These plans shall identify the possible sources (such as loading and storage areas, power outage, waste treatment unit failure, process equipment failure, tank and piping failure) of accidental discharges, untreated or partially treated waste bypass, and polluted drainage.
6. Special Provisions for Municipal Facilities (POTWs Only)
- a. Sludge Disposal Requirements
 - (1) Collected screenings, sludge, and other solids removed from liquid wastes shall be disposed of in a manner that is consistent with State Water Resources Control Board and Integrated Waste Management Board's joint regulations (Title 27) of the California Code of Regulations and approved by the Water Board's Executive Officer.
 - (2) The use and disposal of biosolids shall comply with existing Federal and State laws and regulations, including permitting requirements and technical standards included in 40 CFR 503.
 - (3) Any proposed change in biosolids use or disposal practice from a previously approved practice should be reported to the Executive Officer and EPA Regional Administrator at least 90 days in advance of the change.

- (4) The discharger shall take all reasonable steps to minimize or prevent any discharge or biosolids use or disposal that has the potential of adversely affecting human health or the environment.
- b. The Discharger shall submit a monthly report that validates that recycled water used for recharge is an oxidized and filtered wastewater. The report shall include:
 - (1) Description of when, how often and whether coagulation of the wastewater is employed in the treatment process. If coagulation is not used at all times, the users shall:
 - i. Continuously monitor the turbidity of the influent to the filters. Turbidity exceedances of 10 NTU or above at any time, and of 5 NTU for more than 15 minutes, shall be included in the monthly report;
 - ii. Certify that chemical addition for coagulation has been automatically employed when the filter influent turbidity exceeds 5 NTU for more than 15 minutes.
 - (2) Description of the type and rate of filtration employed in the treatment process.

7. Other Special Provisions

Maximum Benefit Provisions:

- a. The discharger shall implement the maximum benefit commitments specified in Attachment L in accordance with the compliance date(s) stipulated therein⁷.
- b. For the San Timoteo Management Zone, the demonstration of maximum benefit is contingent on the Discharger's effective implementation of the maximum benefit commitments specified in Attachment L and on the effective implementation by the Yucaipa Valley Water District of maximum benefit commitments specified in the Basin Plan (Table 5-9a) (see Attachment F, Fact Sheet, page 17).
- c. If the discharger elects not to implement the maximum benefit commitments specified in Attachment L, or if the Regional Board determines that the maximum benefit commitments are not being implemented effectively by the Discharger in accordance with the schedule prescribed in Attachment L, or if the Regional Board determines that the Yucaipa Valley Water District is not implementing its maximum benefit commitments pertaining to the San Timoteo Management Zone⁸, then the discharger shall mitigate TDS and nitrogen discharges affecting the San Timoteo and Beaumont GMZs that took place in excess of the limits specified in Sections IV.A.1.c. and IV.A.1.d., and

⁷ The Discharger has not yet proposed specific plans to implement recycled water recharge projects. Accordingly, appropriate requirements for such projects are not specified in this Order. Therefore, compliance with the recycled water recharge maximum benefit commitment specified in Attachment L, item 5, is not now required to assure effective maximum benefit implementation. This Order will be reconsidered and revised as necessary to implement relevant maximum benefit commitments if and when the Discharger proposes recycled water projects.

⁸ See Fact Sheet, page F-17

Sections IV.C.1.b. and IV.C.1.c. A proposed mitigation plan and schedule shall be submitted within 60-days of notification by the Regional Board Executive Officer of the need to do so. The discharger shall implement the plan and schedule upon approval by the Regional Board.

VII. COMPLIANCE DETERMINATION

Compliance with the effluent limitations contained in Section IV of this Order will be determined as specified below:

A. Average Monthly Effluent Limitation (AMEL).

If the average of daily discharges over a calendar month exceeds the AMEL for a given parameter, an alleged violation will be flagged and the Discharger will be considered out of compliance for each day of that month for that parameter (e.g., resulting in 31 days of non-compliance in a 31-day month). The average of daily discharges over the calendar month that exceeds the AMEL for a parameter will be considered out of compliance for that month only. If only a single sample is taken during the calendar month and the analytical result for that sample exceeds the AMEL, the Discharger will be considered out of compliance for that calendar month. For any one calendar month during which no sample (daily discharge) is taken, no compliance determination can be made for that calendar month. (See also paragraph VII.M.2., below)

B. Average Weekly Effluent Limitation (AWEL).

If the average of daily discharges over a calendar week exceeds the AWEL for a given parameter, an alleged violation will be flagged and the Discharger will be considered out of compliance for each day of that week for that parameter, resulting in 7 days of non-compliance. The average of daily discharges over the calendar week that exceeds the AWEL for a parameter will be considered out of compliance for that week only. If only a single sample is taken during the calendar week and the analytical result for that sample exceeds the AWEL, the Discharger will be considered out of compliance for that calendar week. For any one calendar week during which no sample (daily discharge) is taken, no compliance determination can be made for that calendar week.

C. Maximum Daily Effluent Limitation (MDEL).

If a daily discharge exceeds the MDEL for a given parameter, an alleged violation will be flagged and the Discharger will be considered out of compliance for that parameter for that 1 day only within the reporting period. For any 1 day during which no sample is taken, no compliance determination can be made for that day.

D. Instantaneous Minimum Effluent Limitation.

If the analytical result of a single grab sample is lower than the instantaneous minimum effluent limitation for a parameter, a violation will be flagged and the Discharger will be considered out of compliance for that parameter for that single sample. Non-compliance for each sample will be considered separately (e.g., the results of two grab samples taken within a calendar day that both are lower than the instantaneous minimum effluent limitation would result in two instances of non-compliance with the instantaneous minimum effluent limitation).

E. Instantaneous Maximum Effluent Limitation.

If the analytical result of a single grab sample is higher than the instantaneous maximum effluent limitation for a parameter, a violation will be flagged and the Discharger will be considered out of compliance for that parameter for that single sample. Non-compliance for each sample will be considered separately (e.g., the results of two grab samples taken within a calendar day that both exceed the instantaneous maximum effluent limitation would result in two instances of non-compliance with the instantaneous maximum effluent limitation).

- F.** Compliance with the 12-month average limit under Discharge Specification IV.A.1.c. and IV.A.1.d. shall be determined by the arithmetic mean of the last twelve monthly averages.
- G.** Compliance determinations for total chlorine residual shall be based on 99% compliance. To determine 99% compliance with the effluent limitation specified in Discharge Specification IV.A.1.a. for total chlorine residual, the following conditions shall be satisfied
- a. The total time during which the total chlorine residual values are above 0.1 mg/l (instantaneous maximum value) shall not exceed 7 hours and 26 minutes in any calendar month;
 - b. No individual excursion from 0.1 mg/l value shall exceed 5 minutes; and
 - c. No individual excursion shall exceed 5.0 mg/l.
- H.** The Discharger shall be considered in compliance with Discharge Specifications IV.A.1.e.(1) if the following conditions are met. If the discharger is using a properly operating backup turbidimeter, the reading of the backup turbidimeter shall be considered in determining whether there has been an actual noncompliance:
- a. There are no excursions above the limits specified in Discharge Specifications IV.A.1.e.(1)(a) and (b);
 - b. Exceedances of the "10 NTU at any time" turbidity requirement do not exceed a duration of one minute.
 - c. The apparent exceedance was caused by interference with, or malfunction of, the monitoring instrument.
- I.** Compliance with the weekly average total coliform limit expressed in Discharge Specification A.1.e.(2) shall be based on a running median of the test results from the previous 7 days. To comply with the weekly average limit, the 7-day median MPN must not exceed 2.2 per 100 milliliters on any day during the week. However, only one violation is recorded for each calendar week, even if the 7-day median MPN value is greater than 2.2 for more than one day in the week.
- J.** Pursuant to 40 CFR 401.17, the discharger shall be in compliance with the pH limitation specified in Discharge Specification IV.A.1.a., above, provided that both of the following conditions are satisfied:
1. The total time during which the pH values are outside the required range of 6.5-8.5 pH values shall not exceed 7 hours and 26 minutes in any calendar month; and

2. No individual excursion from the range of pH values shall exceed 60 minutes.
- K.** Compliance determinations shall be based on available analyses for the time interval associated with the effluent limitation. Where only one sample analysis is available in a specified time interval (e.g., monthly or weekly average), that sample shall serve to characterize the discharge for the entire interval. If quarterly sample results show noncompliance with the average monthly limit and that sample result is used for compliance determinations for each month of the quarter, then three separate violations of the average monthly limit shall be deemed to have occurred.
- L.** Compliance with a single effluent limitation which applies to a group of chemicals (e.g., PCBs), based on a single sample shall be determined by considering the concentrations of individual members of the group to be zero if the analytical response for the individual chemical falls below the method detection limit (MDL or PQL) for that chemical.
- M.** For priority pollutants, the discharger shall be deemed out of compliance with an effluent limitation if the concentration of the priority pollutant in the monitoring sample is greater than the effluent limitation.
1. Compliance determination shall be based on the minimum level (ML)⁸ specified in Attachment "G" of this Order, unless an alternative reporting level is approved by the Regional Water Board's Executive Officer. When there is more than one ML value for a given substance, the discharger shall select the ML value that is below the calculated effluent limitation, and use its associated analytical method, listed in Attachment "G" of this Order. If no ML value is below the effluent limitation, then the Regional Water Board will select as the reporting level the lowest ML value and its associated analytical method.
 2. When determining compliance with an average monthly limit and more than one sample result is available in a month, the discharger shall compute the arithmetic mean unless the data set contains one or more reported determinations of detected but not quantified (DNQ) or not detected (ND). In those cases, the discharger shall compute the median in place of the arithmetic mean in accordance with the following procedure:
 - a. The data set shall be ranked from low to high, reported ND determinations lowest, DNQ determinations next, followed by quantified values (if any). The order of the individual ND or DNQ determinations is unimportant.

⁸ Minimum level is the concentration at which the entire analytical system must give a recognizable signal and acceptable point. The ML is the concentration in a sample that is equivalent to the concentration of the lowest calibration standard analyzed by a specific analytical procedure, assuming that all the method specified sample weights, volumes, and processing steps have been followed.

- b. The median value of the data set shall be determined. If the data set has an odd number of data points, then the median is the middle value. If the data set has an even number of data points, then the median is the average of the two values around the middle unless one or both of the points are ND or DNQ, in which case the median value shall be the lower of the two data points where DNQ is lower than a value and ND is lower than DNQ. If a sample result, or the arithmetic mean or median of multiple sample results, is below the reporting level, and there is evidence that the priority pollutant is present in the effluent above an effluent limitation and the discharger conducts a pollutant minimization program (PMP)⁹, the discharger shall not be deemed out of compliance.
- N. For non-priority pollutants, the discharge shall be considered to be in compliance with an effluent limitation that is less than or equal to the practical quantitation level (PQL)¹⁰ specified in Attachment "I" of this Order if the arithmetic mean of all test results for the monitoring period is less than the constituent effluent limitation. Analytical results that are less than the specified PQL shall be assigned a value of zero.
- O. Compliance with the 12-month average limit under Discharge Specification IV.C.1.b and IV.C.1.c. shall be determined by the arithmetic mean of the last twelve monthly averages.

⁹ The goal of the PMP shall be to reduce all potential sources of a priority pollutant(s) through pollutant minimization (control) strategies, including pollution prevention measures as appropriate, to maintain the effluent concentration at or below the water quality-based effluent limitation.

¹⁰ PQL is the lowest concentration of a substance that can be determined within ± 20 percent of the true concentration by 75 percent of the analytical laboratories tested in a performance evaluation study. Alternatively, if performance data are not available, the PQL is the method detection limit (MDL) x 5 for carcinogens and MDL x 10 for noncarcinogens

ATTACHMENT A – DEFINITIONS

Average Monthly Effluent Limitation (AMEL): the highest allowable average of daily discharges over a calendar month, calculated as the sum of all daily discharges measured during a calendar month divided by the number of daily discharges measured during that month.

Average Weekly Effluent Limitation (AWEL): the highest allowable average of daily discharges over a calendar week (Sunday through Saturday), calculated as the sum of all daily discharges measured during a calendar week divided by the number of daily discharges measured during that week.

Daily Discharge: Daily Discharge is defined as either: (1) the total mass of the constituent discharged over the calendar day (12:00 am through 11:59 pm) or any 24-hour period that reasonably represents a calendar day for purposes of sampling (as specified in the permit), for a constituent with limitations expressed in units of mass or; (2) the unweighted arithmetic mean measurement of the constituent over the day for a constituent with limitations expressed in other units of measurement (e.g., concentration).

The daily discharge may be determined by the analytical results of a composite sample taken over the course of one day (a calendar day or other 24-hour period defined as a day) or by the arithmetic mean of analytical results from one or more grab samples taken over the course of the day.

For composite sampling, if 1 day is defined as a 24-hour period other than a calendar day, the analytical result for the 24-hour period will be considered as the result for the calendar day in which the 24-hour period ends.

Instantaneous Maximum Effluent Limitation: the highest allowable value for any single grab sample or aliquot (i.e., each grab sample or aliquot is independently compared to the instantaneous maximum limitation).

Instantaneous Minimum Effluent Limitation: the lowest allowable value for any single grab sample or aliquot (i.e., each grab sample or aliquot is independently compared to the instantaneous minimum limitation).

Maximum Daily Effluent Limitation (MDEL): the highest allowable daily discharge of a pollutant.

Method Detection Limit (MDL) is the minimum concentration of a substance that can be measured and reported with 99 percent confidence that the analytical concentration is greater than zero, as defined in 40 CFR 136, Appendix B, revised as of May 14, 1999.

Minimum level (ML) is the concentration at which the entire analytical system must give a recognizable signal and acceptable point. The ML is the concentration in a sample that is equivalent to the concentration of the lowest calibration standard analyzed by a specific analytical procedure, assuming that all the method specified sample weights, volumes, and processing steps have been followed.

Practical Quantitation Level (PQL) is the lowest concentration of a substance that can be determined within ± 20 percent of the true concentration by 75 percent of the analytical laboratories tested in a performance evaluation study. Alternatively, if performance data are not available, the PQL is the method detection limit (MDL) x 5 for carcinogens and MDL x 10 for noncarcinogens

Coefficient of Variation (CV): is a measure of the data variability and is calculated as the estimated standard deviation divided by the arithmetic mean of the observed values.

LTAs: Long-Term Averages.

MEC: Maximum pollutant concentration for the effluent.

ECA: Effluent concentration allowance.

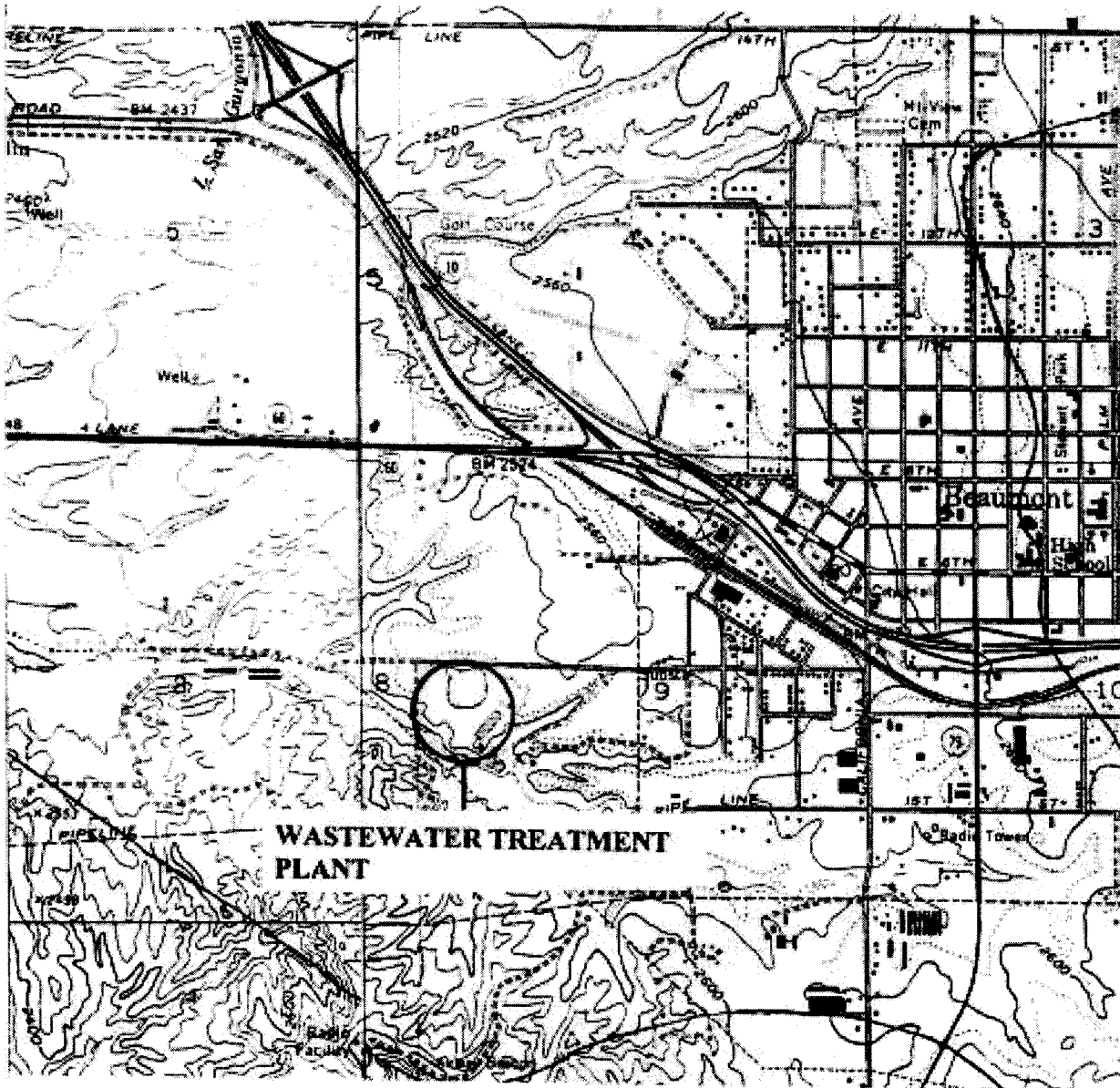
CMC: Criteria maximum concentration.

CCC: Criteria chronic concentration.

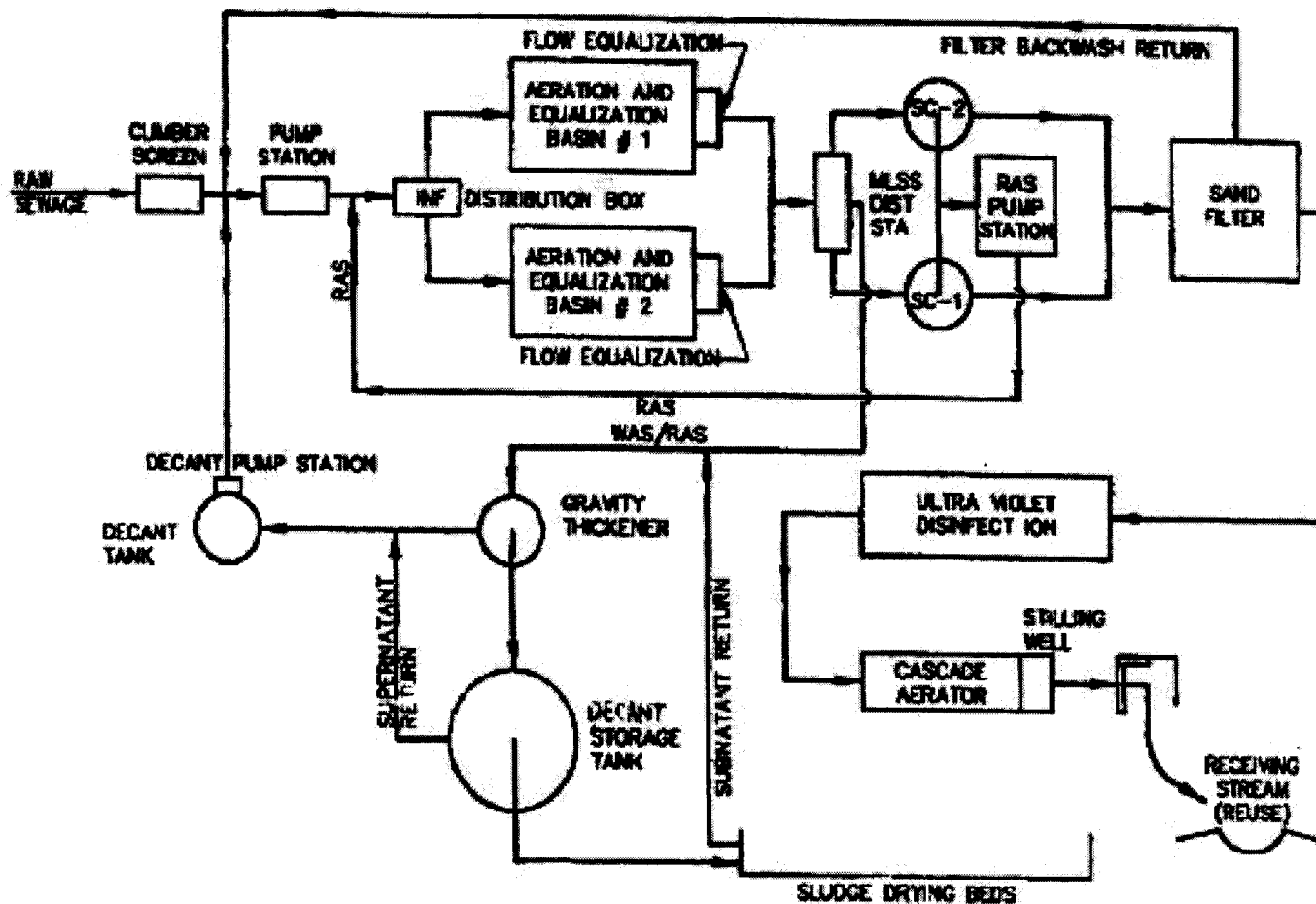
A **"grab" sample** is defined as any individual sample collected in less than 15 minutes.

A **composite sample** is defined as a combination of no fewer than eight individual grab samples obtained over the specified sampling period. The volume of each individual grab sample shall be proportional to the discharge flow rate at the time of sampling. The compositing period shall equal the specific sampling period, or 24 hours, if no period is specified.

ATTACHMENT B – VICINITY MAP



ATTACHMENT C – FLOW SCHEMATIC



ATTACHMENT D – FEDERAL STANDARD PROVISIONS

I. STANDARD PROVISIONS – PERMIT COMPLIANCE

A. Duty to Comply

1. The Discharger must comply with all of the conditions of this Order. Any noncompliance constitutes a violation of the Clean Water Act (CWA) and the California Water Code (CWC) and is grounds for enforcement action, for permit termination, revocation and reissuance, or denial of a permit renewal application [*40 CFR §122.41(a)*].
2. The Discharger shall comply with effluent standards or prohibitions established under Section 307(a) of the Clean Water Act for toxic pollutants and with standards for sewage sludge use or disposal established under Section 405(d) of the CWA within the time provided in the regulations that establish these standards or prohibitions, even if this Order has not been modified to incorporate the requirement [*40 CFR §122.41(a)(1)*].

B. Need to Halt or Reduce Activity Not a Defense

It shall not be a defense for a Discharger in an enforcement action that it would have been necessary to halt or reduce the permitted activity in order to maintain compliance with the conditions of this Order [*40 CFR §122.41(c)*].

C. Duty to Mitigate

The Discharger shall take all reasonable steps to minimize or prevent any discharge or sludge use or disposal in violation of this Order that has a reasonable likelihood of adversely affecting human health or the environment [*40 CFR §122.41(d)*].

D. Proper Operation and Maintenance

The Discharger shall at all times properly operate and maintain all facilities and systems of treatment and control (and related appurtenances) which are installed or used by the Discharger to achieve compliance with the conditions of this Order. Proper operation and maintenance also includes adequate laboratory controls and appropriate quality assurance procedures. This provision requires the operation of backup or auxiliary facilities or similar systems that are installed by a Discharger only when necessary to achieve compliance with the conditions of this Order [*40 CFR §122.41(e)*].

E. Property Rights

1. This Order does not convey any property rights of any sort or any exclusive privileges [*40 CFR §122.41(g)*].

2. The issuance of this Order does not authorize any injury to persons or property or invasion of other private rights, or any infringement of State or local law or regulations [40 CFR §122.5(c)].

F. Inspection and Entry

The Discharger shall allow the Regional Water Quality Control Board (RWQCB), State Water Resources Control Board (SWRCB), United States Environmental Protection Agency (USEPA), and/or their authorized representatives (including an authorized contractor acting as their representative), upon the presentation of credentials and other documents, as may be required by law, to [40 CFR §122.41(i)] [CWC 13383(c)]:

1. Enter upon the Discharger's premises where a regulated facility or activity is located or conducted, or where records are kept under the conditions of this Order [40 CFR §122.41(i)(1)];
2. Have access to and copy, at reasonable times, any records that must be kept under the conditions of this Order [40 CFR §122.41(i)(2)];
3. Inspect and photograph, at reasonable times, any facilities, equipment (including monitoring and control equipment), practices, or operations regulated or required under this Order [40 CFR §122.41(i)(3)];
4. Sample or monitor, at reasonable times, for the purposes of assuring Order compliance or as otherwise authorized by the CWA or the CWC, any substances or parameters at any location [40 CFR §122.41(i)(4)].

G. Bypass

1. Definitions
 - a. "Bypass" means the intentional diversion of waste streams from any portion of a treatment facility [40 CFR §122.41(m)(1)(i)].
 - b. "Severe property damage" means substantial physical damage to property, damage to the treatment facilities, which causes them to become inoperable, or substantial and permanent loss of natural resources that can reasonably be expected to occur in the absence of a bypass. Severe property damage does not mean economic loss caused by delays in production [40 CFR §122.41(m)(1)(ii)].
2. Bypass not exceeding limitations – The Discharger may allow any bypass to occur which does not cause exceedances of effluent limitations, but only if it is for essential maintenance to assure efficient operation. These bypasses are not subject to the provisions listed in Standard Provisions – Permit Compliance I.G.3 and I.G.5 below [40 CFR §122.41(m)(2)].

3. Prohibition of bypass – Bypass is prohibited, and the Regional Water Board may take enforcement action against a Discharger for bypass, unless [40 CFR §122.41(m)(4)(i)]:
 - a. Bypass was unavoidable to prevent loss of life, personal injury, or severe property damage [40 CFR §122.41(m)(4)(A)];
 - b. There were no feasible alternatives to the bypass, such as the use of auxiliary treatment facilities, retention of untreated wastes, or maintenance during normal periods of equipment downtime. This condition is not satisfied if adequate back-up equipment should have been installed in the exercise of reasonable engineering judgment to prevent a bypass that occurred during normal periods of equipment downtime or preventive maintenance [40 CFR §122.41(m)(4)(B)]; and
 - c. The Discharger submitted notice to the Regional Water Board as required under Standard Provision – Permit Compliance I.G.5 below [40 CFR §122.41(m)(4)(C)].
4. The Regional Water Board may approve an anticipated bypass, after considering its adverse effects, if the Regional Water Board determines that it will meet the three conditions listed in Standard Provisions – Permit Compliance I.G.3 above [40 CFR §122.41(m)(4)(ii)].
5. Notice
 - a. Anticipated bypass. If the Discharger knows in advance of the need for a bypass, it shall submit a notice, if possible at least 10 days before the date of the bypass [40 CFR §122.41(m)(3)(i)].
 - b. Unanticipated bypass. The Discharger shall submit notice of an unanticipated bypass as required in Standard Provisions - Reporting V.E below [40 CFR §122.41(m)(3)(ii)].

H. Upset

Upset means an exceptional incident in which there is unintentional and temporary noncompliance with technology based permit effluent limitations because of factors beyond the reasonable control of the permittee. An upset does not include noncompliance to the extent caused by operational error, improperly designed treatment facilities, inadequate treatment facilities, lack of preventive maintenance, or careless or improper operation [40 CFR §122.41(n)(1)].

1. Effect of an upset. An upset constitutes an affirmative defense to an action brought for noncompliance with such technology based permit effluent limitations if the requirements of paragraph H.2 of this section are met. No determination made during administrative review of claims that noncompliance was caused by upset, and before an action for noncompliance, is final administrative action subject to judicial review [40 CFR §122.41(n)(2)].

2. Conditions necessary for a demonstration of upset. A Discharger who wishes to establish the affirmative defense of upset shall demonstrate, through properly signed, contemporaneous operating logs or other relevant evidence that [40 CFR §122.41(n)(3)]:
 - a. An upset occurred and that the Discharger can identify the cause(s) of the upset [40 CFR §122.41(n)(3)(i)];
 - b. The permitted facility was, at the time, being properly operated [40 CFR §122.41(n)(3)(i)];
 - c. The Discharger submitted notice of the upset as required in Standard Provisions – Reporting V.E.2.b [40 CFR §122.41(n)(3)(iii)]; and
 - d. The Discharger complied with any remedial measures required under Standard Provisions – Permit Compliance I.C above [40 CFR §122.41(n)(3)(iv)].
3. Burden of proof. In any enforcement proceeding, the Discharger seeking to establish the occurrence of an upset has the burden of proof [40 CFR §122.41(n)(4)].

II. STANDARD PROVISIONS – PERMIT ACTION

A. General

This Order may be modified, revoked and reissued, or terminated for cause. The filing of a request by the Discharger for modification, revocation and reissuance, or termination, or a notification of planned changes or anticipated noncompliance does not stay any Order condition [40 CFR §122.41(f)].

B. Duty to Reapply

If the Discharger wishes to continue an activity regulated by this Order after the expiration date of this Order, the Discharger must apply for and obtain a new permit [40 CFR §122.41(b)].

C. Transfers

This Order is not transferable to any person except after notice to the Regional Water Board. The Regional Water Board may require modification or revocation and reissuance of the Order to change the name of the Discharger and incorporate such other requirements as may be necessary under the CWA and the CWC [40 CFR §122.41(l)(3)] [40 CFR §122.61].

III. STANDARD PROVISIONS – MONITORING

- A. Samples and measurements taken for the purpose of monitoring shall be representative of the monitored activity [40 CFR §122.41(j)(1)].

- B.** Monitoring results must be conducted according to test procedures under 40 CFR Part 136 or, in the case of sludge use or disposal, approved under 40 CFR Part 136 unless otherwise specified in 40 CFR Part 503 unless other test procedures have been specified in this Order [40 CFR §122.41(j)(4)] [40 CFR §122.44(i)(1)(iv)].

IV. STANDARD PROVISIONS – RECORDS

- A.** Except for records of monitoring information required by this Order related to the Discharger's sewage sludge use and disposal activities, which shall be retained for a period of at least five years (or longer as required by 40 CFR Part 503), the Discharger shall retain records of all monitoring information, including all calibration and maintenance records and all original strip chart recordings for continuous monitoring instrumentation, copies of all reports required by this Order, and records of all data used to complete the application for this Order, for a period of at least three (3) years from the date of the sample, measurement, report or application. This period may be extended by request of the Regional Water Board Executive Officer at any time [40 CFR §122.41(j)(2)].

B. Records of monitoring information shall include:

1. The date, exact place, and time of sampling or measurements [40 CFR §122.41(j)(3)(i)];
2. The individual(s) who performed the sampling or measurements [40 CFR §122.41(j)(3)(ii)];
3. The date(s) analyses were performed [40 CFR §122.41(j)(3)(iii)];
4. The individual(s) who performed the analyses [40 CFR §122.41(j)(3)(iv)];
5. The analytical techniques or methods used [40 CFR §122.41(j)(3)(v)]; and
6. The results of such analyses [40 CFR §122.41(j)(3)(vi)].

C. Claims of confidentiality for the following information will be denied [40 CFR §122.7(b)]:

1. The name and address of any permit applicant or Discharger [40 CFR §122.7(b)(1)]; and
2. Permit applications and attachments, permits and effluent data [40 CFR §122.7(b)(2)].

V. STANDARD PROVISIONS – REPORTING

A. Duty to Provide Information

The Discharger shall furnish to the Regional Water Board, SWRCB, or USEPA within a reasonable time, any information which the Regional Water Board, SWRCB, or USEPA may request to determine whether cause exists for modifying, revoking and reissuing, or terminating this Order or to determine compliance with this Order. Upon request, the Discharger shall also furnish to the Regional Water Board, SWRCB, or USEPA copies of records required to be kept by this Order [40 CFR §122.41(h)] [CWC 13267].

B. Signatory and Certification Requirements

1. All applications, reports, or information submitted to the Regional Water Board, SWRCB, and/or USEPA shall be signed and certified in accordance with paragraph (2.) and (3.) of this provision [40 CFR §122.41(k)].
2. All permit applications shall be signed as follows:
 - a. For a corporation: By a responsible corporate officer. For the purpose of this section, a responsible corporate officer means: (i) A president, secretary, treasurer, or vice-president of the corporation in charge of a principal business function, or any other person who performs similar policy- or decision-making functions for the corporation, or (ii) the manager of one or more manufacturing, production, or operating facilities, provided, the manager is authorized to make management decisions which govern the operation of the regulated facility including having the explicit or implicit duty of making major capital investment recommendations, and initiating and directing other comprehensive measures to assure long term environmental compliance with environmental laws and regulations; the manager can ensure that the necessary systems are established or actions taken to gather complete and accurate information for permit application requirements; and where authority to sign documents has been assigned or delegated to the manager in accordance with corporate procedures [40 CFR §122.22(a)(1)];
 - b. For a partnership or sole proprietorship: by a general partner or the proprietor, respectively [40 CFR §122.22(a)(2)]; or
 - c. For a municipality, State, federal, or other public agency: by either a principal executive officer or ranking elected official. For purposes of this provision, a principal executive officer of a federal agency includes: (i) the chief executive officer of the agency, or (ii) a senior executive officer having responsibility for the overall operations of a principal geographic unit of the agency (e.g., Regional Administrators of USEPA) [40 CFR §122.22(a)(3)].

3. All reports required by this Order and other information requested by the Regional Water Board, SWRCB, or USEPA shall be signed by a person described in paragraph (b) of this provision, or by a duly authorized representative of that person. A person is a duly authorized representative only if:
 - a. The authorization is made in writing by a person described in paragraph (2.) of this provision [40 CFR §122.22(b)(1)];
 - b. The authorization specified either an individual or a position having responsibility for the overall operation of the regulated facility or activity such as the position of plant manager, operator of a well or a well field, superintendent, position of equivalent responsibility, or an individual or position having overall responsibility for environmental matters for the company (a duly authorized representative may thus be either a named individual or any individual occupying a named position) [40 CFR §122.22(b)(2)]; and
 - c. The written authorization is submitted to the Regional Water Board, SWRCB, or USEPA [40 CFR §122.22(b)(3)].
4. If an authorization under paragraph (3.) of this provision is no longer accurate because a different individual or position has responsibility for the overall operation of the facility, a new authorization satisfying the requirements of paragraph (3.) of this provision must be submitted to the Regional Water Board, SWRCB or USEPA prior to or together with any reports, information, or applications, to be signed by an authorized representative [40 CFR §122.22(c)].
5. Any person signing a document under paragraph (2.) or (3.) of this provision shall make the following certification:

“I certify under penalty of law that this document and all attachments were prepared under my direction or supervision in accordance with a system designed to assure that qualified personnel properly gather and evaluate the information submitted. Based on my inquiry of the person or persons who manage the system or those persons directly responsible for gathering the information, the information submitted is, to the best of my knowledge and belief, true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment for knowing violations” [40 CFR §122.22(d)].

C. Monitoring Reports

1. Monitoring results shall be reported at the intervals specified in the Monitoring and Reporting Program in this Order [40 CFR §122.41(l)(4)].
2. Monitoring results must be reported on a Discharge Monitoring Report (DMR) form or forms provided or specified by the Regional Water Board or SWRCB for reporting results of monitoring of sludge use or disposal practices [40 CFR §122.41(l)(4)(i)].

3. If the Discharger monitors any pollutant more frequently than required by this Order using test procedures approved under 40 CFR Part 136 or, in the case of sludge use or disposal, approved under 40 CFR Part 136 unless otherwise specified in 40 CFR Part 503, or as specified in this Order, the results of this monitoring shall be included in the calculation and reporting of the data submitted in the DMR or sludge reporting form specified by the Regional Water Board [40 CFR §122.41(l)(4)(ii)].
4. Calculations for all limitations, which require averaging of measurements, shall utilize an arithmetic mean unless otherwise specified in this Order [40 CFR §122.41(l)(4)(iii)].

D. Compliance Schedules

Reports of compliance or noncompliance with, or any progress reports on, interim and final requirements contained in any compliance schedule of this Order, shall be submitted no later than 14 days following each schedule date [40 CFR §122.41(l)(5)].

E. Twenty-Four Hour Reporting

1. The Discharger shall report any noncompliance that may endanger health or the environment. Any information shall be provided orally within 24 hours from the time the Discharger becomes aware of the circumstances. A written submission shall also be provided within five (5) days of the time the Discharger becomes aware of the circumstances. The written submission shall contain a description of the noncompliance and its cause; the period of noncompliance, including exact dates and times, and if the noncompliance has not been corrected, the anticipated time it is expected to continue; and steps taken or planned to reduce, eliminate, and prevent reoccurrence of the noncompliance [40 CFR §122.41(l)(6)(i)].
2. The following shall be included as information that must be reported within 24 hours under this paragraph [40 CFR §122.41(l)(6)(ii)]:
 - a. Any unanticipated bypass that exceeds any effluent limitation in this Order [40 CFR §122.41(l)(6)(ii)(A)].
 - b. Any upset that exceeds any effluent limitation in this Order [40 CFR §122.41(l)(6)(ii)(B)].
 - c. Violation of a maximum daily discharge limitation for any of the pollutants listed in this Order to be reported within 24 hours [40 CFR §122.41(l)(6)(ii)(C)].
3. The Regional Water Board may waive the above-required written report under this provision on a case-by-case basis if an oral report has been received within 24 hours [40 CFR §122.41(l)(6)(iii)].

F. Planned Changes

The Discharger shall give notice to the Regional Water Board as soon as possible of any planned physical alterations or additions to the permitted facility. Notice is required under this provision only when [40 CFR §122.41(l)(1)]:

1. The alteration or addition to a permitted facility may meet one of the criteria for determining whether a facility is a new source in 40 CFR §122.29(b) [40 CFR §122.41(l)(1)(i)]; or
2. The alteration or addition could significantly change the nature or increase the quantity of pollutants discharged. This notification applies to pollutants which are subject neither to effluent limitations in this Order nor to notification requirements under 40 CFR Part 122.42(a)(1) (see Additional Provisions—Notification Levels VII.A.1) [40 CFR §122.41(l)(1)(ii)].
3. The alteration or addition results in a significant change in the Discharger's sludge use or disposal practices, and such alteration, addition, or change may justify the application of permit conditions that are different from or absent in the existing permit, including notification of additional use or disposal sites not reported during the permit application process or not reported pursuant to an approved land application plan [40 CFR §122.41(l)(1)(iii)].

G. Anticipated Noncompliance

The Discharger shall give advance notice to the Regional Water Board or SWRCB of any planned changes in the permitted facility or activity that may result in noncompliance with General Order requirements [40 CFR §122.41(l)(2)].

H. Other Noncompliance

The Discharger shall report all instances of noncompliance not reported under Standard Provisions – Reporting E.3, E.4, and E.5 at the time monitoring reports are submitted. The reports shall contain the information listed in Standard Provision – Reporting V.E [40 CFR §122.41(l)(7)].

I. Other Information

When the Discharger becomes aware that it failed to submit any relevant facts in a permit application, or submitted incorrect information in a permit application or in any report to the Regional Water Board, SWRCB, or USEPA, the Discharger shall promptly submit such facts or information [40 CFR §122.41(l)(8)].

VI. STANDARD PROVISIONS – ENFORCEMENT

- A. The CWA provides that any person who violates section 301, 302, 306, 307, 308, 318 or 405 of the Act, or any permit condition or limitation implementing any such sections in a permit issued under section 402, or any requirement imposed in a pretreatment program approved under sections 402(a)(3) or 402(b)(8) of the Act, is subject to a civil penalty not to exceed \$25,000 per day for each violation. The CWA provides that any person who negligently violates sections 301, 302, 306, 307, 308, 318, or 405 of the Act, or any condition or limitation implementing any of such sections in a permit issued under section 402 of the Act, or any requirement imposed in a pretreatment program approved under section 402(a)(3) or 402(b)(8) of the Act, is subject to criminal penalties of \$2,500 to \$25,000 per day of violation, or imprisonment of not more than one (1) year, or both. In the case of a second or subsequent conviction for a negligent violation, a person shall be subject to criminal penalties of not more than \$50,000 per day of violation, or by imprisonment of not more than two (2) years, or both. Any person who knowingly violates such sections, or such conditions or limitations is subject to criminal penalties of \$5,000 to \$50,000 per day of violation, or imprisonment for not more than three (3) years, or both. In the case of a second or subsequent conviction for a knowing violation, a person shall be subject to criminal penalties of not more than \$100,000 per day of violation, or imprisonment of not more than six (6) years, or both. Any person who knowingly violates section 301, 302, 303, 306, 307, 308, 318 or 405 of the Act, or any permit condition or limitation implementing any of such sections in a permit issued under section 402 of the Act, and who knows at that time that he thereby places another person in imminent danger of death or serious bodily injury, shall, upon conviction, be subject to a fine of not more than \$250,000 or imprisonment of not more than 15 years, or both. In the case of a second or subsequent conviction for a knowing endangerment violation, a person shall be subject to a fine of not more than \$500,000 or by imprisonment of not more than 30 years, or both. An organization, as defined in section 309(c)(3)(B)(iii) of the Clean Water Act, shall, upon conviction of violating the imminent danger provision, be subject to a fine of not more than \$1,000,000 and can be fined up to \$2,000,000 for second or subsequent convictions [*40 CFR §122.41(a)(2)*] [*CWC 13385 and 13387*].
- B. Any person may be assessed an administrative penalty by the Regional Water Board for violating section 301, 302, 306, 307, 308, 318 or 405 of this Act, or any permit condition or limitation implementing any of such sections in a permit issued under section 402 of this Act. Administrative penalties for Class I violations are not to exceed \$10,000 per violation, with the maximum amount of any Class I penalty assessed not to exceed \$25,000. Penalties for Class II violations are not to exceed \$10,000 per day for each day during which the violation continues, with the maximum amount of any Class II penalty not to exceed \$125,000 [*40 CFR §122.41(a)(3)*].

- C. The CWA provides that any person who falsifies, tampers with, or knowingly renders inaccurate any monitoring device or method required to be maintained under this permit shall, upon conviction, be punished by a fine of not more than \$10,000, or by imprisonment for not more than 2 years, or both. If a conviction of a person is for a violation committed after a first conviction of such person under this paragraph, punishment is a fine of not more than \$20,000 per day of violation, or by imprisonment of not more than 4 years, or both [40 CFR §122.41(j)(5)].
- D. The CWA provides that any person who knowingly makes any false statement, representation, or certification in any record or other document submitted or required to be maintained under this Order, including monitoring reports or reports of compliance or noncompliance shall, upon conviction, be punished by a fine of not more than \$10,000 per violation, or by imprisonment for not more than six months per violation, or by both [40 CFR §122.41(k)(2)].

VII. ADDITIONAL PROVISIONS – NOTIFICATION LEVELS

A. Non-Municipal Facilities

Existing manufacturing, commercial, mining, and silvicultural Dischargers shall notify the Regional Water Board as soon as they know or have reason to believe [40 CFR §122.42(a)]:

1. That any activity has occurred or will occur that would result in the discharge, on a routine or frequent basis, of any toxic pollutant that is not limited in this Order, if that discharge will exceed the highest of the following "notification levels" [40 CFR §122.42(a)(1)]:
 - a. 100 micrograms per liter (µg/L) [40 CFR §122.42(a)(1)(i)];
 - b. 200 µg/L for acrolein and acrylonitrile; 500 µg/L for 2,4-dinitrophenol and 2-methyl-4,6-dinitrophenol; and 1 milligram per liter (mg/L) for antimony [40 CFR §122.42(a)(1)(ii)];
 - c. Five (5) times the maximum concentration value reported for that pollutant in the Report of Waste Discharge [40 CFR §122.42(a)(1)(iii)]; or
 - d. The level established by the Regional Water Board in accordance with 40 CFR §122.44(f) [40 CFR §122.42(a)(1)(iv)].
2. That any activity has occurred or will occur that would result in the discharge, on a non-routine or infrequent basis, of any toxic pollutant that is not limited in this Order, if that discharge will exceed the highest of the following "notification levels" [40 CFR §122.42(a)(2)]:
 - a. 500 micrograms per liter (µg/L) [40 CFR §122.42(a)(2)(i)];
 - b. 1 milligram per liter (mg/L) for antimony [40 CFR §122.42(a)(2)(ii)];

- c. Ten (10) times the maximum concentration value reported for that pollutant in the Report of Waste Discharge [40 CFR §122.42(a)(2)(iii)]; or
- d. The level established by the Regional Water Board in accordance with 40 CFR §122.44(f) [40 CFR §122.42(a)(2)(iv)].

B. Publicly-Owned Treatment Works (POTWs)

All POTWs shall provide adequate notice to the Regional Water Board of the following [40 CFR §122.42(b)]:

- 1. Any new introduction of pollutants into the POTW from an indirect Discharger that would be subject to Sections 301 or 306 of the CWA if it were directly discharging those pollutants [40 CFR §122.42(b)(1)]; and
- 2. Any substantial change in the volume or character of pollutants being introduced into that POTW by a source introducing pollutants into the POTW at the time of adoption of the Order [40 CFR §122.42(b)(2)].

Adequate notice shall include information on the quality and quantity of effluent introduced into the POTW as well as any anticipated impact of the change on the quantity or quality of effluent to be discharged from the POTW [40 CFR §122.42(b)(3)].

ATTACHMENT E – MONITORING AND REPORTING PROGRAM

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ATTACHMENT E – MONITORING AND REPORTING PROGRAM (MRP)

The Code of Federal Regulations (CFR) at 40 CFR §122.48 requires that all NPDES permits specify monitoring and reporting requirements. CWC sections 13267 and 13383 also authorize the Regional Water Quality Control Board (RWQCB) to require technical and monitoring reports. This MRP establishes monitoring and reporting requirements that implement the federal and California regulations.

I. GENERAL MONITORING PROVISIONS

A. General Monitoring Provisions

1. All sampling and sample preservation shall be in accordance with the current edition of *“Standard Methods for the Examination of Water and Wastewater”* (American Public Health Association).
2. All laboratory analyses shall be performed in accordance with test procedures under 40 CFR 136 (revised as of May 14, 1999) "Guidelines Establishing Test Procedures for the Analysis of Pollutants," promulgated by the United States Environmental Protection Agency (EPA), unless otherwise specified in this MRP. For priority pollutants, the test methods must meet the lowest minimum levels (MLs) specified in Attachment E of this Order, where no methods/MLs are specified in Attachment G, then monitoring is to be conducted in accordance with methods/MLs approved by this Regional Water Board or the State Water Board consistent with the State Water Board's Quality Assurance Program. In addition, the Regional Board and/or EPA, at their discretion, may specify test methods that are more sensitive than those specified in 40 CFR 136.
3. Chemical, bacteriological, and bioassay analyses shall be conducted at a laboratory certified for such analyses by the California Department of Health Services or EPA or at laboratories approved by the Regional Water Board's Executive Officer.
4. Whenever the Discharger monitors any pollutant more frequently than is required by this Order, the results of this monitoring shall be included in the calculation and reporting of the data submitted in the discharge monitoring report specified by the Executive Officer.
5. In conformance with federal regulations 40 CFR 122.45(c), analyses to determine compliance with the effluent limitations for metals shall be conducted using the total recoverable method. For Chromium (VI), the dissolved method in conformance with 40 CFR 136 may be used to measure compliance with the Chromium (VI) limitation.

6. For effluent and ambient receiving water monitoring:

- a. The discharger shall require its testing laboratory to calibrate the analytical system down to the minimum level (ML)¹ specified in Attachment “G” for priority pollutants with effluent limitations in this Order, unless an alternative minimum level is approved by the Regional Water Board’s Executive Officer. When there is more than one ML value for a given substance, the discharger shall use the ML values, and their associated analytical methods, listed in Attachment “G” that are below the calculated effluent limitation. The discharger may select any one of those cited analytical methods for compliance determination. If no ML value is below the effluent limitation, then the lowest ML value, and its associated analytical method, listed in Attachment “G” shall be used. Any internal quality control data associated with the sample must be reported when requested by the Executive Officer. The Regional Water Board will reject the quantified laboratory data if quality control data is unavailable or unacceptable.
- b. The discharger shall report the results of analytical determinations for the presence of chemical constituents in a sample using the following reporting protocols:
 - 1) Sample results greater than or equal to the reported ML shall be reported as measured by the laboratory (i.e., the measured chemical concentration in the sample).
 - 2) Sample results less than the reported ML, but greater than or equal to the laboratory’s current Method Detection Limit (MDL)², shall be reported as “Detected, but Not Quantified,” or “DNQ.” The estimated chemical concentration of the sample shall also be reported.
 - 3) Sample results not detected above the laboratory’s MDL shall be reported as “not detected” or “ND.”
- c. The discharger shall submit to the Regional Water Board reports necessary to determine compliance with effluent limitations for priority pollutants in this Order and shall follow the chemical nomenclature and sequential order of constituents shown in Attachment “H” – Priority Pollutant Lists. The discharger shall report with each sample result:
 - 1) The reporting level achieved by the testing laboratory; and

¹ Minimum level is the concentration at which the entire analytical system must give a recognizable signal and acceptable point. The ML is the concentration in a sample that is equivalent to the concentration of the lowest calibration standard analyzed by a specific analytical procedure, assuming that all the method specified sample weights, volumes, and processing steps have been followed.

² MDL is the minimum concentration of a substance that can be measured and reported with 99 percent confidence that the analytical concentration is greater than zero, as defined in 40 CFR 136, Appendix B, revised as of May 14, 1999.

- 2) The laboratory's current MDL, as determined by the procedure found in 40 CFR 136 (revised as of May 14, 1999).
- d. For receiving water monitoring and for those priority pollutants without effluent limitations, the discharger shall require its testing laboratory to quantify constituent concentrations to the lowest achievable MDL as determined by the procedure found in 40 CFR 136 (revised as of May 14, 1999). In situations where the most stringent applicable receiving water objective (freshwater or human health (consumption of organisms only), as specified for that pollutant in 40 CFR 131.38³ is below the minimum level value specified in Attachment "G" and the discharger cannot achieve an MDL value for that pollutant below the ML value, the discharger shall submit justification why a lower MDL value cannot be achieved. Justification shall be submitted together with monthly monitoring reports.
7. For non-priority pollutants monitoring, all analytical data shall be reported with identification of practical quantitation levels and with method detection limits, as determined by the procedure found in 40 CFR 136 (revised as of May 14, 1999).
8. The discharger shall have and implement an acceptable written quality assurance (QA) plan for laboratory analyses. Duplicate chemical analyses must be conducted on a minimum of ten percent (10%) of the samples, or at least one sample per month, whichever is greater. A similar frequency shall be maintained for analyzing spiked samples. When requested by the Regional Water Board or EPA, the discharger will participate in the NPDES discharge monitoring report QA performance study.
9. For every item of monitoring data where the requirements are not met, the monitoring report shall include a statement discussing the reasons for noncompliance, the actions undertaken or proposed that will bring the discharge into full compliance with requirements at the earliest time, and an estimate of the date when the discharger will be in compliance. The discharger shall notify the Regional Water Board by letter when compliance with the time schedule has been achieved.
10. The discharger shall assure that records of all monitoring information are maintained and accessible for a period of at least five years from the date of the sample, report, or application. This period of retention shall be extended during the course of any unresolved litigation regarding this discharge or by the request of the Regional Water Board at any time. Records of monitoring information shall include:
 - a. The information listed in Attachment D- IV Standard Provisions – Records, subparagraph B. of this Order;
 - b. The laboratory which performed the analyses;
 - c. The modification(s) to analytical techniques or methods used;

³ See Federal Register/ Vol. 65, No. 97 / Thursday, May 18, 2000 / Rules and Regulations.

- d. All sampling and analytical results, including
 - 1) Units of measurement used;
 - 2) Reporting level for the analysis (minimum level, practical quantitation level (PQL));
 - 3) Results less than the reporting level but above the method detection limit (MDL);
 - 4) Data qualifiers and a description of the qualifiers;
 - 5) Quality control test results (and a written copy of the laboratory quality assurance plan);
 - 6) Dilution factors, if used; and
 - 7) Sample matrix type; and
 - e. All monitoring equipment calibration and maintenance records;
 - f. All original strip charts from continuous monitoring devices;
 - g. All data used to complete the application for this Order; and,
 - h. Copies of all reports required by this Order.
 - i. Electronic data and information generated by the Supervisory Control and Data Acquisition (SCADA) System.
11. The flow measurement system shall be calibrated at least once per year or more frequently, to ensure continued accuracy.
12. All monitoring instruments and devices used by the discharger to fulfill the prescribed monitoring program shall be properly maintained and calibrated as necessary to ensure their continued accuracy. In the event that continuous monitoring equipment is out of service for greater than a 24-hour period, the discharger shall obtain a representative grab sample each day the equipment is out of service. The discharger shall correct the cause(s) of failure of the continuous monitoring equipment as soon as practicable. In its monitoring report, the discharger shall specify the period(s) during which the equipment was out of service and if the problem has not been corrected, shall identify the steps which the discharger is taking or proposes to take to bring the equipment back into service and the schedule for these actions.

II. MONITORING LOCATIONS

The Discharger shall establish the following monitoring locations to demonstrate compliance with the effluent limitations, discharge specifications, and other requirements in this Order:

Discharge Point Name	Monitoring Location Name	Monitoring Location Description
Influent	M-INF	Grit Chamber
Potable	M-POT	Treatment facility's administrative office
001	M-001	Latitude 33° 55' 24" N and longitude 116°59'34"W, to Cooper's Creek
002	Rec-001	Outfall to recycled water use area
003	Storm-001	V-shaped Concrete Channel
004	Storm-002	V-shaped Concrete Channel
005	Storm-003	V-shaped Concrete Channel
006	Storm-004	V-shaped Concrete Channel

III. INFLUENT MONITORING REQUIREMENTS

A. Monitoring Location

1. The Discharger shall monitor influent at M-INF as follows:

Constituent	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method
Flow	MGD	Recorder/Totalizer	Continuous	See Section I.A.2., above of this MRP
Specific Conductance	µmhos/cm	Recorder	"	"
PH	pH units	Recorder	"	"
BOD	mg/l	24-hr Composite	Daily	"
Suspended Solids	mg/l	24-hr Composite	"	"
COD	mg/l	24-hr Composite	Monthly	"
Total Dissolved Solids	"	"	"	"
Ammonia-Nitrogen	mg/l	Grab	"	"
Total Inorganic Nitrogen (TIN)	"	"	"	"
Cyanide (Total)	µg/l	Grab	Quarterly	"
Boron	mg/l	24-hr Composite	"	"
Chloride	"	"	"	"
Fluoride	"	"	"	"
Sodium	"	"	"	"
Sulfate	"	"	"	"
Total Hardness	"	"	"	"

Constituent	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method
Arsenic	µg/l	24-hr Composite	Quarterly	See Section I.A.2., above of this MRP
Cadmium	"	"	"	"
Copper	"	"	"	"
Lead	"	"	"	"
Manganese	"	"	"	"
Mercury	"	"	"	"
Nickel	"	"	"	"
Selenium	"	"	"	"
Silver	"	"	"	"
Total Chromium	"	"	"	"
Zinc	"	"	"	"
EPA Priority Pollutants (See Attachment "H")	µg/l	24-hr composite	Annually	"

IV. EFFLUENT MONITORING REQUIREMENTS

A. Monitoring Location

1. The Discharger shall sample and monitor M-001 as follows:

Parameter	Units	Sample Type	Minimum Sampling & Testing Frequency	Required Analytical Test Method
Flow	mgd	Recorder/Totalizer	Continuous	---
Specific Conductance	µmhos/cm	Recorder	"	
pH	pH units	Grab	"	See Section I.A.2., above of this MRP
Turbidity Four-hour Results Daily Average Daily 95th Percentile	NTU ⁴	Grab until the continuous monitoring device is installed & operational	Continuous	See Section I.A.2., above of this MRP
Coliform Organisms	MPN per 100 ml ⁵	Grab	Daily ⁶	"

⁴ NTU = Nephelometric Turbidity Units

⁵ MPN/100mL = Most Probable Number per 100 milliliters

Parameter	Units	Sample Type	Minimum Sampling & Testing Frequency	Required Analytical Test Method
Suspended Solids	Mg/l	Composite	Daily ⁷	See Section I.A.2., above of this MRP
BOD	"	"	"	"
COD	"	"	"	"
Total Inorganic Nitrogen	"	"	Monthly	"
Ammonia-Nitrogen	"	"	"	"
Toxicity Monitoring	----	(See Section V, Below)	(See Section V, Below)	"
Total Dissolved Solids	Mg/l	Composite	Monthly	"
Chloride	Mg/l	Composite	Monthly	See Section I.A.2., above of this MRP
Sodium	"	"	"	"
Sulfate	"	"	"	"
Total Hardness	"	"	"	"
Boron	"	"	"	"
Calcium	"	"	"	"
Carbonate	"	"	"	"
Fluoride	"	"	"	"
Magnesium	"	"	"	"
Copper	µg/l	"	"	"
Chromium (VI) or Total Chromium	"	"	"	"
Mercury	"	"	"	"
Selenium	"	"	"	"
Silver	"	"	"	"
Cyanide (Free)	µg/l	Grab	Quarterly (See A.2., below)	See Section I.A.2., above of this MRP
Arsenic	"	Composite	"	"
Barium	"	"	"	"
Cobalt	"	"	"	"
Iron	"	"	"	"
Zinc	"	"	"	"

⁶ Weekdays excluding holidays

⁷ Weekdays excluding holidays

Parameter	Units	Sample Type	Minimum Sampling & Testing Frequency	Required Analytical Test Method
Lead	"	"	"	"
Cadmium	µg/l	Composite	Quarterly (See A.2., below)	See Section I.A.2., above of this MRP
Antimony	"	"	"	"
Manganese	"	"	"	"
Nickel	"	"	"	"
Thallium	"	"	"	"
Chloroform	µg/l	Grab	"	"
Phenolic Compounds	µg/l	Grab	Quarterly (See A.2., below)	See Section I.A.2., above of this MRP
Remaining volatile organic portion of EPA Priority Pollutants (See Attachment "H")	"	"	Annually (See A.3., below)	"
Remaining EPA Priority Pollutants (See Attachment "H")	"	"	Annually (See A.3., below)	"

Notes:

- (1) Samples for total coliform bacteria shall be collected at least daily. Samples shall be taken from the disinfected effluent.
 - (2) Turbidity analysis shall be continuous, performed by a continuous recording turbidimeter. Compliance with the daily average operating filter effluent turbidity shall be determined by averaging the levels of recorded turbidity taken at a minimum of four-hour intervals over a 24-hour period. The results of the daily average turbidity determinations shall be reported monthly.
2. The monitoring frequency for those priority pollutants that are detected during the required semi-annual monitoring at a concentration greater than fifty percent of the most stringent applicable receiving water objectives (freshwater or human health (consumption of organisms only) as specified for that pollutant⁸ in 40 CFR 131.38⁹) shall be accelerated to monthly. To return to the monitoring frequency specified, the discharger shall request and receive approval from the Regional Water Board's Executive Officer or designee.
 3. The monitoring frequency for those priority pollutants that are detected during the required annual monitoring at a concentration greater than fifty percent of the most stringent applicable receiving water objectives (freshwater or human health (consumption of organisms only) as specified for that pollutant¹² in 40 CFR 131.38¹³) shall be accelerated to quarterly for one year.

⁸ For those priority pollutants without specified criteria values, accelerated monitoring is not required.

⁹ See Federal Register/ Vol. 65, No. 97 / Thursday, May 18, 2000 / Rules and Regulations

To return to the monitoring frequency specified, the discharger shall request and receive approval from the Regional Water Board's Executive Officer or designee.

V. WHOLE EFFLUENT TOXICITY TESTING REQUIREMENTS

1. The discharger shall conduct critical life stage chronic toxicity testing in accordance with Method 1001.4 - Survival and Reproduction test for water flea, *Ceriodaphnia dubia* as specified in "Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Freshwater Organisms", Fourth Edition, Environmental Monitoring Systems Laboratory, U.S. Environmental Protection Agency 2002, Cincinnati, Ohio (October 2002, EPA-821-R-02-013).
2. The discharger shall establish procedures to ensure that the toxicity testing laboratory notifies the discharger of the results of toxicity testing within twenty-four hours of completing such tests.
3. A minimum of one monthly chronic toxicity test shall be conducted on 24-hour composite samples.
4. The discharger shall increase the frequency of chronic toxicity testing to every two weeks whenever any test result exceeds 1.0 TUc. The first test under the accelerated schedule shall be conducted within two weeks of receiving notice of the test which exceeds 1.0 TUc, and every two weeks thereafter. The discharger may resume the regular test schedule when two consecutive chronic toxicity tests result in 1.0 TUc, or when the results of the Initial Investigation Reduction Evaluation conducted by the discharger have adequately addressed the identified toxicity problem.
5. The presence of chronic toxicity shall be estimated as specified in Short-Term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Freshwater Organisms. Fourth Edition. EPA-821-R-02-013.
6. Results for both survival and reproduction endpoints shall be reported in TUc, where $TUc = 100/NOEC$ or $100/IC_p$ or EC_p (p is the percent effluent). The no observed effect concentration (NOEC) is the highest concentration of toxicant to which organisms are exposed in a chronic test, that causes no observable adverse effect on the tests organisms (e.g., the highest concentration of toxicant to which the values for the observed responses are not statistically significant different from the controls). The inhibition concentration (IC) is a point estimate of the toxicant concentration that causes a given percent reduction in a non-quantal biological measurement (e.g., reproduction or growth) calculated from a continuous model (the EPA Interpolation Method). The effective concentration (EC) is a point estimate of the toxicant concentration that would cause a given percent reduction in quantal biological measurement (e.g., larval development, survival) calculated from a continuous model (e.g., probit).

7. Additional Testing Requirements.

- a. A series of at least five dilutions and a control will be tested. The series shall be within 60% to 100% effluent concentration.
- b. If organisms are not cultured in-house, concurrent testing with reference toxicants shall be conducted. Where organisms are cultured in-house, monthly reference toxicant testing is sufficient. Reference toxicants shall also be conducted using the same test conditions as the effluent toxicity test (e.g., same test duration, etc).
- c. If either of the reference toxicant test or the effluent tests do not meet all test acceptability criteria as specified in the manual¹⁰, then the discharger must re-sample and re-test within 14 days or as soon as the discharger receives notification of failed tests.
- d. Control and dilution water should be receiving water or lab water, as appropriate, as described in the manual. If the dilution water used is different from the culture water, a second control, using culture water shall also be used.

8. Quality Assurance/Control:

- a. A quality assurance/quality control (QA/QC) program shall be instituted to verify the results of the effluent toxicity monitoring program. The QA/QC program shall include but shall not be limited to the following: (1) Selection of an independent testing laboratory; (2) Approval by the Regional Water Board's Executive Officer or Executive Officer's designee of the independent testing laboratory; (3) Once during the year, the discharger shall split samples with the independent laboratory for conducting chronic toxicity testing; (4) Results from the independent laboratory shall be submitted to the Regional Water Board and the discharger for evaluation; (5) The discharger shall review the test acceptability criteria in accordance with the EPA test protocols, EPA-821-R-02-013.
- b. Results from the independent laboratory of the annual QA/QC split samples are to be used for Quality Assurance/Quality Control (QA/QC) purposes only and not for purposes of determining compliance with other requirements of this Order.

9. The use of alternative methods for measuring chronic toxicity may be considered by the Executive Officer on a case-by-case basis. The use of a different test species, in lieu of conducting the required test species may be considered/approved by the Executive Officer on a case-by case basis upon submittal of the documentation supporting discharger's determination that a different species is more sensitive and appropriate.

¹⁰ Refers to USEPA Manual "Short-Term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Freshwater Organisms. Fourth Edition. EPA-821-R-02-013.

10. Reporting: Results of all toxicity testing conducted within the month following the reporting period shall be submitted monthly in accordance with "Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Freshwater Organisms", Fourth Edition, Environmental Monitoring Systems Laboratory, U.S. Environmental Protection Agency 2002, Cincinnati, Ohio (October 2002, EPA-821-R-02-013). The report shall include a determination of the median value of all chronic toxicity testing results conducted during the two previous months.
11. Whenever an Initial Investigation Reduction Evaluation is conducted, the results of the evaluation shall be submitted upon completion. In addition, monthly status reports shall be submitted as part of the discharger's monitoring report for the previous month.
12. If there are no wastewater discharges into the creek during any semi-annual toxicity monitoring period, toxicity monitoring is not necessary for that period and a note to that effect shall be reported to the Regional Board in lieu of the toxicity data. If the result of any toxicity monitoring triggers an accelerated monitoring program pursuant to Section V. 4, above and discharge into the creek has stopped prior to the completion of the accelerated monitoring, the monitoring shall commence/resume immediately upon the next discharge into the creek. The discharger may return to the semi-annual monitoring frequency once the requirements specified in Section V.4. of the monitoring and reporting program are satisfied.

VI. LAND DISCHARGE MONITORING REQUIREMENTS – NOT APPLICABLE

VII. RECLAMATION MONITORING REQUIREMENTS

1. The Discharger shall sample and monitor REC-001 as follows:

Parameter	Units	Sample Type	Minimum Sampling & Testing Frequency	Required Analytical Test Method
Flow	mgd	Recorder/Totalizer	Continuous	---
BOD	mg/l	Composite	"	"
Suspended Solids	"	"	"	"
PH	pH units	Grab	"	See Section I.A.2., above of this MRP
Total Dissolved Solids				
Total Inorganic Nitrogen				
Coliform Organisms	MPN per 100 ml ¹¹	Grab	Daily ¹²	"

¹¹ MPN/100mL = Most Probable Number per 100 milliliters

2. Whenever recycled water is supplied to a user, the volume of recycled water, the user of recycled water, the locations of those sites including the names of the groundwater subbasins underlying the recycled water use sites, type of use (e.g. irrigation, industrial, etc) and the dates at which water is supplied shall be recorded on a permanent log. A summary report of water use by groundwater subbasins shall be submitted annually.

VIII. RECEIVING WATER MONITORING REQUIREMENTS – SURFACE WATER AND GROUNDWATER

A. Surface Water

1. The following receiving water stations shall be monitored for the indicated constituents:

Station A: Cooper's Creek, within 100 feet upstream of the point of discharge.				
Station B: Cooper's Creek, within 100 feet downstream of the point of discharge.				
Station	Constituent	Unit	Type of Sample	Minimum Frequency of Sampling & Analysis
A and B	Dissolved Oxygen	mg/l	Grab	Weekly
A and B	Temperature	°C	"	"
A check for the presence of any color changes, foam, deposition of material, or odor in the receiving water from the discharge shall be made daily at station B.				

2. At station A, all the priority pollutants listed in Attachment "H" shall be monitored semi-annually and reported by the last day of the month following the monitoring period.
3. Unless otherwise directed by the Regional Water Board Executive Officer, the discharger shall implement the approved plan for the annual sampling and testing of mercury levels in fish flesh samples collected from the Santa Ana River. The frequency of monitoring and submission of reports shall be as stipulated in the approved plan.

B. Groundwater Monitoring: Not applicable

IX. OTHER MONITORING REQUIREMENTS

A. Biosolids Monitoring:

1. Biosolids monitoring shall be conducted as follows:

¹² Weekdays excluding holidays

Biosolids Monitoring	Units	Type of Sample	Minimum Frequency of Sampling & Testing
Priority Pollutants	mg/kg	A composite of six grab samples	Quarterly
Moisture Content (% solid)	mg/kg	Grab	"

- The discharger shall maintain a permanent log of solids hauled away from the treatment facilities for use/disposal elsewhere, including the date hauled, the volume or weight (in dry tons), type (screening, grit, raw sludge, biosolids), application (agricultural, composting, etc.), and destination. This information shall be reported annually.

B. Water Supply Monitoring:

- In January of each year, a sample of each source of the water supplied to the sewer area shall be obtained and analyzed for total dissolved solids concentration expressed in "mg/l".
- Monthly reports shall be submitted stating the amount (in percentage or acre-feet) supplied to the sewer area from each source of water and the resulting flow-weighted water supply quality for total dissolved solids.

C. Storm Water Monitoring – See Attachment K

D. Monitoring Pursuant To Maximum Benefit Commitments

- The discharger shall comply with the monitoring and reporting requirement specified in Attachment L, as appropriate.

X. REPORTING REQUIREMENTS

A. General Monitoring and Reporting Requirements

- The Discharger shall comply with all Standard Provisions (Attachment D) related to monitoring, reporting, and recordkeeping.
- The monthly reports for June and December shall include a roster of plant personnel, including job titles, duties, and level of State certification for each individual.

3. By January 1 of each year, the discharger shall submit an annual report to the Regional Water Board. The annual report shall include the following:
 - a. Tabular and graphical summaries of the monitoring data obtained during the previous year;
 - b. A discussion of the compliance record and the corrective actions taken or planned, which may be needed to bring the discharge into full compliance with the waste discharge requirements;
 - c. A summary of the quality assurance (QA) activities for the previous year; and
 - d. For priority pollutant constituents that do not have effluent limitations but are required to be monitored, the Discharger shall evaluate the monitoring data obtained during the previous year and determine whether detected constituents are at levels that would warrant reopening the permit to include effluent limitations for such constituent(s). To conduct this evaluation, the concentration of detected constituents shall be compared to the most stringent applicable receiving water objectives (freshwater or human health (consumption of organisms only) as specified for that pollutant¹³ in 40 CFR 131.38¹⁴). The Discharger shall include a discussion of the corrective actions taken or planned to address values above receiving water objectives.
4. At any time during the term of this Order when electronic submittal of monitoring reports has become the norm, the State or Regional Water Board may notify the Discharger to discontinue submittal of hard copies of reports. When such notification is given, the Discharger shall stop submitting hard copies of required monitoring reports.
5. The Discharger shall report monitoring results for specific parameters in accordance with the following table:

Parameter	Measurement
Flow	Daily total flow
PH	Daily High and daily low
Electrical Conductivity	Daily High
Turbidity	Daily maximum
Total Residual Chlorine	Daily Maximum

B. Self Monitoring Reports (SMRs)

1. At any time during the term of this Order, the State or Regional Water Board may notify the Discharger to electronically submit self-monitoring reports. Until such notification is given, the Discharger shall submit self-monitoring reports in accordance with the requirements described below.

¹³ For those priority pollutants without specified criteria values, accelerated monitoring is not required.

¹⁴ See Federal Register/ Vol. 65, No. 97 / Thursday, May 18, 2000 / Rules and Regulations

2. The Discharger shall submit monthly, semi-annual, and annual Self Monitoring Reports including the results of all required monitoring using USEPA-approved test methods or other test methods specified in this Order. Monthly reports shall be due on the 1st day of the second month following the end of each calendar month; Semi-annual reports shall be due on August 1 and February 1 following each semi-annual period; Annual reports shall be due on February 1 following each calendar year.
3. Monitoring periods and reporting for all required monitoring shall be completed according to the following schedule:

Sampling Frequency	Monitoring Period Begins On...	Monitoring Period	SMR Due Date
Continuous	Permit effective date	All	First day of second calendar month following month of sampling
Daily	Permit effective date	(Midnight through 11:59 PM) or any 24-hour period that reasonably represents a calendar day for purposes of sampling.	First day of second calendar month following month of sampling
Weekly	Sunday following permit effective date or on permit effective date if on a Sunday	Sunday through Saturday	First day of second calendar month following month of sampling
Monthly	First day of calendar month following permit effective date or on permit effective date if that date is first day of the month	1 st day of calendar month through last day of calendar month	First day of second calendar month following month of sampling
Quarterly	Closest of January 1, April 1, July 1, or October 1 following (or on) permit effective date	January 1 through March 31 April 1 through June 30 July 1 through September 30 October 1 through December 31	May 1 August 1 November 1 February 1
Semi-Annually	Closest of January 1 or July 1 following (or on) permit effective date	January 1 through June 30 July 1 through December 31	August 1 February 1
Annually	January 1 following (or on) permit effective date	January 1 through December 31	February 1

4. The Discharger shall report with each sample result the applicable Minimum Level (ML) and the current Method Detection Limit (MDL), as determined by the procedure in 40 CFR Part 136.
5. The Discharger shall arrange all reported data in a tabular format. The data shall be summarized to clearly illustrate whether the facility is operating in compliance with interim and/or final effluent limitations.
6. The Discharger shall attach a cover letter to the SMR. The information contained in the cover letter shall clearly identify violations of the WDRs; discuss corrective actions taken or planned; and the proposed time schedule for corrective actions. Identified violations must include a description of the requirement that was violated and a description of the violation.
7. Discharge monitoring data shall be submitted in a format acceptable to the Regional Water Board and EPA. Specific reporting format may include preprinted forms and/or electronic

media. The results of all monitoring required by this Order shall be reported to the Regional Water Board, and shall be submitted in such a format as to allow direct comparison with the limitations and requirements of this Order. The hard copy of submitted reports shall serve as the official submittal.

8. SMRs must be submitted to the Regional Water Board, signed and certified as required by the standard provisions (Attachment D), to the address listed below:

Regional Water Quality Control Board
Santa Ana Region
3737 Main Street, Suite 500
Riverside, CA 92501-3348

C. Discharge Monitoring Reports (DMRs)

1. As described in Section X.B.1 above, at any time during the term of this Order, the State or Regional Water Board may notify the Discharger to electronically submit self-monitoring reports. Until such notification is given, the Discharger shall submit discharge monitoring reports (DMRs) in accordance with the requirements described below.
2. DMRs must be signed and certified as required by the standard provisions (Attachment D). The Discharge shall submit the original DMR and one copy of the DMR to the address listed below:

State Water Resources Control Board
Discharge Monitoring Report Processing Center
Post Office Box 671
Sacramento, CA 95812

3. All discharge monitoring results must be reported on the official USEPA pre-printed DMR forms (EPA Form 3320-1). Forms that are self-generated or modified cannot be accepted.

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ATTACHMENT F – FACT SHEET

As described in Section II of this Order, this Fact Sheet includes the legal requirements and technical rationale that serve as the basis for the requirements of this Order.

I. PERMIT INFORMATION

A. The following table summarizes administrative information related to the facility.

WDID	8 330101001
Discharger	City of Beaumont
Name of Facility	Wastewater Treatment Plant No. 1
Facility Address	715 W. 4th Street
	Beaumont, CA 92223
	Riverside County
Facility Contact, Title and Phone	Alan Kapanicas, City Manager, (951) 769-8534
Authorized Person to Sign and Submit Reports	Alan Kapanicas, City Manager, (951) 769-8534
Mailing Address	550 E 6th Street, Beaumont, CA 92223
Billing Address	SAME
Type of Facility	POTW
Major or Minor Facility	Major
Threat to Water Quality	1
Complexity	A
Pretreatment Program	N
Reclamation Requirements	Producer
Facility Permitted Flow	4 mgd
Facility Design Flow	4 mgd
Watershed	Santa Ana River Watershed
Receiving Water	Cooper's Creek, San Timoteo Creek, Santa Ana River, San Timoteo Groundwater Management Zone, and Beaumont Groundwater Management Zone
Receiving Water Type	Creek, river and groundwater

B. The City of Beaumont (hereinafter Discharger) owns Wastewater Treatment Plant No. 1 (hereinafter Facility), a tertiary treatment facility. Urban Logic Consultants, a private contractor, operates the Facility for the City. The facility receives and treats domestic and commercial/industrial wastewater generated within the City of Beaumont and Highland Springs (portions of the unincorporated area of Cherry Valley). The discharges from the facility are currently to Cooper's Creek, a tributary of San Timoteo Creek, Reach 3 and regulated under Order No. 00-10, NPDES No. CA80105376. The Discharger proposes to expand its Facility's treatment capacity from 2 to 4.0 million gallons per day (MGD). The expansion will also include the construction of a regional recycled water system. Order No. 00-10 is being renewed to update and include this proposed increase in the Facility's

treatment capacity and increase in recycled water use. With the construction of the recycled water system, most of the tertiary treated wastewater will be recycled, and tertiary treated wastewater from the Facility will be discharged only intermittently to Cooper's Creek.

- C. The Discharger filed a report of waste discharge and submitted an application for renewal of its Waste Discharge Requirements (WDRs) and National Pollutant Discharge Elimination System (NPDES) permit on October 28, 2004. Supplemental Information was requested and received on July 21, 2005.

II. FACILITY DESCRIPTION

A. Description of Wastewater and Biosolids Treatment or Controls

As discussed above, the Discharger proposes to expand the existing treatment capacity of the facility from 2 MGD to 4.0 MGD average dry weather flow. The current plant expansion design incorporates the same treatment processes as the existing plant. The wastewater treatment system consists of bar screens, activated sludge extended aeration, equalization, clarification, dual media sand filtration, UV disinfection and sludge thickening/drying. Sludge from the facility is dewatered (15-20 percent solids) and stored in drying beds prior to disposal. The Facility produces tertiary treated water that complies with requirements established in Title 22 of the California Code of Regulations for unrestricted non-potable water reuse.

The existing Facility utilizes sludge thickeners, sludge centrifuge (for mechanical drying and thickening) and drying beds to dewater sludge.

B. Discharge Points and Receiving Waters

The treatment facility and discharge points are located in an unincorporated area of Riverside County, within the portions of Section 9, T3S, R1W, SBB&M. The discharger currently discharges tertiary treated effluent to Cooper's Creek at latitude 33°55'24"N and longitude 116°59'34"W. This unlined reach of the Creek overlies and recharges the San Timoteo Groundwater Management Zone (GMZ). While the discharge is to Cooper's Creek, it is considered a *de facto* discharge to San Timoteo Creek and the San Timoteo Management Zone.

The discharger is proposing to construct a 0.2 to 0.5 MGD capacity recycled water reservoir at the existing wastewater treatment facility and is currently working with the Beaumont-Cherry Valley Water District (BCVWD) to determine the final size of this reservoir. A pump station will be constructed to pump recycled water from this reservoir into the BCVWD non-potable water distribution system. The pump station will be designed and constructed so that it can be expanded over time. The location of the proposed recycled water reservoir and pump station will be approximately 33°55'25"N and 116°59'38"W, which is just northeast of the existing point of discharge to Cooper's Creek.

BCVWD estimates that there will be an average demand for non-potable water of about 2.9 MGD by 2010 and 5.0 MGD by 2025. BCVWD will take all the recycled water that is available and blend it with untreated state project water. BCVWD will then deliver this water for irrigation to Oak Valley Golf Course, Highland Springs North Golf Course, PGA Oak Valley Golf Courses, Sunny Cal Egg Ranch, and for irrigation of parkways, medians, cemeteries, and parks.

The Discharger also plans to construct a 4 million gallon recycled water reservoir. The site for this reservoir has not been finalized yet.

C. As previously described (I.B., above), with the construction of the recycled water system, most of the Discharger's recycled water will be recycled and there will be only intermittent discharges to Cooper's Creek. The Discharger currently proposes to use the recycled water for landscape irrigation and other similar uses in areas overlying the Beaumont Management Zone. The Discharger may propose recycled water recharge projects in the future¹. Summary of Existing Requirements and Self-Monitoring Report (SMR) Data

1. Effluent limitations contained in the previous Order 00-10 are as follows:

Parameter (units)	Effluent Limitation					
	Average Monthly	Average Weekly	Maximum Daily	Instantaneous Minimum	Instantaneous Maximum	12-month Average
Discharge Flow (MGD)	--	--	1.4	--	--	--
BOD (mg/L)	20	30	--	--	--	--
TSS (mg/L)	20	30	--	--	--	--
Total Inorganic Nitrogen (mg/L)	--	--	--	--	--	10
pH (pH units)	--	--	--	6.5	8.5	--
Total Dissolved Solids (mg/l)	--	--	--	--	--	490
Total Coliform	MPN of 23/100 ml. in any Calendar Month	MPN of 2.2/100 ml.	--	--	--	--

2. Self-Monitoring Report (SMR) Data for previous Order 00-10 are as follows:

Parameter (units)	Monitoring Data from 1-2000 To 5-2005		
	Highest Average Monthly Discharge	Highest Average Weekly Discharge	Highest Daily Discharge
Discharge Flow (MGD)	1.835	1.975	2.354
BOD (mg/L)	9.2	15.6	24
TSS (mg/L)	7.1	12.3	22

¹ This Order would need to be reopened and revised to specify appropriate recharge requirements, unless such requirements are specified in separate waste discharge requirements.

Parameter (units)	Monitoring Data from 1-2000 To 5-2005			
	Highest Monthly Discharge	Average	Highest Weekly Discharge	Average Daily Discharge
Total Inorganic Nitrogen (mg/L)				12
pH (pH units)	8		8.15	8.5
Total Dissolved Solids (mg/l)				490
Total Coliform	27.5		101.6	500
Selenium	10			

D. Compliance Summary

Based on a review of effluent monitoring data submitted by the discharger for the period from 2000 through 2005, the wastewater discharged from the wastewater treatment facility was in violation of the following effluent limitations:

Parameter	Unit	Date	Value	Permit Limit ²	Reason for Violation	Corrective Measures
Chronic Toxicity Reproduction	TUc	7/11/2000	10.00	1.7	An investigation conducted by staff found a local orange juice factory was discharging an organic phosphate into the collection system.	The orange juice factory was directed to cease discharging the organic phosphate.
Chronic Toxicity Reproduction	TUc	9/7/2000	3.13	1.7	An investigation conducted by staff found a local orange juice factory was discharging an organic phosphate into the collection system.	The orange juice factory was directed to cease discharging the organic phosphate.
Chronic Toxicity Reproduction	TUc	9/19/2000	3.13	1.7	An investigation conducted by staff found a local orange juice factory was discharging an organic phosphate into the collection system.	The orange juice factory was directed to cease discharging the organic phosphate.
Chronic Toxicity Reproduction	TUc	10/4/2000	5.56	1.7	An investigation conducted by staff found a local orange juice factory was discharging an organic phosphate into the collection system.	The orange juice factory was directed to cease discharging the organic phosphate.
Chronic Toxicity Reproduction	TUc	10/25/2000	1.79	1.7	An investigation conducted by staff found a local orange juice factory was discharging an organic phosphate into the collection system.	The orange juice factory was directed to cease discharging the organic phosphate.
Chronic Toxicity Reproduction	TUc	1/9/2001	5.56	1.7	Investigation by staff found that the effluent sampler tubing was replaced with non-food grade vinyl tubing.	The tubing was replaced with food grade tubing.
Chronic Toxicity Reproduction	TUa	4/3/2001	1.79	1.7	Investigation by staff found that the effluent sampler tubing was replaced with non-food grade vinyl tubing.	The tubing was replaced with food grade tubing.
Chronic Toxicity Reproduction	TUa	4/17/2001	3.13	1.7	Investigation by staff found that the effluent sampler tubing was replaced with non-food grade vinyl tubing.	The tubing was replaced with food grade tubing.

² The 1.7 TUc is not a permit limit but a trigger to conduct accelerated chronic toxicity testing.
Attachment F – Fact Sheet

Parameter	Unit	Date	Value	Permit Limit ²	Reason for Violation	Corrective Measures
Coliform Bacteria	MPN/100ml	1/10/2002	30	23	Violation occurred as a result of improper sampling techniques.	Operators were instructed on proper sampling procedures.
Coliform Bacteria	MPN/100ml	1/23/2002	500	23	Violation occurred as a result of improper sampling techniques.	Operators were instructed on proper sampling procedures.
Average Turbidity in 24 hour period	NTU	5/16/2002	>2	2	Investigation by staff found excessive septage dumping by vacuum truck haulers.	Limits were placed to minimize receiving of septic waste.
Chronic Toxicity Reproduction	TUc	6/4/2002	1.79	1.7	Cause of violation unknown.	Sampling techniques were reviewed and related treatment processes were evaluated.
Coliform Bacteria 7 Day Average	MPN/100ml	9/3-9/9/2002	>2.2	2.2	UV-3000 Reactor chamber needed to be cleaned and operators needed to be re-instructed as to the proper sampling procedure for UV systems.	Re-instructed operators and re-scheduled the cleaning of the UV Reactor chamber to the week prior to Chronic toxicity testing.
Discharge of partially treated effluent and turbidity > 10 NTUs	NTU	2/12/2003	>10	10	Hydraulic overload from heavy rains and plugged return activated sludge line coming from the south Clarifier.	Adjusted aeration basins effluent valves to reduce flow to clarifiers and filters. Need to expand the Treatment facility to except greater flows during rain events.
Discharge of partially treated effluent and turbidity > 10 NTUs	NTU	2/3/2004	>10	10	Violation was due to plugged return activated sludge (RAS) lines resulting from tumble weeds and other debris blowing into process tanks.	Wire fence was erected around perimeter of basins to catch flying debris. Reduced flow from basins, by adjustment of pinch values, minimized effects resulting from these discharges.
Discharge of partially treated effluent and turbidity > 10 NTUs	NTU	4/25/2004	>10	10	Violation was due to plugged return activated sludge (RAS) lines resulting from tumble weeds and other debris blowing into process tanks.	Wire fence was erected around perimeter of basins to catch flying debris. Reduced flow from basins, by adjustment of pinch values, minimized effects resulting from these discharges.
Stream dissolved oxygen	mg/L	7/15-7/23/2004	<5	>5	Violation occurred as a result of an interruption of electrical power to the sludge handling systems due to construction.	Additional air was introduced to the basins in order to handle the increased solids loading.
Average Turbidity in 24 hour period	NTU	8/20-9/2/2004	>2	2	Aeration basins plugged with construction debris, thus plugging the sand filters.	Restored system by pumping to new aeration basins and cleaning the filters.
Coliform Bacteria	MPN/100ml	9/1-9/14/2004	300	23	Investigation by plant staff found that coliform violations were the result of defective UV lamps and improper sampling techniques.	The UV system lamps were replaced and operators were briefed on proper coliform sampling procedures.
Stream dissolved oxygen	mg/L	9/17-9/24/2004	<5	>5	Violation occurred as a result of an interruption of electrical power to the sludge handling systems due to construction.	Additional air was introduced to the basins in order to handle the increased solids loading.
Coliform Bacteria	MPN/100ml	12/16/2004	240	23	Investigation by plant staff found that coliform violations were the result of defective UV lamps and improper sampling techniques.	The UV system lamps were replaced and operators were briefed on proper coliform sampling procedures.

Parameter	Unit	Date	Value	Permit Limit ²	Reason for Violation	Corrective Measures
Coliform Bacteria	MPN/100ml	1/8/2005	130	23	Investigation by plant staff found no explanation for violation.	UV system lamps were cleaned as a precaution.
Average monthly ammonia	mg/L	Jan-05	7.3	5	Investigation by plant staff found no explanation for violation.	UV system lamps were cleaned as a precaution.

E. Planned Changes:

The discharger is proposing to increase wastewater treatment plant capacity from 2 mgd to 4.0 mgd. The discharger is working with Beaumont-Cherry Valley Water District (BCVWD) to design and construct a recycled water pump station and a transmission system to deliver recycled water for irrigation of landscaping and golf courses. The discharger and BCVWD are considering design and construction of recycled water storage facilities. Implementation of the recycled water system will result in changes in the volume and frequency of surface water discharges (see I.B and II.B., above)

III. APPLICABLE PLANS, POLICIES, AND REGULATIONS

The requirements contained in the proposed Order are based on the requirements and authorities described in this section.

A. Legal Authorities

This Order is issued pursuant to section 402 of the Federal Clean Water Act (CWA) and implementing regulations adopted by the U.S. Environmental Protection Agency (USEPA) and Chapter 5.5, Division 7 of the California Water Code (CWC). It shall serve as a NPDES permit for point source discharges from this facility to surface waters. This Order also serves as Waste Discharge Requirements (WDRs) pursuant to Article 4, Chapter 4 of the CWC.

B. California Environmental Quality Act (CEQA)

This action to adopt an NPDES permit is exempt from the provisions of the California Environmental Quality Act (Public Resources Code Section 21100, et seq.) in accordance with Section 13389 of the CWC.

C. State and Federal Regulations, Policies, and Plans

1. **Water Quality Control Plans.** A revised Water Quality Control Plan (Basin Plan) became effective on January 24, 1995. The Basin Plan specifies beneficial uses and water quality objectives for waters in the Santa Ana Region. On January 22, 2004, the Regional Water Board adopted Resolution No. R8-2004-0001, amending the Basin Plan to incorporate revised boundaries for groundwater subbasins, now termed

“management zones”, new nitrate-nitrogen and TDS objectives for the new management zones, and new nitrogen and TDS management strategies applicable to both surface and ground waters. The State Water Resources Control Board and Office of Administrative Law (OAL) approved the N/TDS Amendment on September 30, 2004 and December 23, 2004, respectively. The surface water components of the N/TDS Amendment are awaiting EPA approval. This Order implements those surface water provisions, which, for the City of Beaumont, are as or more stringent than those in the Basin Plan. The groundwater-related components of the N/TDS Amendment became effective upon approval by OAL. Accordingly, these waste discharge requirements also implement relevant, groundwater-related components of the N/TDS Amendment.

Tertiary treated wastewater from the treatment plant is discharged to Cooper’s Creek, a tributary of San Timoteo Creek, Reach 3. San Timoteo Creek is tributary to the Santa Ana River, Reach 5. The Santa Ana River, Reach 5 is tributary to the Santa Ana River, Reach 4, thence Reach 3 and downstream reaches. The beneficial uses of receiving waters affected by the discharge from the Facility are tabulated as follows:

Discharge Point	Receiving Water Name	Beneficial Use(s)
M-001	Copper’s Creek and San Timoteo Creek, Reach 3 ³	a. Wildlife habitat, b. Warm freshwater habitat, c. Groundwater recharge, d. Water contact recreation, and e. Non-contact water recreation.
	Santa Ana River, Reach 5 ⁴	a. Agricultural supply, b. Groundwater recharge, c. Water contact recreation, d. Non-contact water recreation, e. Warm freshwater habitat, f. Wildlife habitat, and g. Rare, threatened, or endangered species.
	San Timoteo Groundwater Management Zone	a. Municipal and domestic supply, b. Agricultural Supply, c. Industrial process supply, and d. Industrial service supply
M-002	Beaumont Groundwater Management Zone	a. Municipal and domestic supply, b. Agricultural Supply, c. Industrial process supply, and d. Industrial service supply

³ Excepted from municipal and domestic supply (MUN)

⁴ Excepted from municipal and domestic supply downstream of Orange Avenue (Redlands)

2. **National Toxics Rule (NTR) and California Toxics Rule (CTR).** USEPA adopted the NTR on December 22, 1992, which was amended on May 4, 1995 and November 9, 1999, and the CTR on May 18, 2000, which was amended on February 13, 2001. These rules include water quality criteria for priority pollutants and are applicable to this discharge.
3. **State Implementation Policy.** On March 2, 2000, State Water Board adopted the Policy for Implementation of Toxics Standards for Inland Surface Waters, Enclosed Bays, and Estuaries of California (State Implementation Policy or SIP). The SIP became effective on April 28, 2000, with respect to the priority pollutant criteria promulgated for California by the USEPA through the NTR and to the priority pollutant objectives established by the Regional Water Boards in their basin plans, with the exception of the provision on alternate test procedures for individual discharges that have been approved by USEPA Regional Administrator. The alternate test procedures provision was effective on May 22, 2000. The SIP became effective on May 18, 2000. The SIP includes procedures for determining the need for and calculating water quality-based effluent limitations (WQBELs), and requires Dischargers to submit data sufficient to do so. On February 24, 2005, the State Water Board amended the SIP. The Office of Administrative Law (OAL) approved the amendments on May 31, 2005. On July 13, 2005, the United States Environmental Protection Agency (USEPA) approved the amendments.
4. **Antidegradation Policy.** Section 131.12 of 40 CFR requires that State water quality standards include an antidegradation policy consistent with the federal policy. The State Water Board established California's antidegradation policy in State Water Board Resolution 68-16, which incorporates the requirements of the federal antidegradation policy. Resolution 68-16 requires that existing water quality is maintained unless degradation is justified based on specific findings. The permitted discharge is consistent with the antidegradation provision of 40 CFR §131.12 and State Water Board Resolution 68-16.
5. **Anti-Backsliding Requirements.** Sections 402(o)(2) and 303(d)(4) of the CWA and 40 CFR §122.44(l) prohibit backsliding in NPDES permits. These anti-backsliding provisions require that effluent limitations in a reissued permit must be as stringent as those in the previous permit, with some exceptions in which limitations may be relaxed. All effluent limitations in the Order are at least as stringent as the effluent limitations in the previous Order.
6. **Monitoring and Reporting Requirements.** Section 122.48 of 40 CFR requires that all NPDES permits specify requirements for recording and reporting monitoring results. Sections 13267 and 13383 of the CWC authorize the Regional Water Boards to require technical and monitoring reports. The Monitoring and Reporting Program (MRP) establishes monitoring and reporting requirements to implement federal and State requirements. This MRP is provided in Attachment E.

7. **Pretreatment.** The expanded treatment plant capacity is only 4 mgd and there are no significant industrial users within the service areas. Consequently, this Order does not contain requirements for the implementation of an effective pretreatment program pursuant to Section 307 of the Federal Clean Water Act; Parts 35 and 403 of Title 40, Code of Federal Regulations (40 CFR 35 and 40 CFR 403); and/or Section 2233, Title 23, California Code of Regulations.
8. **Biosolids.** On February 19, 1993, the USEPA issued a final rule for the use and disposal of sewage sludge, 40 CFR, Part 503. This rule requires that producers of sewage sludge meet certain reporting, handling, and disposal requirements. The State of California has not been delegated the authority to implement this program, therefore, the U.S. Environmental Protection Agency is the implementing agency.

D. Impaired Water Bodies on CWA 303(d) List (Not Applicable)

IV. RATIONALE FOR EFFLUENT LIMITATIONS AND DISCHARGE SPECIFICATIONS

The CWA requires point source discharges to control the amount of conventional, non-conventional, and toxic pollutants that are discharged into the waters of the United States. The control of pollutants discharged is established through effluent limitations; and other requirements in NPDES permits. There are two principal bases for effluent limitations: 40 CFR §122.44(a) requires that permits include applicable technology-based limitations and standards; and 40 CFR §122.44(d) requires that permits include water quality-based effluent limitations to attain and maintain applicable numeric and narrative water quality criteria to protect the beneficial uses of the receiving water. Where numeric water quality objectives have not been established. Three options exist to protect water quality: 1) 40 CFR §122.44(d) specifies that WQBELs may be established using USEPA criteria guidance under CWA section 304(a); 2) proposed State criteria or a State policy interpreting narrative criteria supplemented with other relevant information may be used; or 3) an indicator parameter may be established.

A. Discharge Prohibitions

Discharge Prohibitions in this Board Order are based on the Federal Clean Water Act, Basin Plan, State Water Resources Control Board's plans and policies, and U.S. Environmental Protection Agency guidance and regulations.

B. Technology-Based Effluent Limitations

1. Scope and Authority

The CWA requires that technology-based effluent limitations be established based on several levels of controls:

- a. Best practicable treatment control technology (BPT) represents the average of the best performance by plants within an industrial category or subcategory. BPT standards apply to toxic, conventional, and nonconventional pollutants.
- b. Best available technology economically achievable (BAT) represents the best existing performance of treatment technologies that are economically achievable within an industrial point source category. BAT standards apply to toxic and nonconventional pollutants.
- c. Best conventional pollutant control technology (BCT) represents the control from existing industrial point sources of conventional pollutants including BOD, TSS, fecal coliform, pH, and oil and grease. The BCT standard is established after considering the "cost reasonableness" of the relationship between the cost of attaining a reduction in effluent discharge and the benefits that would result, and also the cost effectiveness of additional industrial treatment beyond BPT.
- d. New source performance standards (NSPS) represent the best available demonstrated control technology standards. The intent of NSPS guidelines is to set limitations that represent state-of-the-art treatment technology for new sources.

The CWA requires EPA to develop effluent limitations, guidelines and standards (ELGs) representing application of BPT, BAT, BCT, and NSPS. Section 402(a)(1) of the CWA and 40 CFR §125.3 of the NPDES regulations authorize the use of best professional judgment (BPJ) to derive technology-based effluent limitations on a case-by-case basis where ELGs are not available for certain industrial categories and/or pollutants of concern. Where BPJ is used, the permit writer must consider specific factors outlined in 40 CFR §125.3.

2. Applicable Technology-Based Effluent Limitations

Basis for Limitations

Constituents	Basis for Limitations
Biochemical Oxygen Demand (BOD)	Discharges to waters that support aquatic life, that is dependent on oxygen. Organic matter in the discharge may consume oxygen as it breaks down.
Total Suspended Solids (TSS)	High levels of suspended solids can adversely impact aquatic habitat. Untreated or improperly treated wastewater can contain high amounts of suspended solids.
Hydrogen Ion (pH)	Hydrogen Ion (pH) is a measure of Hydrogen Ion concentration in the water. A range specified between 6.5 to 8.5 ensures suitability of biological life. This limitation has been adopted in the Basin Plan of the Santa Ana Region.
Total Dissolved Solids	High levels of TDS can adversely impact aquatic life. The TDS limit is based on evaluation of plant performance data and consistent with the Basin Plan.

Constituents	Basis for Limitations
Flow	The proposed design capacity of the treatment plant is 4.0 MGD.

C. Water Quality-Based Effluent Limitations (WQBELs)

1. Scope and Authority

- a. Effluent discharged from this facility could contain pollutants in sufficient quantities to affect receiving water quality. Pursuant to Section 13263, Article 4, Chapter 4 of the Porter Cologne Water Quality Control Act, the Regional Water Board is required to issue Waste Discharge Requirements for discharges that could affect the quality of the State's waters. Furthermore, Federal Regulation 40 CFR 122.1 requires the issuance of NPDES permits for pollutants discharged from a point source to the waters of the United States.
- b. The U.S. Environmental Protection Agency published the adopted California Toxics Rule (CTR) (40 CFR §131.38). The CTR promulgates new criteria for both human health protection and protection of aquatic life. New numeric aquatic life criteria for 23 priority toxic pollutants and numeric human health criteria for 57 priority toxic pollutants are listed. In addition, the CTR contains a compliance schedule provision, which authorizes the State to issue schedules of compliance for new or revised NPDES permit limits based on the federal criteria when certain conditions are met.

2. Applicable Water Quality Criteria and Objectives

Table, below summarizes the applicable water quality criteria/objective for priority pollutants reported in detectable concentrations in the effluent or receiving water. These criteria were used in conducting the Reasonable Potential Analysis for this Order.

CTR No.	Parameter	Water Quality Criteria			
		Freshwater		Human Health for Consumption of:	
		Acute	Chronic	Water & Organisms	Organisms only
		µg/L	µg/L	µg/L	µg/L
1	Selenium		5		

3. Determining the need for WQBELs

In accordance with Section 1.3 of the SIP, the Regional Water Board conducted a reasonable potential analysis (RPA) for each priority pollutant with an applicable criterion or objective to determine if a WQBEL is required in the Order. The Regional Water Board analyzed effluent data to determine if a pollutant in a discharge has the

reasonable potential to cause or contribute to an excursion above a state water quality standard. For all parameters that have the reasonable potential to cause or contribute to an excursion above a water quality standard, numeric WQBELs are required. The RPA considers criteria from the CTR, and when applicable, water quality objectives specified in the Basin Plan.

Sufficient data are needed to conduct a complete RPA. If data are not sufficient, the discharger will be required to gather the appropriate data for the Regional Water Board to conduct the RPA. Upon review of the data, and if the Regional Water Board determines that WQBELs are needed to protect the beneficial uses, the permit will be reopened for appropriate modification.

The RPA was performed for the priority pollutants for which effluent data were available. These data were used in the RPA and are summarized in the following Table:

Permit limit implementing California Toxics Rule for freshwater discharges

Hardness Dependent Trace Metals to determine CMC and CCC, then to put into MEC table form comparison

Effluent Hardness =		170		LN Hardness =		5.136						Total Recoverable	
California Toxics Rule							Dissolved						
Metal	m _A	b _A	m _C	b _C	CF _A	CF _C	CMC	CCC	T/D Ratio	WER	Acute	Chronic	
No Published Aquatic Criterion Value							(b)	(b)	No CMC, CCC use Human Health Criteria				
Antimony					1.000	1.000	340	150	1.0	1.0	340	150	
Arsenic	No Published Aquatic Criterion Value						(b)	(b)	No CMC, CCC or Human Health Criteria				
Beryllium													
Cadmium	1.128	-3.6867	0.7852	-2.715	0.922	0.887	7.58	3.31	2.60	1.00	19.70	8.61	
Chromium (III)	0.819	3.688	0.819	1.561	0.316	0.860	847.4	274.9	1.0	1.0	2682	320	
Chromium (VI)	---	---	---	---	0.982	0.962	16.0	11.0	1/C _{fa}	1/CF _c	16.3	11.4	
Copper	0.9422	-1.7	0.8545	-1.702	0.960	0.960	22.2	14.1	2.6	1.0	57.6	36.6	
Cyanide											22.0	5.2	
Lead	1.273	-1.46	1.273	-4.705	0.714	0.714	114.5	4.5	6.1	1.0	698	27	
Mercury	No CMC or CCC use Human Health Criteria for organisms only							0.051	1/C _{fa}	1/CF _c		0.051	
Nickel	0.846	2.255	0.846	0.0584	0.998	0.997	734	81	1.0	1.0	735	82	
Selenium	---	---	---	---	(a)	(a)		5.0				5.0	
Silver	1.72	-6.52	---	---	0.850	(b)	8.6	---	1/C _{fa}	1.0	10.1	---	
Thallium	---	---	---	---	(b)	(b)	---	No CMC or CCC use Human Health Criteria					
Zinc	0.8473	0.884	0.8473	0.884	0.978	0.986	184	185	1.0	1.0	188	188	

(a) Bioaccumulative compound and inappropriate to adjust to percent dissolved

(b) EPA has not published an aquatic life criterion value

unit: ug/l

Equation used :

CMC = (exp{m_A{ln(hardness)} + b_A})

Acute Value = CMC x WER x Acute Conversion Value (CFA) x T/D Ratio (1/CFA)

CCC = (exp{m_C{ln(hardness)} + b_C})

Chronic Value = CCC x WER x Chronic Conversion Factor (CFC) x T/D Ratio (or 1/CFC)

Notes:

1. The water effect ratio being used is 1

2. The site specific total to dissolved ratio for cadmium, copper and lead are as follows:

a. Cadmium 2.6 : 1

b. Copper 2.6 : 1

c. Lead 6.1 : 1

3. For those metals without site specific t/d ratio developed, the total to dissolved ratio for these metals is either 1/CFA for CMC or 1/CFC for CCC

4. No mixing zone and dilution considered in the calculation.

5. Permit limit calculations is based on the procedures stipulated in the Policy for Implementation of Toxics Standards for Inland Surface Waters, Enclosed Bays, and Estuaries.

6. Effluent hardness is calculated median of effluent data.

PERMIT LIMIT CALCULATION AND DETERMINATION OF THE MOST APPROPRIATE ML VALUE CONSIDERING CV
unit in ug/l

Constituent	Caltoxis				CV = 0.6		LTA	Aquatic		Permit Limit	
	Freshwater		Human Health		Acute M	Chronic M		Objective/limits		Concentration Limit	
					0.321	0.527		3.11	1.55		
	CMC	CCC	H2O+Org	Org only	Acute LTA	Chronic LTA		MDEL	AMEL	MDEL	AMEL
Selenium		5.000			0.00	2.64	2.64	8.19	4.08	8.19	4.08

Comments: Reviewing RDLs of emwd indicated that some RDLs are greater than MLs recommended by SWRCB, especially for VOCs.

4. WQBEL Calculations

- a. Water quality based effluent limits (final) are based on monitoring results and following the calculation process outlined in Section 1.4 of the California Toxic Rule and the Policy for Implementation of Toxics Standards for Inland Surface Waters, Enclosed Bays and Estuaries of California. The final WQBELs were calculated for this Order using the process described below.

- b. WQBELS Calculation Example

Using selenium as an example, the following methodology demonstrates how water quality based effluent limits were established for this Order. The process for developing these limits is in accordance with Section 1.4 of the SIP.

Step 1:

For selenium the applicable freshwater water quality criteria is:

$$ECA_{\text{chronic}} = 5.00 \mu\text{g/l}$$

Step 2: For each ECA based on aquatic life criterion/objective, determine the long-term average discharge condition (LTA) by multiplying the ECA by a factor (multiplier). The multiplier is a statistically based factor that adjusts the ECA to account for effluent variability. The value of the multiplier varies depending on the coefficient of variation (CV) of the data set and whether it is an acute or chronic criterion/objective. Table 1 of SIP provides pre-calculated values for the multipliers based on the value of the CV. Equations to develop the multipliers in place of using values in the tables are provided in Section 1.4, Step 3 of the SIP and will not be repeated here.

$$LTA_{\text{acute}} = ECA_{\text{acute}} \times \text{Multiplier}_{\text{acute}}$$

$$LTA_{\text{chronic}} = ECA_{\text{chronic}} \times \text{Multiplier}_{\text{chronic}}$$

The CV for the data set must be determined before the multipliers can be selected and will vary depending on the number of samples and the standard deviation of a data set. If the data set is less than 10 samples, or at least 80% of the samples in the data set are reported as non-detect, the CV shall be set equal to 0.6.

For selenium, the following data was used to develop the acute and chronic LTA using Table 1 of the SIP:

No. of Samples	CV	Multiplier _{acute}	Multiplier _{chronic}
16	0.6	0.321	0.527

$$LTA_{\text{chronic}} = 5.00 \mu\text{g/l} \times 0.527 = 2.64 \mu\text{g/l}$$

Step 3: Select the most limiting (lowest) of the LTA.

LTA = most limiting of LTA_{acute} or LTA_{chronic}

For selenium, the most limiting LTA was the LTA_{acute}

$$LTA = 2.64 \mu\text{g/l}$$

Step 4: Calculate the water quality based effluent limits by multiplying the LTA by a factor (multiplier). Water quality-based effluent limits are expressed as Average Monthly Effluent Limitations (AMEL) and Maximum Daily Effluent Limitation (MDEL). The multiplier is a statistically based factor that adjusts the LTA for the averaging periods and exceedance frequencies of the criteria/objectives and the effluent limitations. The value of the multiplier varies depending on the probability basis, the coefficient of variation (CV) of the data set, the number of samples (for AMEL) and whether it is monthly or daily limit. Table 2 of the SIP provides pre-calculated values for the multipliers based on the value of the CV and the number of samples. Equations to develop the multipliers in place of using values in the tables are provided in Section 1.4, Step 5 of the SIP and will not be repeated here.

$$AMEL_{\text{aquatic life}} = LTA \times AMEL_{\text{multiplier}}$$

$$MDEL_{\text{aquatic life}} = LTA \times MDEL_{\text{multiplier}}$$

AMEL multipliers are based on a 95th percentile occurrence probability, and the MDEL multipliers are based on the 99th percentile occurrence probability. If the number of samples is less than four (4), the default number of samples to be used is four (4).

For selenium, the following data was used to develop the AMEL and MDEL for aquatic life using Table 2 of the SIP:

No. of Samples	CV	MultiplierMDEL	MultiplierAMEL
16	0.6	3.11	1.55

$$\text{AMELaquatic life} = 2.64 \times 1.55 = 4.08 \text{ } \mu\text{g/l}$$

$$\text{MDELaquatic life} = 2.64 \times 3.11 = 8.21 \text{ } \mu\text{g/l}$$

5. Whole Effluent Toxicity (WET) - (Not Applicable)

D. Final Effluent Limitations

Final effluent limitations required by this Order are shown in Section IV, Effluent Limitations and Discharge Specifications of the Order.

Proposed effluent limitations for Biochemical Oxygen Demand 5-day @ 20°C, total suspended solids and total coliform organisms are based on tertiary treatment standards. This Order implements relevant portions of the N/TDS Amendment by specifying effluent limitations and other requirements that pertain to both the “maximum benefit” and “antidegradation” management zones/water quality objectives. Provided that the maximum benefit commitments shown in Attachment L of this Order are satisfied, then the requirements of the Order that address the Beaumont Groundwater Management Zone and the “maximum benefit” objectives apply. (see also discussion in F. Land Discharge Specifications, below)

1. Mass-based Effluent Limitations

Mass-based effluent limitations are established using the following formula:

$$\text{Mass (lbs/day)} = \text{flow rate (MGD)} \times 8.34 \times \text{effluent limitation (mg/L)}$$

where: Mass = mass limitation for a pollutant (lbs/day)

Effluent limitation = concentration limit for a pollutant (mg/L)

Flow rate = discharge flow rate (MGD)

E. Interim Effluent Limitations (Not Applicable)

F. Land Discharge Specifications

As shown in Chapter 4 of the Basin Plan as amended by the N/TDS Amendment, two sets of TDS and nitrate-nitrogen objectives have been adopted for both the San Timoteo and Beaumont Groundwater Management Zones (GMZ): the “maximum benefit” objectives and more stringent objectives based on historic ambient quality (the “antidegradation” objectives). The application of the “maximum benefit” objectives for these Management Zones is contingent on the implementation of commitments by the City of Beaumont and the San Timoteo Watershed Management Authority (STWMA) (and, in the case of the San Timoteo Management Zone, by the Yucaipa Valley Water District (YVWD)) to implement specific water and wastewater resources management programs. These programs are delineated in Chapter 5 of the Basin Plan, as amended by

the N/TDS Amendment and include compliance dates for the implementation of specific activities. These programs are part of a coordinated effort by the member agencies (the City of Beaumont, the Beaumont-Cherry Valley Water District and the South Mesa Water Company) of STWMA to develop and implement projects that will assure reliable water supplies to meet rapidly increasing demands in this area. The San Timoteo Watershed Management Program (STWMP) developed by STWMA entails enhanced recharge of native and recycled water, maximizing the direct use of recycled water, optimizing the direct use of imported water, recharge and conjunctive use.

This Order implements relevant portions of the N/TDS Amendment by specifying effluent limitations and other requirements that pertain to both the “maximum benefit” and “antidegradation” management zones/water quality objectives. Provided that the maximum benefit commitments shown in the N/TDS Amendment are satisfied, then the requirements of the Order that address the “maximum benefit” objectives for the San Timoteo and Beaumont GMZ apply. If the Regional Board finds that the maximum benefit commitments are not being met, then the requirements of the Order that addresses the “antidegradation” TDS and nitrate-nitrogen objectives for these GMZ apply. Although the maximum benefit commitments for the Beaumont and San Timoteo GMZ were made jointly by both the City of Beaumont and STWMA, this Order recognizes the City of Beaumont as the responsible party to mitigate the effects of TDS and TIN discharges recharge that took place in excess of the limits based on the “antidegradation” objectives if the Regional Board makes the finding that maximum benefit is not demonstrated. Again, for the San Timoteo GMZ, the finding of maximum benefit is contingent on the implementation of maximum benefit commitments by both the City of Beaumont (and STWMA) and the Yucaipa Valley Water District.

G. Reclamation Specifications

Section 13523 of the California Water Code provides that a Regional Water Board, after consulting with and receiving the recommendations from the CDHS and any party who has requested in writing to be consulted, and after any necessary hearing, shall prescribe water reclamation requirements for water which is used or proposed to be used as recycled water, if, in the judgment of the Board, such requirements are necessary to protect the public health, safety, or welfare. Section 13523 further provides that such requirements shall include, or be in conformance with, the statewide uniform water recycling criteria established by the CDHS pursuant to California Water Code Section 13521.

Reclamation specifications in the proposed Order are based upon the recycling criteria contained in Title 22, Division 4, Chapter 3, Sections 60301 through 60355, California Code of Regulations, “Guidelines for Use of Reclaimed Water” by the California Department of Health Services, and Pursuant to the California Water Code Section 13521.

V. RATIONALE FOR RECEIVING WATER LIMITATIONS

A. Surface Water

1. The surface water receiving water limitations in the proposed Order are based upon the water quality objectives contained in the Basin Plan. As such, they are a required part of the proposed Order.

The proposed TDS limit (490 mg/l) for direct discharges into Cooper's Creek is based on the maximum benefit waste load allocation specified in Table 5-5 of the Basin Plan, as amended by the N/TDS Amendment. The Order also includes a TDS limit based on the quality of the water supplied to the service area plus a reasonable use increment. The more restrictive of the two TDS limits applies to the discharges. The proposed TDS limit is more restrictive than the waste load allocation for the discharge that was specified in the 1995 Basin Plan (540 mg/l). However, the discharger has demonstrated that compliance with the more restrictive limit is achievable.

This Order also includes TDS limits based on the antidegradation wasteload allocation in Table 5-5 of the Basin Plan, as amended by the N/TDS amendment. This TDS limit will become effective if maximum benefit is not being demonstrated (see preceding discussion in IV. D. Final Effluent Limitations and IV. F. Land Discharge Specifications) If maximum benefit is not demonstrated, this Order requires the discharger to mitigate TDS discharges in excess of the anti-degradation TDS objectives.

This Order also includes two sets of TIN limits: one limit is based on the maximum benefit wasteload allocation specified in Table 5-5 of the Basin Plan, as amended by the N/TDS Amendment; the other limit is based on the antidegradation wasteload allocation also specified in Table 5-5. The TIN limit based on the maximum benefit wasteload allocation is effective provided that maximum benefit is demonstrated. If maximum benefit is not demonstrated, then the limit based on the antidegradation wasteload allocation applies.

2. Requirement to meet 2.2 total coliform bacteria limit in the effluent:
 - a. Article 3, Section 60305 of Title 22, Chapter 3, "Reclamation Criteria" of the California Code of Regulations specifies that recycled water used as a source supply for nonrestricted recreational impoundments shall be at all times an adequately disinfected, oxidized, coagulated, clarified, filtered wastewater (tertiary treated). The degree of treatment specified represents an approximately 5-log reduction in the virus content of the water. The California Department of Health Services (CDHS) has determined that this degree of virus removal is necessary to protect the health of people using these impoundments for water contact recreation.

- b. The CDHS has developed wastewater disinfection guidelines ("Wastewater Disinfection for Health Protection", Department of Health Services, Sanitary Engineering Branch, February 1987) for discharges of wastewater to surface waters where water contact recreation (REC-1) is a beneficial use. The disinfection guidelines recommend the same treatment requirements for wastewater discharges to REC-1 waters as those stipulated in Title 22 for supply of recycled water to nonrestricted recreational impoundments, since the public health risks under both scenarios are analogous. The disinfection guidelines are based on sound science and are widely used as guidance to assure public health and beneficial use protection.
- c. The Santa Ana River, and Cooper's/San Timoteo Creeks, are not "nonrestricted recreational impoundments," nor is "recycled water", as defined in the Reclamation Criteria, being used as a supply source for the River or Creeks. However, except during major storms, most of the flow in the River and Creeks is composed of treated municipal wastewater discharges. The River and Creeks are used for water contact recreation and, accordingly, are designated REC-1 (water contact beneficial use). People recreating in the River or Creeks face an exposure similar to those coming in contact with recycled water in an impoundment. Therefore, to protect the water contact recreation beneficial use and to prevent nuisance and health risk, it is necessary and appropriate to require the same degree of treatment for wastewater discharges to the River and Creeks as would be required for the use of recycled water in a nonrestricted recreational impoundment.

B. Groundwater

Discharges from the Facility to Cooper's Creek and thence San Timoteo Creek, Reach 3 affect the San Timoteo Groundwater Management Zone. Currently expected recycled water use in the Discharger's service area will affect the Beaumont Groundwater Management Zone. For recycled water use, this Order specifies TDS and TIN limits based on the maximum benefit water quality objectives for the Beaumont GMZ. The Order also includes TDS and TIN limits based on the antidegradation water quality objectives for the Beaumont GMZ. The limits based on the antidegradation objectives will become effective if maximum benefit is not demonstrated.

Selenium and pH limits for recycled water use are based on the Basin Plan water quality objectives.

BOD and TSS limits for recycled water use are based on secondary treatment standards.

Total Coliform bacteria and turbidity limits are based on Title 22 regulations for the use of recycled water.

VI. RATIONALE FOR MONITORING AND REPORTING REQUIREMENTS

Section 122.48 of 40 CFR requires all NPDES permits to specify recording and reporting of monitoring results. Sections 13267 and 13383 of the California Water Code authorize the Water Boards to require technical and monitoring reports. The Monitoring and Reporting Program, Attachment E of this Order, establishes monitoring and reporting requirements to implement federal and state requirements. The following provides the rationale for the monitoring and reporting requirements contained in the Monitoring and Reporting Program for this facility.

A. Influent Monitoring

This Order carries forward the treatment plant influent monitoring requirements without change.

B. Effluent Monitoring

The Discharger is required to conduct monitoring of the permitted discharges in order to evaluate compliance with permit conditions. Monitoring requirements are given in the proposed monitoring and reporting program (Attachment E). This provision requires compliance with the monitoring and reporting program, and is based on 40 CFR 122.44(i), 122.62, 122.63 and 124.5. The SMP is a standard requirement in almost all NPDES permits (including the proposed Order) issued by the Regional Water Board. In addition to containing definitions of terms, it specifies general sampling/analytical protocols and the requirements of reporting of spills, violations, and routine monitoring data in accordance with NPDES regulations, the California Water Code, and Regional Water Board's policies. The monitoring and reporting program also contains sampling program specific for the Discharger's wastewater treatment plant. It defines the sampling stations and frequency, pollutants to be monitored, and additional reporting requirements. Pollutants to be monitored include all pollutants for which effluent limitations are specified. Further, in accordance with Section 1.3 of the SIP, periodic monitoring is required for all priority pollutants defined by the CTR, for which criteria apply and for which no effluent limitations have been established, to evaluate reasonable potential to cause or contribute to an excursion above a water quality standard.

C. Whole Effluent Toxicity Testing

Whole effluent toxicity (WET) protects the receiving water quality from the aggregate toxic effect of a mixture of pollutants in the effluent. WET tests measure the degree of response of exposed aquatic test organisms to an effluent. The WET approach allows for protection of the narrative "no toxics in toxic amounts" criterion while implementing numeric criteria for toxicity. There are two types of WET tests: acute and chronic. An acute toxicity test is conducted over a shorter time period and measures mortality. A chronic toxicity test is conducted over a longer period of time and may measure mortality, reproduction, and growth.

The Basin Plan specifies a narrative objective for toxicity, requiring that all waters be maintained free of toxic substances in concentrations that are lethal to or produce other detrimental response on aquatic organisms. Detrimental response includes but is not limited to decreased growth rate, decreased reproductive success of resident or indicator species, and/or significant alterations in population, community ecology, or receiving water biota.

In addition to the Basin Plan requirements, Section 4 of the SIP states that a chronic toxicity effluent limitation is required in permits for all discharges that will cause, have the reasonable potential to cause, or contribute to chronic toxicity in receiving waters. Therefore, in accordance with the SIP, this Order requires the Discharger to conduct chronic toxicity testing. In addition, the Order establishes thresholds that when exceeded requires the Discharger to conduct accelerated toxicity testing and/or conduct toxicity identification evaluation (TIE) studies.

This Order requires the discharger to conduct chronic toxicity testing of the effluent on a monthly basis. The Order also requires the discharger to conduct an Initial Investigation Toxicity Reduction Evaluation (IITRE) program when either the two-month median of toxicity test results exceeds 1 TUc or any single test exceeds 1.7 TUc for survival endpoint. Based on the results of this investigation program and at the discretion of the Executive Officer, a more rigorous Toxicity Reduction Evaluation/Toxicity Identification Evaluation (TRE/TIE) may be required. A re-opener provision is included in the Order to incorporate a chronic toxicity effluent limitation if warranted by the toxicity test results.

D. Receiving Water Monitoring

1. Surface Water

Receiving water monitoring is required to determine compliance with receiving water limitations and to characterize the water quality of the receiving water. Requirements are based on the Basin Plan.

2. Groundwater

The Discharger is required to submit a program for conducting groundwater monitoring at recycled water use sites for three years. Monitoring results of this groundwater-monitoring program will be used to develop groundwater quality objectives for total dissolved solids and total inorganic nitrogen for the Beaumont Groundwater Management Zone.

E. Other Monitoring Requirements

1. Water Supply Monitoring

The Discharger will be required to collect a sample of each source of water supplied and analyze for total dissolved solids. The result of this monitoring will to show compliance with TDS limitations in the Order.

2. Biosolids/Sludge Monitoring

To track where, how much and what quality of biosolids is disposed, the Order requires the Discharger to maintain a permanent log of all solids hauled away from the treatment facility for use/disposal elsewhere and to provide a summary of the volume, type (screenings, grit, raw sludge, digested sludge), use (agricultural, composting, etc.), and the destination in accordance with the Monitoring and Reporting Program of this Order. The sludge that is stockpiled at the treatment facility will be sampled and analyzed for those constituents listed in the sludge monitoring section of the Monitoring and Reporting Program of this Order.

VII. RATIONALE FOR PROVISIONS

A. Standard Provisions

Standard Provisions, which in accordance with 40 CFR §§122.41 and 122.42, apply to all NPDES discharges and must be included in every NPDES permit, are provided in Attachment D to the Order.

B. Special Provisions

1. Reopener Provisions

This provision is based on 40 CFR Part 123. The Regional Water Board may reopen the permit to modify permit conditions and requirements. Causes for modifications include the promulgation of new regulations, modification in sludge use or disposal practices, or adoption of new regulations by the State Board or Regional Water Board, including revisions to the Basin Plan.

2. Special Studies and Additional Monitoring Requirements

- a. Toxicity Identification Evaluations or Toxicity Reduction Evaluations. This provision is based on the SIP, Section 4, Toxicity Control Provisions.
- b. Antidegradation Analysis. This provision is based on State Water Resources Control Board Resolution No. 68-16, which requires the Board in regulating the discharge of waste to maintain high quality waters of the state (the Discharger must demonstrate that it has implemented adequate controls (e.g., adequate treatment capacity) to ensure that high quality waters will be maintained.

3. Best Management Practices and Pollution Prevention

In accordance with Section 402 (p) of the Federal Clean Water Act, EPA published the final regulations for storm water runoff on November 16, 1990 (40 CFR Parts 122, 123 and 124). Industrial facilities, including POTW sites, are required to obtain NPDES Permits for storm water discharges. On April 17, 1997, the State Board adopted a General Industrial Storm Water Permit, Order No. 97-03-DWQ, NPDES No. CAS000001. This Order includes pertinent provisions of the General Industrial Storm Water Permit appropriate for this discharge.

4. Compliance Schedules (Not Applicable)

5. Special Provisions for Municipal Facilities (POTWs Only)

- a. Sludge Disposal Requirements. Requirements are based on the previous Order.

6. Other Special Provisions

Maximum Benefit Provisions:

These provisions are based on the Basin Plan, as amended by the N/TDS amendment. Chapter 5, Section VI of the amended Basin Plan specifies "Maximum Benefit Implementation Plans for Salt Management", including plans for the Yucaipa Valley Water District that apply to the Yucaipa and San Timoteo Groundwater Management Zones, and for the City of Beaumont/San Timoteo Watershed Management Authority that apply to the San Timoteo and Beaumont Groundwater Management Zones. The plans specify tasks and projects, with schedules, that the responsible parties have committed to implement. Provided that these commitments are met, then maximum benefit groundwater objectives and wasteload allocations for TDS and TIN apply and are used as the basis for establishing effluent limitations. If the commitments are not met, then antidegradation groundwater objectives and wasteload allocations for TDS and TIN apply and are the basis of effluent limitations. The maximum benefit and antidegradation objectives and wasteload allocations are specified in Chapter 4 (Table 4-1) and Chapter 5 (Table 5-5) of the amended Basin Plan, respectively.

VIII. PUBLIC PARTICIPATION

The California Regional Water Quality Control Board, Santa Ana Region (Regional Water Board) is considering the issuance of waste discharge requirements (WDRs) that will serve as a National Pollutant Discharge Elimination System (NPDES) permit for the City of Beaumont's Wastewater Treatment Plant No. 1. As a step in the WDR adoption process, the Regional Water Board staff has developed tentative WDRs. The Regional Water Board encourages public participation in the WDR adoption process.

A. Notification of Interested Parties

The Regional Water Board has notified the Discharger and interested agencies and persons of its intent to prescribe waste discharge requirements for the discharge and has provided them with an opportunity to submit their written comments and recommendations. Notification was provided through: posting of the Notice of Public Hearing at the City Hall and publication of the Notice in the local newspaper; and, posting the Notice and draft Order on the Regional Water Board website: <http://www.waterboards.ca.gov/santaana> on December 19, 2005.

B. Written Comments

The staff determinations are tentative. Interested persons are invited to submit written comments concerning these tentative WDRs. Comments should be submitted either in person or by mail to the Executive Office at the Regional Water Board at the address above on the cover page of this Order.

To be fully responded to by staff and considered by the Regional Water Board, written comments should be received at the Regional Water Board offices by 5:00 p.m. on December 27, 2005

C. Public Hearing

The Regional Water Board will hold a public hearing on the tentative WDRs during its regular Board meeting on the following date and time and at the following location:

Date: January 18, 2006
Time: 9:00 a.m.
Location: City of Loma Linda
25541 Barton Road
Loma Linda

Interested persons are invited to attend. At the public hearing, the Regional Water Board will hear testimony, if any, pertinent to the discharge, WDRs, and permit. Oral testimony will be heard; however, for accuracy of the record, important testimony should be in writing.

Please be aware that dates and venues may change. Our web address is www.waterboards.ca.gov/santaana where you can access the current agenda for changes in dates and locations.

D. Waste Discharge Requirements Petitions

Any aggrieved person may petition the State Water Resources Control Board to review the decision of the Regional Water Board regarding the final WDRs. The petition must be submitted within 30 days of the Regional Water Board's action to the following address:

State Water Resources Control Board
Office of Chief Counsel
P.O. Box 100, 1001 I Street
Sacramento, CA 95812-0100

E. Information and Copying

The Report of Waste Discharge (RWD), related documents, tentative effluent limitations and special provisions, comments received, and other information are on file and may be inspected at the address above at any time between 8:30 a.m. and 4:45 p.m., Monday through Friday. Copying of documents may be arranged through the Regional Water Board by calling (951) 782-4130.

F. Register of Interested Persons

Any person interested in being placed on the mailing list for information regarding the WDRs and NPDES permit should contact the Regional Water Board, reference this facility, and provide a name, address, and phone number.

G. Additional Information

Requests for additional information or questions regarding this Order should be directed to J. Shami at (951) 782-3288.

ATTACHMENT G – MINIMUM LEVELS

MINIMUM LEVELS IN PPB (µg/l)

Table 1 - VOLATILE SUBSTANCES ¹	GC	GCMS
Acrolein	2.0	5
Acrylonitrile	2.0	2
Benzene	0.5	2
Bromoform	0.5	2
Carbon Tetrachloride	0.5	2
Chlorobenzene	0.5	2
Chlorodibromomethane	0.5	2
Chloroethane	0.5	2
Chloroform	0.5	2
Dichlorobromomethane	0.5	2
1,1 Dichloroethane	0.5	1
1,2 Dichloroethane	0.5	2
1,1 Dichloroethylene	0.5	2
1,2 Dichloropropane	0.5	1
1,3 Dichloropropylene (volatile)	0.5	2
Ethylbenzene	0.5	2
Methyl Bromide (<i>Bromomethane</i>)	1.0	2
Methyl Chloride (<i>Chloromethane</i>)	0.5	2
Methylene Chloride (<i>Dichloromethane</i>)	0.5	2
1,1,2,2 Tetrachloroethane	0.5	1
Tetrachloroethylene	0.5	2
Toluene	0.5	2
trans-1,2 Dichloroethylene	0.5	1
1,1,1 Trichloroethane	0.5	2
1,1,2 Trichloroethane	0.5	2
Trichloroethylene	0.5	2
Vinyl Chloride	0.5	2
1,2 Dichlorobenzene (volatile)	0.5	2
1,3 Dichlorobenzene (volatile)	0.5	2
1,4 Dichlorobenzene (volatile)	0.5	2

SELECTION AND USE OF APPROPRIATE ML VALUE:

ML Selection: When there is more than one ML value for a given substance, the discharger may select any one of those ML values, and their associated analytical methods, listed herein that are below the calculated effluent limitation for compliance determination. If no ML value is below the effluent limitation, then the discharger shall select the lowest ML value, and its associated analytical method, listed in this Attachment "A".

ML Usage: The ML value listed herein represents the lowest quantifiable concentration in a sample based on the proper application of all method-based analytical procedures and the absence of any matrix interferences. Assuming that all method-specific analytical steps are followed, the ML value will also represent, after the appropriate application of method-specific factors, the lowest standard in the calibration curve for that specific analytical technique. Common analytical practices sometimes require different treatment of the sample relative to calibration standards.

Note: chemical names in parenthesis and italicized is another name for the constituent.

¹ The normal method-specific factor for these substances is 1, therefore, the lowest standard concentration in the calibration curve is equal to the above ML value for each substance.

MINIMUM LEVELS IN PPB (µg/l)

Table 2 – Semi-Volatile Substances ²	GC	GCMS	LC
2-Chloroethyl vinyl ether	1	1	
2 Chlorophenol	2	5	
2,4 Dichlorophenol	1	5	
2,4 Dimethylphenol	1	2	
4,6 Dinitro-2-methylphenol	10	5	
2,4 Dinitrophenol	5	5	
2- Nitrophenol		10	
4- Nitrophenol	5	10	
4 Chloro-3-methylphenol	5	1	
2,4,6 Trichlorophenol	10	10	
Acenaphthene	1	1	0.5
Acenaphthylene		10	0.2
Anthracene		10	2
Benidine		5	
Benzo (a) Anthracene (1,2 Benzanthracene)	10	5	
Benzo(a) pyrene (3,4 Benzopyrene)		10	2
Benzo (b) Flouranthene (3,4 Benzofluoranthene)		10	10
Benzo(g,h,i)perylene		5	0.1
Benzo(k)fluoranthene		10	2
bis 2-(1-Chloroethoxyl) methane		5	
bis(2-chloroethyl) ether	10	1	
bis(2-Chloroisopropyl) ether	10	2	
bis(2-Ethylhexyl) phthalate	10	5	
4-Bromophenyl phenyl ether	10	5	
Butyl benzyl phthalate	10	10	
2-Chloronaphthalene		10	
4-Chlorophenyl phenyl ether		5	
Chrysene		10	5
Dibenzo(a,h)-anthracene		10	0.1
1,2 Dichlorobenzene (semivolatile)	2	2	
1,3 Dichlorobenzene (semivolatile)	2	1	
1,4 Dichlorobenzene (semivolatile)	2	1	
3,3' Dichlorobenzidine		5	
Diethyl phthalate	10	2	
Dimethyl phthalate	10	2	
di-n-Butyl phthalate		10	
2,4 Dinitrotoluene	10	5	
2,6 Dinitrotoluene		5	
di-n-Octyl phthalate		10	
1,2 Diphenylhydrazine		1	
Fluoranthene	10	1	0.05
Fluorene		10	0.1
Hexachloro-cyclopentadiene	5	5	
1,2,4 Trichlorobenzene	1	5	

MINIMUM LEVELS IN PPB (µg/l)

Table 2 - SEMI-VOLATILE SUBSTANCES ²	GC	GCMS	LC	COLOR
Pentachlorophenol	1	5		
Phenol ³	1	1		50
Hexachlorobenzene	5	1		
Hexachlorobutadiene	5	1		
Hexachloroethane	5	1		
Indeno(1,2,3,cd)-pyrene		10	0.05	
Isophorone	10	1		
Naphthalene	10	1	0.2	
Nitrobenzene	10	1		
N-Nitroso-dimethyl amine	10	5		
N-Nitroso -di n-propyl amine	10	5		
N-Nitroso diphenyl amine	10	1		
Phenanthrene		5	0.05	
Pyrene		10	0.05	

Table 3 – INORGANICS ⁴	FAA	GFAA	ICP	ICPMS	SPGFAA	HYDRIDE	CVA	COLOR	DCP
Antimony	10	5	50	0.5	5	0.5			1000
Arsenic		2	10	2	2	1		20	1000
Beryllium	20	0.5	2	0.5	1				1000
Cadmium	10	0.5	10	0.25	0.5				1000
Chromium (total)	50	2	10	0.5	1				1000
Chromium VI	5							10	
Copper		5	10	0.5	2				
Lead	20	5	5	0.5	2				10000
Mercury				0.5			0.2		
Nickel	50	5	20	1	5				1000
Selenium		5		2	5	1			
Silver	10	1	10	0.25	2				1000
Thallium	10	2	10	1	5				1000
Zinc	20		20	1	10				1000
Cyanide								5	

² With the exception of phenol by colorimetric technique, the normal method-specific factor for these substances is 1000, therefore, the lowest standards concentration in the calibration curve is equal to the above ML value for each substance multiplied by 1000.

³ Phenol by colorimetric technique has a factor of 1

⁴ The normal method-specific factor for these substances is 1, therefore, the lowest standard concentration in the calibration curve is equal to the above ML value for each substance.

MINIMUM LEVELS IN PPB (µg/l)

Table 4 - PESTICIDES – PCBs ⁵	GC
Aldrin	0.005
alpha-BHC (<i>a</i> -Hexachloro-cyclohexane)	0.01
beta-BHC (<i>b</i> -Hexachloro-cyclohexane)	0.005
Gamma-BHC (<i>Lindane</i> ; <i>g</i> -Hexachloro-cyclohexane)	0.02
Delta-BHC (<i>d</i> -Hexachloro-cyclohexane)	0.005
Chlordane	0.1
4,4'-DDT	0.01
4,4'-DDE	0.05
4,4'-DDD	0.05
Dieldrin	0.01
Alpha-Endosulfan	0.02
Beta-Endosulfan	0.01
Endosulfan Sulfate	0.05
Endrin	0.01
Endrin Aldehyde	0.01
Heptachlor	0.01
Heptachlor Epoxide	0.01
PCB 1016	0.5
PCB 1221	0.5
PCB 1232	0.5
PCB 1242	0.5
PCB 1248	0.5
PCB 1254	0.5
PCB 1260	0.5
Toxaphene	0.5

Techniques:

GC - Gas Chromatography

GCMS - Gas Chromatography/Mass Spectrometry

HRGCMS - High Resolution Gas Chromatography/Mass Spectrometry (i.e., EPA 1613, 1624, or 1625)

LC - High Pressure Liquid Chromatography

FAA - Flame Atomic Absorption

GFAA - Graphite Furnace Atomic Absorption

HYDRIDE - Gaseous Hydride Atomic Absorption

CVAA - Cold Vapor Atomic Absorption

ICP - Inductively Coupled Plasma

ICPMS - Inductively Coupled Plasma/Mass Spectrometry

SPGFAA - Stabilized Platform Graphite Furnace Atomic Absorption (i.e., EPA 200.9)

DCP - Direct Current Plasma

COLOR – Colorimetric

⁵ The normal method-specific factor for these substances is 100, therefore, the lowest standard concentration in the calibration curve is equal to the above ML value for each substance multiplied by 100.

ATTACHMENT H – EPA PRIORITY POLLUTANT LIST

CTR Number	Parameter	CAS Number	Suggested Analytical Methods
1	Antimony	7440360	EPA 6020/200.8
2	Arsenic	7440382	EPA 1632
3	Beryllium	7440417	EPA 6020/200.8
4	Cadmium	7440439	EPA 1638/200.8
5a	Chromium (III)	16065831	EPA 6020/200.8
5a	Chromium (VI)	18540299	EPA 7199/1636
6	Copper	7440508	EPA 6020/200.8
7	Lead	7439921	EPA 1638
8	Mercury	7439976	EPA 1669/1631
9	Nickel	7440020	EPA 6020/200.8
10	Selenium	7782492	EPA 6020/200.8
11	Silver	7440224	EPA 6020/200.8
12	Thallium	7440280	EPA 6020/200.8
13	Zinc	7440666	EPA 6020/200.8
14	Cyanide	57125	EPA 9012A
15	Asbestos	1332214	EPA/600/R-93/116(PCM)
16	2,3,7,8-TCDD	1746016	EPA 8290 (HRGC) MS
17	Acrolein	107028	EPA 8260B
18	Acrylonitrile	107131	EPA 8260B
19	Benzene	71432	EPA 8260B
20	Bromoform	75252	EPA 8260B
21	Carbon Tetrachloride	56235	EPA 8260B
22	Chlorobenzene	108907	EPA 8260B
23	Chlorodibromomethane	124481	EPA 8260B
24	Chloroethane	75003	EPA 8260B
25	2-Chloroethylvinyl Ether	110758	EPA 8260B
26	Chloroform	67663	EPA 8260B
27	Dichlorobromomethane	75274	EPA 8260B
28	1,1-Dichloroethane	75343	EPA 8260B
29	1,2-Dichloroethane	107062	EPA 8260B
30	1,1-Dichloroethylene	75354	EPA 8260B
31	1,2-Dichloropropane	78875	EPA 8260B
32	1,3-Dichloropropylene	542756	EPA 8260B
33	Ethylbenzene	100414	EPA 8260B
34	Methyl Bromide	74839	EPA 8260B
35	Methyl Chloride	74873	EPA 8260B
36	Methylene Chloride	75092	EPA 8260B
37	1,1,2,2-Tetrachloroethane	79345	EPA 8260B
38	Tetrachloroethylene	127184	EPA 8260B
39	Toluene	108883	EPA 8260B
40	1,2-Trans-Dichloroethylene	156605	EPA 8260B

CTR Number	Parameter	CAS Number	Suggested Analytical Methods
41	1,1,1-Trichloroethane	71556	EPA 8260B
42	1,1,2-Trichloroethane	79005	EPA 8260B
43	Trichloroethylene	79016	EPA 8260B
44	Vinyl Chloride	75014	EPA 8260B
45	2-Chlorophenol	95578	EPA 8270C
46	2,4-Dichlorophenol	120832	EPA 8270C
47	2,4-Dimethylphenol	105679	EPA 8270C
48	2-Methyl-4,6-Dinitrophenol	534521	EPA 8270C
49	2,4-Dinitrophenol	51285	EPA 8270C
50	2-Nitrophenol	88755	EPA 8270C
51	4-Nitrophenol	100027	EPA 8270C
52	3-Methyl-4-Chlorophenol	59507	EPA 8270C
53	Pentachlorophenol	87865	EPA 8270C
54	Phenol	108952	EPA 8270C
55	2,4,6-Trichlorophenol	88062	EPA 8270C
56	Acenaphthene	83329	EPA 8270C
57	Acenaphthylene	208968	EPA 8270C
58	Anthracene	120127	EPA 8270C
59	Benzidine	92875	EPA 8270C
60	Benzo(a)Anthracene	56553	EPA 8270C
61	Benzo(a)Pyrene	50328	EPA 8270C
62	Benzo(b)Fluoranthene	205992	EPA 8270C
63	Benzo(ghi)Perylene	191242	EPA 8270C
64	Benzo(k)Fluoranthene	207089	EPA 8270C
65	Bis(2-Chloroethoxy)Methane	111911	EPA 8270C
66	Bis(2-Chloroethyl)Ether	111444	EPA 8270C
67	Bis(2-Chloroisopropyl)Ether	108601	EPA 8270C
68	Bis(2-Ethylhexyl)Phthalate	117817	EPA 8270C
69	4-Bromophenyl Phenyl Ether	101553	EPA 8270C
70	Butylbenzyl Phthalate	85687	EPA 8270C
71	2-Chloronaphthalene	91587	EPA 8270C
72	4-Chlorophenyl Phenyl Ether	7005723	EPA 8270C
73	Chrysene	218019	EPA 8270C
74	Dibenzo(a,h)Anthracene	53703	EPA 8270C
75	1,2-Dichlorobenzene	95501	EPA 8260B
76	1,3-Dichlorobenzene	541731	EPA 8260B
77	1,4-Dichlorobenzene	106467	EPA 8260B
78	3,3'-Dichlorobenzidine	91941	EPA 8270C
79	Diethyl Phthalate	84662	EPA 8270C
80	Dimethyl Phthalate	131113	EPA 8270C
81	Di-n-Butyl Phthalate	84742	EPA 8270C
82	2,4-Dinitrotoluene	121142	EPA 8270C
83	2,6-Dinitrotoluene	606202	EPA 8270C
84	Di-n-Octyl Phthalate	117840	EPA 8270C
85	1,2-Diphenylhydrazine	122667	EPA 8270C
86	Fluoranthene	206440	EPA 8270C
87	Fluorene	86737	EPA 8270C

CTR Number	Parameter	CAS Number	Suggested Analytical Methods
88	Hexachlorobenzene	118741	EPA 8260B
89	Hexachlorobutadiene	87863	EPA 8260B
90	Hexachlorocyclopentadiene	77474	EPA 8270C
91	Hexachloroethane	67721	EPA 8260B
92	Indeno(1,2,3-cd)Pyrene	193395	EPA 8270C
93	Isophorone	78591	EPA 8270C
94	Naphthalene	91203	EPA 8260B
95	Nitrobenzene	98953	EPA 8270C
96	N-Nitrosodimethylamine	62759	EPA 8270C
97	N-Nitrosodi-n-Propylamine	621647	EPA 8270C
98	N-Nitrosodiphenylamine	86306	EPA 8270C
99	Phenanthrene	85018	EPA 8270C
100	Pyrene	129000	EPA 8270C
101	1,2,4-Trichlorobenzene	120821	EPA 8260B
102	Aldrin	309002	EPA 8081A
103	alpha-BHC	319846	EPA 8081A
104	beta-BHC	319857	EPA 8081A
105	gamma-BHC	58899	EPA 8081A
106	delta-BHC	319868	EPA 8081A
107	Chlordane	57749	EPA 8081A
108	4,4'-DDT	50293	EPA 8081A
109	4,4'-DDE	72559	EPA 8081A
110	4,4'-DDD	72548	EPA 8081A
111	Dieldrin	60571	EPA 8081A
112	alpha-Endosulfan	959988	EPA 8081A
113	beta-Endosulfan	33213659	EPA 8081A
114	Endosulfan Sulfate	1031078	EPA 8081A
115	Endrin	72208	EPA 8081A
116	Endrin Aldehyde	7421934	EPA 8081A
117	Heptachlor	76448	EPA 8081A
118	Heptachlor Epoxide	1024573	EPA 8081A
119	PCB-1016	12674112	EPA 8082
120	PCB-1221	11104282	EPA 8082
121	PCB-1232	11141165	EPA 8082
122	PCB-1242	53469219	EPA 8082
123	PCB-1248	12672296	EPA 8082
124	PCB-1254	11097691	EPA 8082
125	PCB-1260	11096825	EPA 8082
126	Toxaphene	8001352	EPA 8081A

ATTACHMENT I – PRACTICAL QUANTITATION LEVELS FOR COMPLIANCE

PRACTICAL QUANTITATION LEVELS FOR COMPLIANCE DETERMINATION		
Constituent	PQL µg/l	Analysis Method
1 Arsenic	7.5	GF/AA
2 Barium	20.0	ICP/GFAA
3 Cadmium	15.0	ICP
4 Chromium (VI)	15.0	ICP
5 Cobalt	10.0	GF/AA
6 Copper	19.0	GF/ICP
7 Cyanide	50.0	335.2/335.3
8 Iron	100.0	ICP
9 Lead	26.0	GF/AA
10 Manganese	20.0	ICP
11 Mercury	0.50	CV/AA
12 Nickel	50.0	ICP
13 Selenium	2.0	EPA Method 1638, 1640 or 7742
14 Silver	16.0	ICP
15 Zinc	20.0	ICP
16 1,2 - Dichlorobenzene	5.0	601/602/624
17 1,3 - Dichlorobenzene	5.0	601
18 1,4 - Dichlorobenzene	5.0	601
18 2,4 - Dichlorophenol	10.0	604/625
20 4 - Chloro -3- methylphenol	10.0	604/625
21 Aldrin	0.04	608
22 Benzene	1.0	602/624
23 Chlordane	0.30	608
24 Chloroform	5.0	601/624
25 DDT	0.10	608
26 Dichloromethane	5.0	601/624
27 Dieldrin	0.10	608
28 Fluorantene	10.0	610/625
29 Endosulfan	0.50	608
30 Endrin	0.10	608
31 Halomethanes	5.0	601/624
32 Heptachlor	0.03	608
33 Heptachlor Epoxide	0.05	608
34 Hexachlorobenzene	10.0	625
35 Hexachlorocyclohexane		
Alpha	0.03	608
Beta	0.03	608
Gamma	0.03	608
36 PAH's	10.0	610/625
37 PCB	1.0	608
38 Pentachlorophenol	10.0	604/625
39 Phenol	10.0	604/625
40 TCDD Equivalent	0.05	8280
41 Toluene	1.0	602/625
42 Toxaphene	2.0	608
43 Tributyltin	0.02	GC
44 2,4,6-Trichlorophenol	10.0	604/625

ATTACHMENT J - STORM WATER POLLUTION PREVENTION PLAN REQUIREMENTS

1. Implementation Schedule

The storm water pollution prevention plan (SWPPP) shall be updated and implemented in a timely manner, but in no case later than March 1, 2006.

2. Objectives

The SWPPP has two major objectives: (a) to identify and evaluate sources of pollutants associated with industrial activities that may affect the quality of storm water discharges and authorized non-storm water discharges from the facility; and (b) to identify and implement site-specific best management practices (BMPs) to reduce or prevent pollutants associated with industrial activities in storm water discharges and authorized non-storm water discharges. BMPs may include a variety of pollution prevention measures or other low-cost pollution control measures. They are generally categorized as non-structural BMPs (activity schedules, prohibitions of practices, maintenance procedures, and other low-cost measures) and as structural BMPs (treatment measures, run-off controls, over-head coverage). To achieve these objectives, dischargers should consider the five phase process for SWPPP development and implementation as shown in Table A, below.

The SWPPP requirements are designed to be sufficiently flexible to meet the various needs of the facility. SWPPP requirements that are not applicable to the facility should not be included in the SWPPP.

A facility's SWPPP is a written document that shall contain a compliance activity schedule, a description of industrial activities and pollutant sources, descriptions of BMPs, drawings, maps, and relevant copies or references of parts of other plans. The SWPPP shall be revised whenever appropriate and shall be readily available for review by facility employees or Regional Water Board inspectors.

3. Planning and Organization

a. Pollution Prevention Team

The SWPPP shall identify a specific individual or individuals and their positions within the facility organization as members of a storm water pollution prevention team responsible for developing the SWPPP, assisting the facility manager in SWPPP implementation and revision, and conducting all monitoring program activities required in the Stormwater monitoring program of Order No. R8-2006-0003. The SWPPP shall clearly identify the storm water pollution prevention related responsibilities, duties, and activities of each team member.

b. Review Other Requirements and Existing Facility Plans

The SWPPP may incorporate or reference the appropriate elements of other regulatory requirements. The discharger shall review all local, state, and federal requirements that impact, complement, or are consistent with the requirements of Order No. R8-2006-0003. The discharger shall identify any existing facility plans that contain storm water pollutant control measures or relate to the requirements of Order No. R8-2006-0003. As examples, dischargers whose facilities are subject to Federal Spill Prevention Control and Countermeasures' requirements should already have instituted a plan to control spills of certain hazardous materials. Similarly, the discharger whose facilities are subject to air quality related permits and regulations may already have evaluated industrial activities that generate dust or particulates.

4. Site Map

The SWPPP shall include a site map. The site map shall be provided on an 8-1/2 x 11 inch or larger sheet and include notes, legends, and other data as appropriate to ensure that the site map is clear and understandable. If necessary, the discharger may provide the required information on multiple site maps. The following information shall be included on the site map:

- a. The facility boundaries; the outline of all storm water drainage areas within the facility boundaries; portions of the drainage area impacted by run-on from surrounding areas; and direction of flow of each drainage area, on-site surface water bodies, and areas of soil erosion. The map shall also identify nearby water bodies (such as rivers, lakes, ponds) and municipal storm drain inlets where the facility's storm water discharges and authorized non-storm water discharges may be received.
- b. The location of the storm water collection and conveyance system, associated points of discharge, and direction of flow. Include any structural control measures that affect storm water discharges, authorized non-storm water discharges, and run-on. Examples of structural control measures are catch basins, berms, detention ponds, secondary containment, oil/water separators, diversion barriers, etc.
- c. An outline of all impervious areas of the facility, including paved areas, buildings, covered storage areas, or other roofed structures.
- d. Locations where materials are directly exposed to precipitation and the locations where significant spills or leaks identified in Section 6.a.(4)., below, have occurred.
- e. Areas of industrial activity. This shall include the locations of all storage areas and storage tanks, shipping and receiving areas, fueling areas, vehicle and equipment storage/maintenance areas, material handling and processing areas, waste treatment and disposal areas, dust or particulate generating areas, cleaning and rinsing areas, and other areas of industrial activity which are potential pollutant sources.

5. List of Significant Materials

The SWPPP shall include a list of significant materials handled and stored at the site. For each material on the list, describe the locations where the material is being stored, received, shipped, and handled, as well as the typical quantities and frequency. Materials shall include raw materials, intermediate products, final or finished products, recycled materials, and waste or disposed materials.

6. Description of Potential Pollutant Sources

- a. The SWPPP shall include a narrative description of the facility's industrial activities, as identified in Section 4.e., above, associated potential pollutant sources, and potential pollutants that could be discharged in storm water discharges or authorized non-storm water discharges. At a minimum, the following items related to a facility's industrial activities shall be considered:

(1) Industrial Processes

Describe each industrial process, the type, characteristics, and quantity of significant materials used in or resulting from the process, and a description of the processes (manufacturing or treatment), cleaning, rinsing, recycling, disposal, or other activities related to the process. Where applicable, areas protected by containment structures and the corresponding containment capacity shall be described.

(2) Material Handling and Storage Areas

Describe each handling and storage area, type, characteristics, and quantity of significant materials handled or stored, description of the shipping, receiving, and loading procedures, and the spill or leak prevention and response procedures. Where applicable, areas protected by containment structures and the corresponding containment capacity shall be described.

(3) Dust and Particulate Generating Activities

Describe all industrial activities that generate dust or particulates that may be deposited within the facility's boundaries and identify their discharge locations; the characteristics of dust and particulate pollutants; the approximate quantity of dust and particulate pollutants that may be deposited within the facility boundaries; and a description of the primary areas of the facility where dust and particulate pollutants would settle.

(4) Significant Spills and Leaks

Describe materials that have spilled or leaked in significant quantities in storm water discharges or non-storm water discharges. Include toxic chemicals (listed in 40 Code of Federal Regulations [CFR] Part 302) that have been discharged to storm water as reported on U.S. Environmental Protection Agency (U.S. EPA) Form R, and oil and hazardous substances in excess of reportable quantities (see 40 CFR, Parts 110, 117, and 302).

The description shall include the type, characteristics, and approximate quantity of the material spilled or leaked, the cleanup or remedial actions that have occurred or are planned, the approximate remaining quantity of materials that may be exposed to storm water or non-storm water discharges, and the preventative measures taken to ensure spills or leaks do not reoccur. Such list shall be updated as appropriate during the term of Order No. R8-2006-0003.

(5) Non-Storm Water Discharges

The discharger shall investigate the facility to identify all non-storm water discharges and their sources. As part of this investigation, all drains (inlets and outlets) shall be evaluated to identify whether they connect to the storm drain system.

All non-storm water discharges shall be described. This shall include the source, quantity, frequency, and characteristics of the non-storm water discharges and associated drainage area.

Non-storm water discharges that contain significant quantities of pollutants or that do not meet the conditions of Order No. R8-2006-0003 are prohibited. (Examples of prohibited non-storm water discharges are contact and non-contact cooling water, boiler blowdown, rinse water, wash water, etc.). The SWPPP must include BMPs to prevent or reduce contact of non-storm water discharges with significant materials or equipment.

(6) Soil Erosion

Describe the facility locations where soil erosion may occur as a result of industrial activity, storm water discharges associated with industrial activity, or authorized non-storm water discharges.

- b. The SWPPP shall include a summary of all areas of industrial activities, potential pollutant sources, and potential pollutants. This information should be summarized similar to Table B, below. The last column of Table B, "Control Practices", should be completed in accordance with Section 8., below.

7. Assessment of Potential Pollutant Sources

- a. The SWPPP shall include a narrative assessment of all industrial activities and potential pollutant sources as described in Section 6., above, to determine:
 - (1) Which areas of the facility are likely sources of pollutants in storm water discharges and authorized non-storm water discharges, and
 - (2) Which pollutants are likely to be present in storm water discharges and authorized non-storm water discharges. The discharger shall consider and evaluate various factors when performing this assessment such as current storm water BMPs; quantities of significant materials handled, produced, stored, or disposed of; likelihood of exposure to storm water or authorized non-storm water discharges; history of spill or leaks; and run-on from outside sources.

- b. The discharger shall summarize the areas of the facility that are likely sources of pollutants and the corresponding pollutants that are likely to be present in storm water discharges and authorized non-storm water discharges.

The discharger is required to develop and implement additional BMPs as appropriate and necessary to prevent or reduce pollutants associated with each pollutant source. The BMPs will be narratively described in Section 8., below.

8. Storm Water Best Management Practices

The SWPPP shall include a narrative description of the storm water BMPs to be implemented at the facility for each potential pollutant and its source identified in the site assessment phase (Sections 6. and 7., above). The BMPs shall be developed and implemented to reduce or prevent pollutants in storm water discharges and authorized non-storm water discharges. Each pollutant and its source may require one or more BMPs. Some BMPs may be implemented for multiple pollutants and their sources, while other BMPs will be implemented for a very specific pollutant and its source.

The description of the BMPs shall identify the BMPs as (1) existing BMPs, (2) existing BMPs to be revised and implemented, or (3) new BMPs to be implemented. The description shall also include a discussion on the effectiveness of each BMP to reduce or prevent pollutants in storm water discharges and authorized non-storm water discharges. The SWPPP shall provide a summary of all BMPs implemented for each pollutant source. This information should be summarized similar to Table B.

The discharger shall consider the following BMPs for implementation at the facility:

- a. Non-Structural BMPs: Non-structural BMPs generally consist of processes, prohibitions, procedures, schedule of activities, etc., that prevent pollutants associated with industrial activity from contacting with storm water discharges and authorized non-storm water discharges. They are considered low technology, cost-effective measures. The discharger should consider all possible non-structural BMPs options before considering additional structural BMPs (see Section 8.b., below). Below is a list of non-structural BMPs that should be considered:
 - (1) Good Housekeeping: Good housekeeping generally consist of practical procedures to maintain a clean and orderly facility.
 - (2) Preventive Maintenance: Preventive maintenance includes the regular inspection and maintenance of structural storm water controls (catch basins, oil/water separators, etc.) as well as other facility equipment and systems.
 - (3) Spill Response: This includes spill clean-up procedures and necessary clean-up equipment based upon the quantities and locations of significant materials that may spill or leak.
 - (4) Material Handling and Storage: This includes all procedures to minimize the potential for spills and leaks and to minimize exposure of significant materials to storm water and authorized non-storm water discharges.

- (5) **Employee Training:** This includes training of personnel who are responsible for (a) implementing activities identified in the SWPPP, (b) conducting inspections, sampling, and visual observations, and (c) managing storm water. Training should address topics such as spill response, good housekeeping, and material handling procedures, and actions necessary to implement all BMPs identified in the SWPPP. The SWPPP shall identify periodic dates for such training. Records shall be maintained of all training sessions held.
 - (6) **Waste Handling/Recycling:** This includes the procedures or processes to handle, store, or dispose of waste materials or recyclable materials.
 - (7) **Record Keeping and Internal Reporting:** This includes the procedures to ensure that all records of inspections, spills, maintenance activities, corrective actions, visual observations, etc., are developed, retained, and provided, as necessary, to the appropriate facility personnel.
 - (8) **Erosion Control and Site Stabilization:** This includes a description of all sediment and erosion control activities. This may include the planting and maintenance of vegetation, diversion of run-on and runoff, placement of sandbags, silt screens, or other sediment control devices, etc.
 - (9) **Inspections:** This includes, in addition to the preventative maintenance inspections identified above, an inspection schedule of all potential pollutant sources. Tracking and follow-up procedures shall be described to ensure adequate corrective actions are taken and SWPPPs are made.
 - (10) **Quality Assurance:** This includes the procedures to ensure that all elements of the SWPPP and Monitoring Program are adequately conducted.
- b. **Structural BMPs:** Where non-structural BMPs as identified in Section 8.a., above, are not effective, structural BMPs shall be considered. Structural BMPs generally consist of structural devices that reduce or prevent pollutants in storm water discharges and authorized non-storm water discharges. Below is a list of structural BMPs that should be considered:
- (1) **Overhead Coverage:** This includes structures that provide horizontal coverage of materials, chemicals, and pollutant sources from contact with storm water and authorized non-storm water discharges.
 - (2) **Retention Ponds:** This includes basins, ponds, surface impoundments, bermed areas, etc., that do not allow storm water to discharge from the facility.
 - (3) **Control Devices:** This includes berms or other devices that channel or route run-on and runoff away from pollutant sources.
 - (4) **Secondary Containment Structures:** This generally includes containment structures around storage tanks and other areas for the purpose of collecting any leaks or spills.

- (5) Treatment: This includes inlet controls, infiltration devices, oil/water separators, detention ponds, vegetative swales, etc., that reduce the pollutants in storm water discharges and authorized non-storm water discharges.

9. Annual Comprehensive Site Compliance Evaluation

The discharger shall conduct one comprehensive site compliance evaluation in each reporting period (July 1-June 30). Evaluations shall be conducted within 8-16 months of each other. The SWPPP shall be revised, as appropriate, and the revisions implemented within 90 days of the evaluation. Evaluations shall include the following:

- a. A review of all visual observation records, inspection records, and sampling and analysis results.
- b. A visual inspection of all potential pollutant sources for evidence of, or the potential for, pollutants entering the drainage system.
- c. A review and evaluation of all BMPs (both structural and non-structural) to determine whether the BMPs are adequate, properly implemented and maintained, or whether additional BMPs are needed. A visual inspection of equipment needed to implement the SWPPP, such as spill response equipment, shall be included.
- d. An evaluation report that includes, (1) identification of personnel performing the evaluation, (2) the date(s) of the evaluation, (3) necessary SWPPP revisions, (4) schedule, as required in Section 10.e, below, for implementing SWPPP revisions, (5) any incidents of non-compliance and the corrective actions taken, and (6) a certification that the discharger is in compliance with Order No. R8-2006-0003. If the above certification cannot be provided, explain in the evaluation report why the discharger is not in compliance with this order. The evaluation report shall be submitted as part of the annual report, retained for at least five years, and signed and certified.

10. SWPPP General Requirements

- a. The SWPPP shall be retained on site and made available upon request by a representative of the Regional Water Board and/or local storm water management agency (local agency) which receives the storm water discharges.
- b. The Regional Water Board and/or local agency may notify the discharger when the SWPPP does not meet one or more of the minimum requirements of this section. As requested by the Regional Water Board and/or local agency, the discharger shall submit a SWPPP revision and implementation schedule that meets the minimum requirements of this section to the Regional Water Board and/or local agency that requested the SWPPP revisions. Within 14 days after implementing the required SWPPP revisions, the discharger shall provide written certification to the Regional Water Board and/or local agency that the revisions have been implemented.

- c. The SWPPP shall be revised, as appropriate, and implemented prior to changes in industrial activities which (1) may significantly increase the quantities of pollutants in storm water discharge, (2) cause a new area of industrial activity at the facility to be exposed to storm water, or (3) begin an industrial activity which would introduce a new pollutant source at the facility.
- d. The SWPPP shall be revised and implemented in a timely manner, but in no case more than 90 days after a discharger determines that the SWPPP is in violation of any requirement(s) of Order No. R8-2006-0003.
- e. When any part of the SWPPP is infeasible to implement by the deadlines specified in Order No. R8-2006-0003, due to proposed significant structural changes, the discharger shall submit a report to the Regional Water Board prior to the applicable deadline that (1) describes the portion of the SWPPP that is infeasible to implement by the deadline, (2) provides justification for a time extension, (3) provides a schedule for completing and implementing that portion of the SWPPP, and (4) describes the BMPs that will be implemented in the interim period to reduce or prevent pollutants in storm water discharges and authorized non-storm water discharges. Such reports are subject to Regional Water Board approval and/or modifications. The discharger shall provide written notification to the Regional Water Board within 14 days after the SWPPP revisions are implemented.
- f. The SWPPP shall be provided, upon request, to the Regional Water Board. The SWPPP is considered a report that shall be available to the public by the Regional Water Board under Section 308(b) of the Clean Water Act.

TABLE A

**FIVE PHASES FOR DEVELOPING AND IMPLEMENTING INDUSTRIAL
STORM WATER POLLUTION PREVENTION PLANS**

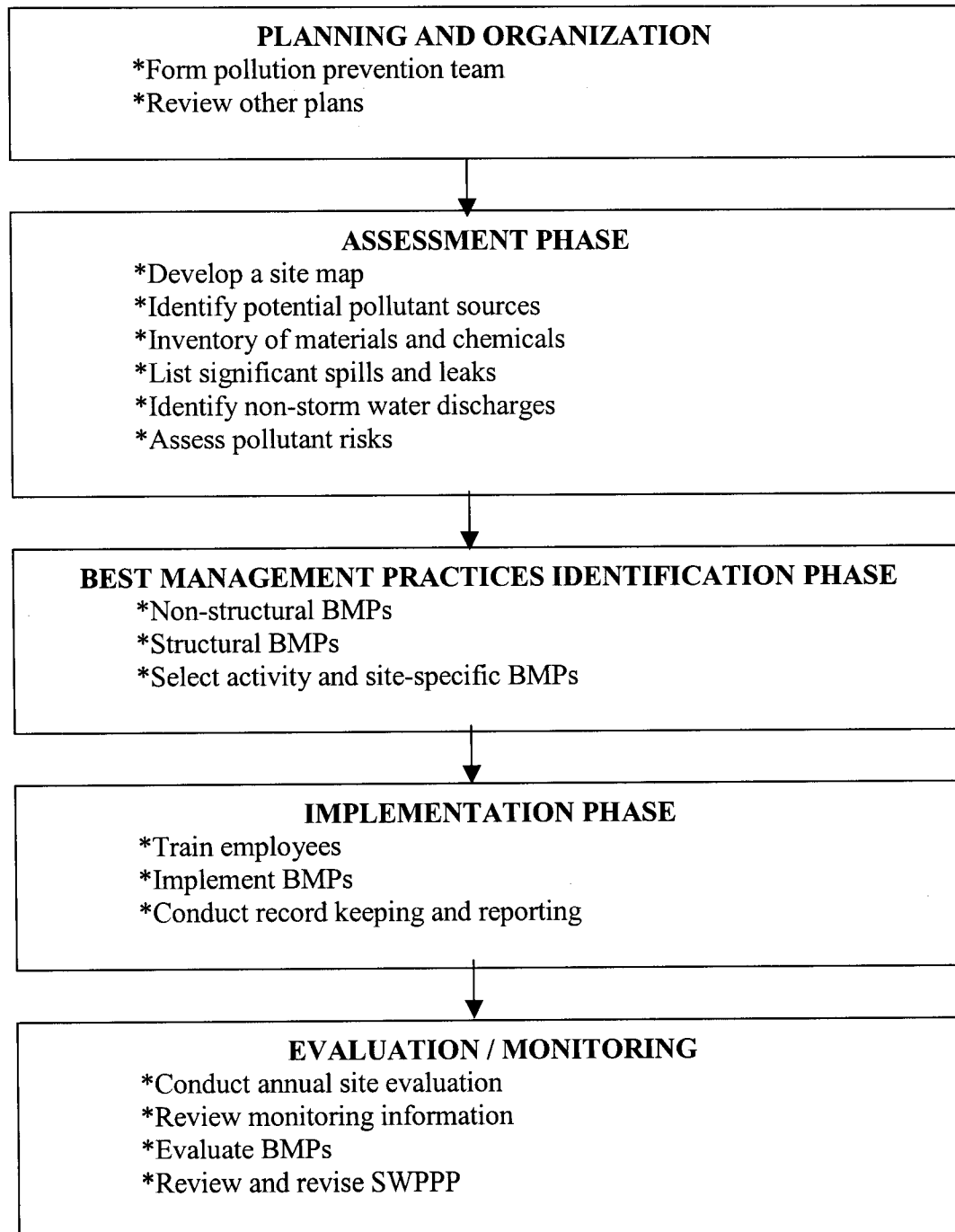


TABLE B

EXAMPLE

**ASSESSMENT OF POTENTIAL POLLUTION SOURCES AND
CORRESPONDING BEST MANAGEMENT PRACTICES
SUMMARY**

AREA	ACTIVITY	POLLUTANT SOURCE	POLLUTANT	BEST MANAGEMENT PRACTICES
Vehicle & equipment fueling	Fueling	Spills and leaks during delivery	Fuel oil	- Use spill and overflow protection - Minimize run-on of storm water into the fueling area - Cover fueling area - Use dry cleanup methods rather than hosing down area - Implement proper spill prevention control program - Implement adequate preventative maintenance program to prevent tank and line leaks - Inspect fueling areas regularly to detect problems before they occur - Train employees on proper fueling, cleanup, and spill response techniques.
		Spills caused by topping off fuel oil	Fuel oil	
		Hosing or washing down fuel area	Fuel oil	
		Leaking storage tanks	Fuel oil	
		Rainfall running off fueling areas, and rainfall running onto and off fueling area	Fuel oil	

ATTACHMENT K - STORMWATER MONITORING PROGRAM AND REPORTING REQUIREMENTS

1. Implementation Schedule

The discharger shall continue to implement their existing Stormwater monitoring program and implement any necessary revisions to their Stormwater monitoring program in a timely manner, but in no case later than March 1, 2006. The discharger may use the monitoring results conducted in accordance with their existing Stormwater monitoring program to satisfy the pollutant/parameter reduction requirements in Section 5.c., below, and Sampling and Analysis Exemptions and Reduction Certifications in Section 10, below.

2. Objectives

The objectives of the monitoring program are to:

- a. Ensure that storm water discharges are in compliance with waste discharge requirements specified in Order No. R8-2006-0003.
- b. Ensure practices at the facility to reduce or prevent pollutants in storm water discharges and authorized non-storm water discharges are evaluated and revised to meet changing conditions.
- c. Aid in the implementation and revision of the SWPPP required by Attachment "J" Stormwater Pollution Prevention Plan of Order No. R8-2006-0003.
- d. Measure the effectiveness of best management practices (BMPs) to prevent or reduce pollutants in storm water discharges and authorized non-storm water discharges. Much of the information necessary to develop the monitoring program, such as discharge locations, drainage areas, pollutant sources, etc., should be found in the Storm Water Pollution Prevention Plan (SWPPP). The facility's monitoring program shall be a written, site-specific document that shall be revised whenever appropriate and be readily available for review by employees or Regional Water Board inspectors.

3. Non-Storm Water Discharge Visual Observations

- a. The discharger shall visually observe all drainage areas within their facility for the presence of unauthorized non-storm water discharges;
- b. The discharger shall visually observe the facility's authorized non-storm water discharges and their sources;

- c. The visual observations required above shall occur quarterly, during daylight hours, on days with no storm water discharges, and during scheduled facility operating hours¹. Quarterly visual observations shall be conducted in each of the following periods: January-March, April-June, July-September, and October-December. The discharger shall conduct quarterly visual observations within 6-18 weeks of each other.
- d. Visual observations shall document the presence of any discolorations, stains, odors, floating materials, etc., as well as the source of any discharge. Records shall be maintained of the visual observation dates, locations observed, observations, and response taken to eliminate unauthorized non-storm water discharges and to reduce or prevent pollutants from contacting non-storm water discharges. The SWPPP shall be revised, as necessary, and implemented in accordance with Attachment "J" Stormwater Pollution Prevention Plan of Order No. R8-2006-0003.

4. Storm Water Discharge Visual Observations

- a. With the exception of those facilities described in Section 4.d., below, the discharger shall visually observe storm water discharges from one storm event per month during the wet season (October 1-May 30). These visual observations shall occur during the first hour of discharge and at all discharge locations. Visual observations of stored or contained storm water shall occur at the time of release.
- b. Visual observations are only required of storm water discharges that occur during daylight hours that are preceded by at least three (3) working days² without storm water discharges and that occur during scheduled facility operating hours.
- c. Visual observations shall document the presence of any floating and suspended material, oil and grease, discolorations, turbidity, odor, and source of any pollutants. Records shall be maintained of observation dates, locations observed, observations, and response taken to reduce or prevent pollutants in storm water discharges. The SWPPP shall be revised, as necessary, and implemented in accordance with Attachment "J" Stormwater Pollution Prevention Plan of Order No. R8-2006-0003.
- d. The discharger with storm water containment facilities shall conduct monthly inspections of their containment areas to detect leaks and ensure maintenance of adequate freeboard. Records shall be maintained of the inspection dates, observations, and any response taken to eliminate leaks and to maintain adequate freeboard.

¹ "Scheduled facility operating hours" are the time periods when the facility is staffed to conduct any function related to industrial activity, but excluding time periods where only routine maintenance, emergency response, security, and/or janitorial services are performed.

² Three (3) working days may be separated by non-working days such as weekends and holidays provided that no storm water discharges occur during the three (3) working days and the non-working days.

5. Sampling and Analysis

- a. The discharger shall collect storm water samples during the first hour of discharge from (1) the first storm event of the wet season, and (2) at least one other storm event in the wet season. All storm water discharge locations shall be sampled. Sampling of stored or contained storm water shall occur at the time the stored or contained storm water is released. The discharger that does not collect samples from the first storm event of the wet season are still required to collect samples from two other storm events of the wet season and shall explain in the "Annual Stormwater Report" (see Section 12, below) why the first storm event was not sampled.
- b. Sample collection is only required of storm water discharges that occur during scheduled facility operating hours and that are preceded by at least (3) three working days without storm water discharge.
- c. The samples shall be analyzed for:
 - (1) Total suspended solids (TSS) pH, specific conductance, and total organic carbon (TOC). Oil and grease (O&G) may be substituted for TOC;
 - (2) Toxic chemicals and other pollutants that are likely to be present in storm water discharges in significant quantities. If these pollutants are not detected in significant quantities after two consecutive sampling events, the discharger may eliminate the pollutant from future sample analysis until the pollutant is likely to be present again;
 - (3) The discharger is not required to analyze a parameter when either of the two following conditions are met: (a) the parameter has not been detected in significant quantities from the last two consecutive sampling events, or (b) the parameter is not likely to be present in storm water discharges and authorized non-storm water discharges in significant quantities based upon the discharger's evaluation of the facilities industrial activities, potential pollutant sources, and SWPPP; and
 - (4) Other parameters as required by the Regional Water Board.

6. Sample Storm Water Discharge Locations

- a. The discharger shall visually observe and collect samples of storm water discharges from all drainage areas that represent the quality and quantity of the facility's storm water discharges from the storm event.
- b. If the facility's storm water discharges are commingled with run-on from surrounding areas, the discharger should identify other visual observation and sample collection locations that have not been commingled by run-on and that represent the quality and quantity of the facility's storm water discharges from the storm event.

- c. If visual observation and sample collection locations are difficult to observe or sample (e.g., sheet flow, submerged outfalls), the discharger shall identify and collect samples from other locations that represent the quality and quantity of the facility's storm water discharges from the storm event.
- d. The discharger that determines that the industrial activities and BMPs within two or more drainage areas are substantially identical may either (1) collect samples from a reduced number of substantially identical drainage areas, or (2) collect samples from each substantially identical drainage area and analyze a combined sample from each substantially identical drainage area. The discharger must document such a determination in the annual Stormwater report.

7. Visual Observation and Sample Collection Exceptions

The discharger is required to be prepared to collect samples and conduct visual observations at the beginning of the wet season (October 1) and throughout the wet season until the minimum requirements of Sections 4. and 5., above, are completed with the following exceptions:

- a. The discharger is not required to collect a sample and conduct visual observations in accordance with Section 4 and Section 5, above, due to dangerous weather conditions, such as flooding, electrical storm, etc., when storm water discharges begin after scheduled facility operating hours or when storm water discharges are not preceded by three working days without discharge. Visual observations are only required during daylight hours. The discharger that does not collect the required samples or visual observations during a wet season due to these exceptions shall include an explanation in the "Annual Stormwater Report" why the sampling or visual observations could not be conducted.
- b. The discharger may conduct visual observations and sample collection more than one hour after discharge begins if the discharger determines that the objectives of this section will be better satisfied. The discharger shall include an explanation in the "Annual Stormwater Report" why the visual observations and sample collection should be conducted after the first hour of discharge.

8. Alternative Monitoring Procedures

The discharger may propose an alternative monitoring program that meets Section 2, above, monitoring program objectives for approval by the Regional Water Board's Executive Officer. The discharger shall continue to comply with the monitoring requirements of this section and may not implement an alternative monitoring plan until the alternative monitoring plan is approved by the Regional Water Board's Executive Officer. Alternative monitoring plans are subject to modification by the Regional Water Board's Executive Officer.

9. Monitoring Methods

- a. The discharger shall explain how the facility's monitoring program will satisfy the monitoring program objectives of Section 2., above. This shall include:
 - (1) Rationale and description of the visual observation methods, location, and frequency;
 - (2) Rationale and description of the sampling methods, location, and frequency; and
 - (3) Identification of the analytical methods and corresponding method detection limits used to detect pollutants in storm water discharges. This shall include justification that the method detection limits are adequate to satisfy the objectives of the monitoring program.
- b. All sampling and sample preservation shall be in accordance with the current edition of "Standard Methods for the Examination of Water and Wastewater" (American Public Health Association). All monitoring instruments and equipment (including the discharger's own field instruments for measuring pH and Electro-conductivity) shall be calibrated and maintained in accordance with manufacturers' specifications to ensure accurate measurements. All laboratory analyses must be conducted according to test procedures under 40 CFR Part 136, unless other test procedures have been specified in Order No. R8-2006-0003 or by the Regional Water Board's Executive Officer. All metals shall be reported as total recoverable metals or unless otherwise specified in Order No. R8-2006-0003. With the exception of analysis conducted by the discharger, all laboratory analyses shall be conducted at a laboratory certified for such analyses by the State Department of Health Services. The discharger may conduct their own sample analyses if the discharger has sufficient capability (qualified employees, laboratory equipment, etc.) to adequately perform the test procedures.

10. Sampling and Analysis Exemptions and Reductions

A discharger who qualifies for sampling and analysis exemptions, as described below in Section 10.a.(1) or who qualifies for reduced sampling and analysis, as described below in Section 10.b., must submit the appropriate certifications and required documentation to the Regional Water Board prior to the wet season (October 1) and certify as part of the annual Stormwater report submittal. A discharger that qualifies for either the Regional Water Board or local agency certification programs, as described below in Section 10.a.(2) and (3), shall submit certification and documentation in accordance with the requirements of those programs. The discharger who provides certification(s) in accordance with this section are still required to comply with all other monitoring program and reporting requirements. The discharger shall prepare and submit their certification(s) using forms and instructions provided by the State Water Board, Regional Water Board, or local agency or shall submit their information on a form that contains equivalent information. The discharger whose facility no longer meets the certification conditions must notify the Regional Water Board's Executive Officer (and local agency) within 30 days and immediately comply with Section 5., Sampling and Analysis requirements. Should a Regional Water Board (or local agency) determine that a certification does not meet the conditions set forth below, the discharger must immediately comply with the Section 5., Sampling and Analysis requirements.

a. Sampling and Analysis Exemptions

A discharger is not required to collect and analyze samples in accordance with Section 5., above, if the discharger meets all of the conditions of one of the following certification programs:

(1) No Exposure Certification (NEC)

This exemption is designed primarily for those facilities where all industrial activities are conducted inside buildings and where all materials stored and handled are not exposed to storm water. To qualify for this exemption, the discharger must certify that their facilities meet all of the following conditions:

- (a) All prohibited non-storm water discharges have been eliminated or otherwise permitted.
- (b) All authorized non-storm water discharges have been identified and addressed in the SWPPP.
- (c) All areas of past exposure have been inspected and cleaned, as appropriate.
- (d) All significant materials related to industrial activity (including waste materials) are not exposed to storm water or authorized non-storm water discharges.
- (e) All industrial activities and industrial equipment are not exposed to storm water or authorized non-storm water discharges.
- (f) There is no exposure of storm water to significant materials associated with industrial activity through other direct or indirect pathways such as from industrial activities that generate dust and particulates.
- (g) There is periodic re-evaluation of the facility to ensure conditions (a), (b), (d), (e), and (f) above are continuously met. At a minimum, re-evaluation shall be conducted once a year.

(2) Regional Water Board Certification Programs

The Regional Water Board may grant an exemption to the Section 5. Sampling and Analysis requirements if it determines a discharger has met the conditions set forth in a Regional Water Board certification program. Regional Water Board certification programs may include conditions to (a) exempt the discharger whose facilities infrequently discharge storm water to waters of the United States, and (b) exempt the discharger that demonstrate compliance with the terms and conditions of Order No. R8-2006-0003.

(3) Local Agency Certifications

A local agency may develop a local agency certification program. Such programs must be approved by the Regional Water Board. An approved local agency program may either grant an exemption from Section 5. Sampling and Analysis requirements or reduce the frequency of sampling if it determines that a discharger has demonstrated compliance with the terms and conditions of the Industrial Activities Storm Water General Permit Order No. 97-03-DWQ which was adopted by the State Water Resources Control Board on April 17, 1997.

b. Sampling and Analysis Reduction

- (1) A discharger may reduce the number of sampling events required to be sampled for the remaining term of Order No. R8-2006-0003 if the discharger provides certification that the following conditions have been met:
 - (a) The discharger has collected and analyzed samples from a minimum of six storm events from all required drainage areas;
 - (b) All prohibited non-storm water discharges have been eliminated or otherwise permitted;
 - (c) The discharger demonstrates compliance with the terms and conditions of the Order No. R8-2006-0003 for the previous two years (i.e., completed Annual Stormwater Reports, performed visual observations, implemented appropriate BMPs, etc.);
 - (d) The discharger demonstrates that the facility's storm water discharges and authorized non-storm water discharges do not contain significant quantities of pollutants; and
 - (e) Conditions (b), (c), and (d) above are expected to remain in effect for a minimum of one year after filing the certification.

11. Records

Records of all storm water monitoring information and copies of all reports (including the Annual Stormwater Reports) required by Order No. R8-2006-0003 shall be retained for a period of at least five years. These records shall include:

- a. The date, place, and time of site inspections, sampling, visual observations, and/or measurements;
- b. The individual(s) who performed the site inspections, sampling, visual observations, and or measurements;

- c. Flow measurements or estimates;
- d. The date and approximate time of analyses;
- e. The individual(s) who performed the analyses;
- f. Analytical results, method detection limits, and the analytical techniques or methods used;
- g. Quality assurance/quality control records and results;
- h. Non-storm water discharge inspections and visual observations and storm water discharge visual observation records (see Sections 3. and 4., above);
- i. Visual observation and sample collection exception records (see Section 5.a, 6.d, 7, and 10.b.(2), above);
- j. All calibration and maintenance records of on-site instruments used;
- k. All Sampling and Analysis Exemption and Reduction certifications and supporting documentation (see Section 10);
- l. The records of any corrective actions and follow-up activities that resulted from the visual observations.

12. Annual Report

The discharger shall submit an Annual Stormwater Report by July 1 of each year to the Executive Officer of the Regional Water Board and to the local agency (if requested). The report shall include a summary of visual observations and sampling results, an evaluation of the visual observation and sampling and analysis results, laboratory reports, the Annual Comprehensive Site Compliance Evaluation Report required in Section 9. of Attachment "J" of Order No. R8-2006-0003, an explanation of why a facility did not implement any activities required by Order No. R8-2006-0003 (if not already included in the Evaluation Report), and records specified in Section 11., above. The method detection limit of each analytical parameter shall be included. Analytical results that are less than the method detection limit shall be reported as "less than the method detection limit". The discharger shall prepare and submit their Annual Stormwater Reports using the annual report forms provided by the State Water Board or Regional Water Board or shall submit their information on a form that contains equivalent information.

13. Watershed Monitoring Option

Regional Water Boards may approve proposals to substitute watershed monitoring for some or all of the requirements of this section if the Regional Water Board finds that the watershed monitoring will provide substantially similar monitoring information in evaluating discharger compliance with the requirements of Order No. R8-2006-0003.

**ATTACHMENT L - City of Beaumont and San Timoteo Watershed Management Authority
 Maximum Benefit Commitments**

Description of Commitment	Compliance Date – as soon as possible, but no later than
1. Surface Water Monitoring Program a. Submit Draft Monitoring Program to Regional Board b. Implement Monitoring Program c. Quarterly data report submittal d. Annual data report submittal	a. February 24, 2005 (submitted) b. Within 30 days from Regional Board approval of monitoring plan c. April 15, July 15, October 15, January 15 d. February 15th
2. Groundwater Monitoring Program a. Submit Draft Monitoring Program to Regional Board b. Implement Monitoring Program c. Annual data report submittal	a. February 24, 2005 (submitted) b. Within 30 days from Regional Board approval of monitoring plan c. February 15th
3. Desalter(s) and Brine Disposal Facilities a. Submit plan and schedule for construction of desalter(s) and brine disposal facilities. Facilities are to be operational as soon as possible but no later than 7 years from date of Regional Board approval of plan/schedule. b. Implement the plan and schedule	a. Within 6 months of either of the following: i. When Beaumont's effluent 5-year running average TDS exceeds 480 mg/L; and/or ii. When volume weighted average concentration in the Beaumont MZ of TDS exceeds 320 mg/L b. Within 30 days from Regional Board approval of plan/schedule
4. Non-potable water supply Implement non-potable water supply system to serve water for irrigation purposes. The non-potable supply shall comply with a 10-year running average TDS concentration of 390 mg/L or less	<i>January 24, 2015</i>

Description of Commitment	Compliance Date – as soon as possible, but no later than
5. Recycled water recharge ¹ The recharge of recycled water in the Beaumont or San Timoteo Management Zones shall be limited to the amount that can be blended with other recharge sources to achieve a 5-year running average equal to or less than the “maximum benefit” objectives for TDS and nitrate-nitrogen for the relevant Management Zone(s). a. Submit baseline report of amount, locations, and TDS and nitrogen quality of stormwater/imported water recharge. b. Submit documentation of amount, TDS and nitrogen quality of all sources of recharge and recharge locations. For stormwater recharge used for blending, submit documentation that the recharge is the result of City of Beaumont/STWMA enhanced recharge facilities/programs	Compliance must be achieved by end of 5th year after initiation of recycled water use/recharge operations. a. Prior to initiation of construction of basins/other facilities to support enhanced stormwater/imported water recharge. b. Annually, by January 15th, after initiation construction of facilities/implementation of programs to support enhanced recharge.
6. Ambient groundwater quality determination	July 1, 2005 (submitted) and every 3 years thereafter
7. Replace denitrification facilities (if necessary to comply with TIN wasteload allocation specified in CITATION FOR EFFLUENT LIMIT)	Compliance with 6 mg/L TIN limitation to be achieved by January 24, 2008
8. City of Beaumont recycled water quality improvement plan and schedule a. Submit plan and schedule	a. 60 days after the TDS 12-month running average effluent quality equals or exceeds 480 mg/L for 3 consecutive months and/or the 12-month running average TIN concentration equals or exceeds 6 mg/L in any month (once facility/operational changes needed to achieve 6 mg/L TIN are in place)

¹ The Discharger has not yet proposed specific plans to implement recycled water recharge projects. Accordingly, appropriate requirements for such projects are not specified in this Order. Therefore, compliance with the recycled water recharge maximum benefit commitment is not now required to assure effective maximum benefit implementation. This Order will be reconsidered and revised as necessary to implement relevant maximum benefit commitments if and when the Discharger proposes recycled water projects.

Description of Commitment	Compliance Date – as soon as possible, but no later than
b. Implement plan and schedule	b. Upon approval by Regional Board
9. Remove/reduce the discharge of Beaumont's effluent from the unlined portion of San Timoteo Creek a. Submit proposed plan/schedule b. Implement plan/schedule	a. July 24, 2005 b. Upon Regional Board approval

California Regional Water Quality Control Board
Santa Ana Region

January 18, 2006

ITEM: 7

SUBJECT: Reissuance of Waste Discharge Requirements for the City of Beaumont's Wastewater Treatment Plant No. 1, Order No. R8-2006-0003, NPDES No. CA0105376, Riverside County

DISCUSSION:

See attached Fact Sheet

RECOMMENDATIONS:

Adopt Order No. R8-2006-0003, NPDES No. CA0105376 as presented.

COMMENT SOLICITATION:

Comments were solicited from the discharger and the following agencies:

U.S. Environmental Protection Agency, Permits Issuance Section (WTR-5) – Dough Eberhardt
U.S. Army District, Los Angeles, Corps of Engineers, Regulatory Branch
U.S. Fish and Wildlife Service – Carlsbad
State Water Resources Control Board, Office of the Chief Counsel – Jorge Leon
State Water Resources Control Board, Division of Water Quality – Jim Maughan
California Department of Health Services, San Diego – Steve Williams
State Department of Water Resources - Glendale
State Department of Fish and Game – Long Beach
Orange County Water District – Nira Yamachika
Riverside County Department of Environmental Health Services
Riverside County Transportation/Flood Control Department
Santa Ana River Dischargers Association
City of Beaumont – Dayle Keller
Orange County Coastkeeper – Garry Brown
Lawyers for Clean Water C/c San Francisco Baykeeper
Wildermuth Environmental, Inc. – Kristal Davis